

Camunda BPM Platform
Loan Assessment Process Lab

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Camunda BPM Platform - Loan Assessment Process lab

The lab is built using the Loan Assessment process as described in Chapter 9 of the textbook "Fundamentals of Business Process Management" (Springer, 2013) authored by M. Dumas, M. La Rosa, J. Mendling and H.A. Reijers. The complete process description can be downloaded from http://fundamentals-of-bpm.org/wp-content/uploads/LoanAssessmentProcessDescription.zip.

Camunda is an open source Java based BPM platform used primarily to automate Business Process Model and Notation (BPMN) 2.0 processes. It is built around the process engine component. The software is still in the development stage and some features are currentl being implemented. There are certain limitations to the use of BPMN 2.0 elements in Camunda. Some elements are dependent on specific modelling criteria, failing to satisfy which doesn't produce the expected outputs. Keeping these hindrances in mind, the loan assessment process has been redesigned as and when needed to overcome them.

Constraints observed have been listed with necessary references and inputs from Camunda development team. Java and Groovy programming languages have been used to integrate some tasks into the loan assessment process.

This lab uses Camunda BPM Platform V7.1.0-Final on Apache Tomcat 7 server. A java development kit (JDK) version 6.0 or greater must be installed to run the software. Eclipse IDE is also required for the process designer.

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1.0 Camunda BPM Introduction

Camunda BPM is a lightweight, open source platform for Business Process Management. It is dedicated to Java developers and their typical software development infrastructure, while providing Business IT-Alignment during process design and runtime using the BPMN 2.0 standard.

Camunda's core is a native BPMN 2.0 process engine that runs inside the Java Virtual Machine. It integrates with Java EE 6 and is a perfect match for the Spring Framework. On top of the process engine, one can also choose from a stack of tools for human workflow management, operations and monitoring.

The tutorial uses Camunda version V7.1.0-Final.

1.1 Process Engine and Infrastructure

- Process Engine: The process engine is a java library responsible for executing BPMN 2.0 processes and workflows. It has a lightweight POJO core and uses a relational database for persistence. ORM mapping is provided by the mybatis mapping framework.
- Spring Framework Integration
- CDI/Java EE Integration
- Runtime Container Integration (Integration with application server infrastructure)

1.2 Web Applications

- REST API: The REST API allows using the process engine from a remote application or a Java Script application.
- Camunda Tasklist: A web application for human workflow management and user tasks. The task list allows process participants to inspect their workflow tasks and navigate to task forms in order to work on the tasks and provide data input.
- Camunda Cockpit: A web application for process monitoring and operations. Camunda Cockpit allows searching for process instanced, inspecting their state and repairing broken instances.
- Camunda Cycle: A web application for synchronizing BPMN 2.0 process models between different modelling tools and modellers

1.3 Additional Tools

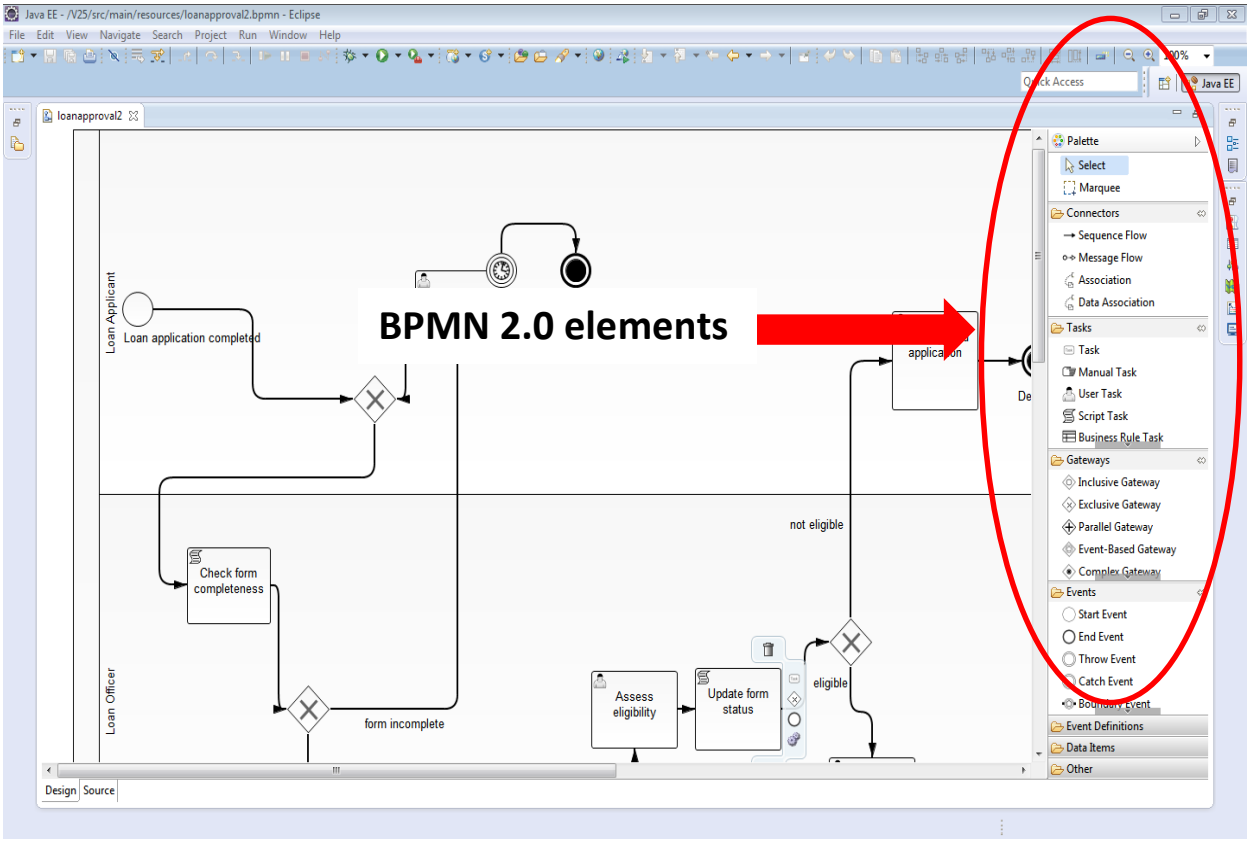
- Camunda Modeller: Eclipse plug-in for process modelling.
- Camunda-bpmn.js: JavaScript framework for parsing, rendering and executing BPMN 2.0 from XML source.

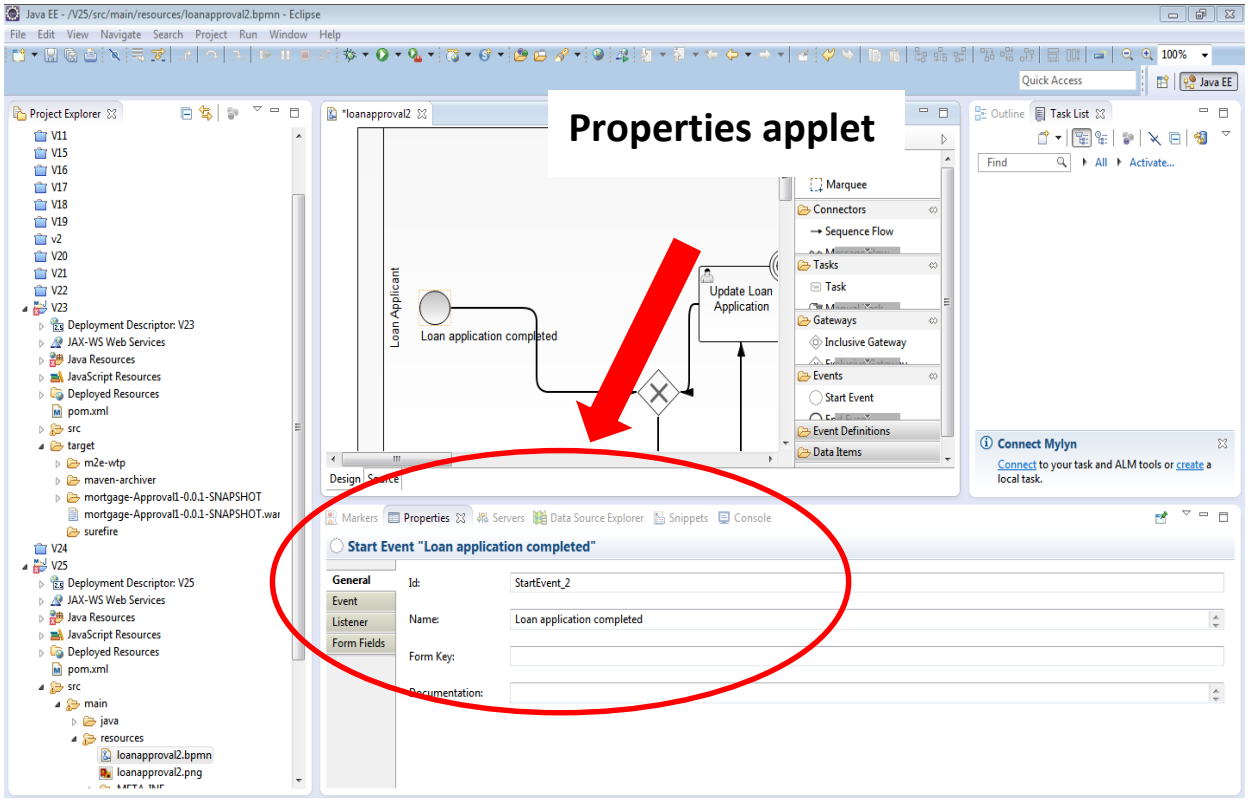
1.4 Eclipse Kepler

Eclipse is an integrated development environment (IDE). It acts as a platform to develop applications for different programming languages such as: Ada, ABAP, C, C++, COBOL, JavaScript, Java, Python etc. A unique feature of Eclipse is that it provides an extensible plug-in system for customizing the workspace as per the work environment needs.

Camunda provides a plugin called 'Camunda Modeller' which is integrated into Eclipse (Indigo/Juno/Kepler) to design the BPMN model and deploy it on Camunda Process Engine as a war file. Camunda Modeller has been integrated into Eclipse Kepler as part of designing and building the lab.

1.5 Overview of Camunda BPMN design tool in Eclipse Kepler:



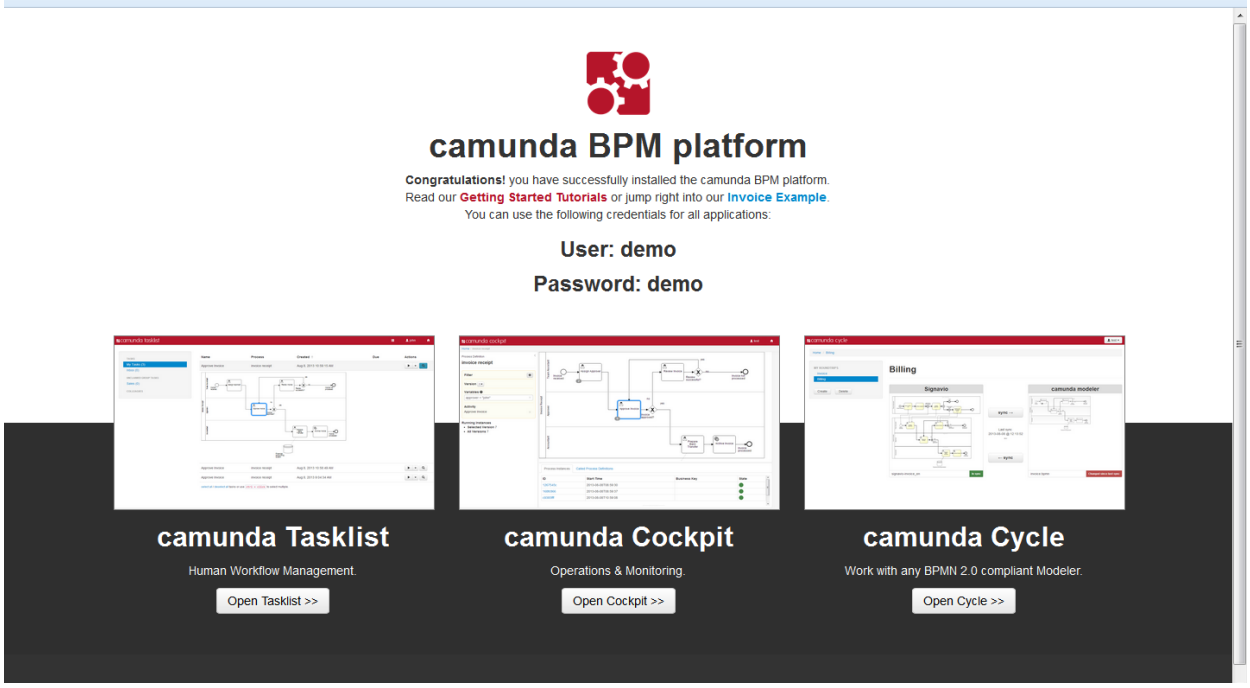


The two main features of Camunda Modeller plug-in is it provides a range of BPMN 2.0 elements to build the BPMN models which can be further specialized as per the requirements by entering the relevant details in 'Properties Applet'.

Features supported by Properties Applet:

- Define expressions, java delegates and classes to automate the BPMN process models
- Create user forms

1.6 Overview of Camunda BPM Platform:



Camunda Cockpit: With camunda BPM Cockpit you can monitor and administrate your running process instances. The Cockpit architecture allows you to use plugins to extend the functionality, so you can individually adapt the tool to your personal requirements.

Camunda Tasklist: The Tasklist is a demo web application to provide you with the possibility to work on User Tasks. The Tasklist is part of camunda BPM distribution and ready to use by opening http://localhost:8080/camunda/app/tasklist.

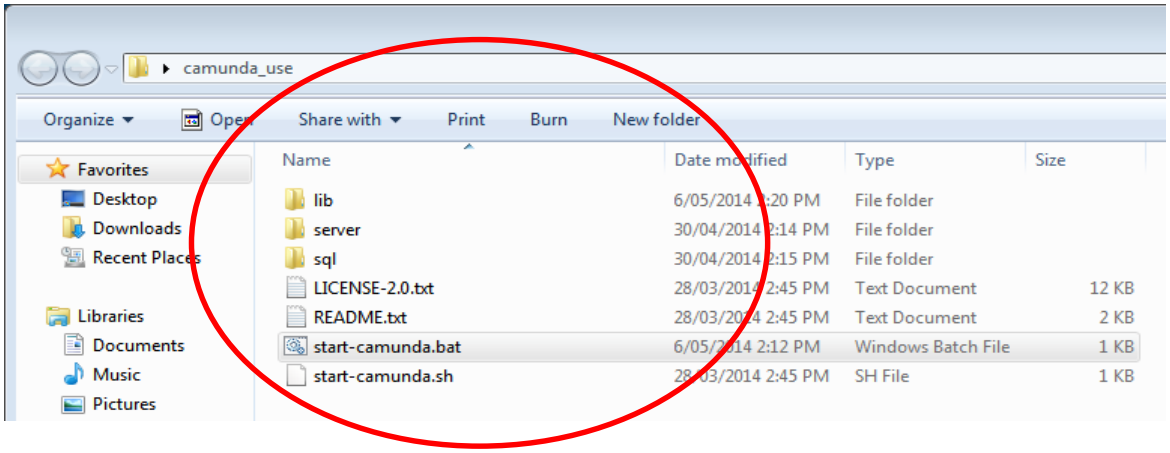
Camunda Cycle: With Cycle you can synchronize the BPMN diagrams in your business analyst's BPMN tool with the technically executable BPMN 2.0 XML files your developers edit with their modeler (example in Eclipse). Depending on your tool we can realize forward- and a reverse engineering, while you can store your BPMN 2.0 XML files in different repositories (example SVN, file system or FTP servers).

2.0 Set up requirements

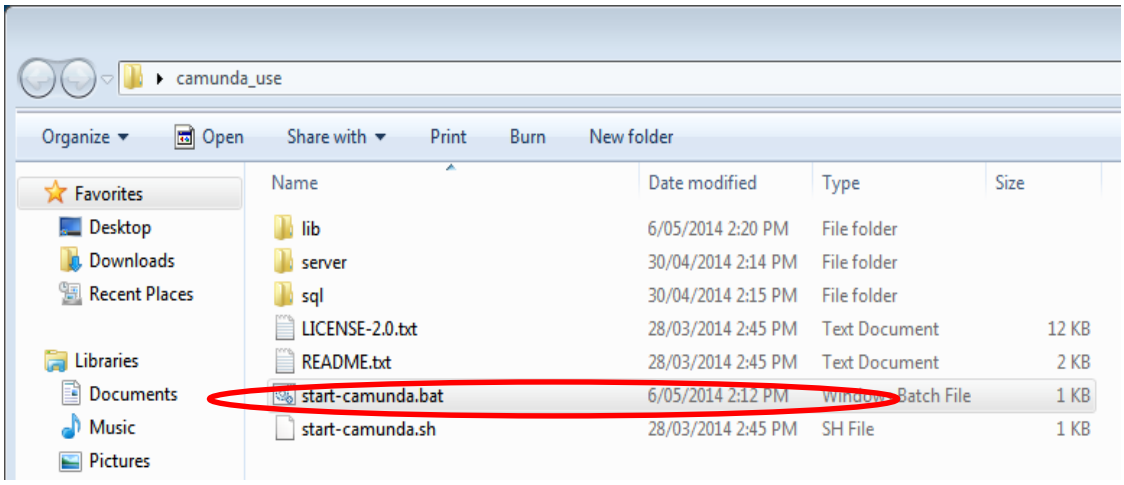
2.1 Installing the applications

Step 1: Install Camunda BPM platform

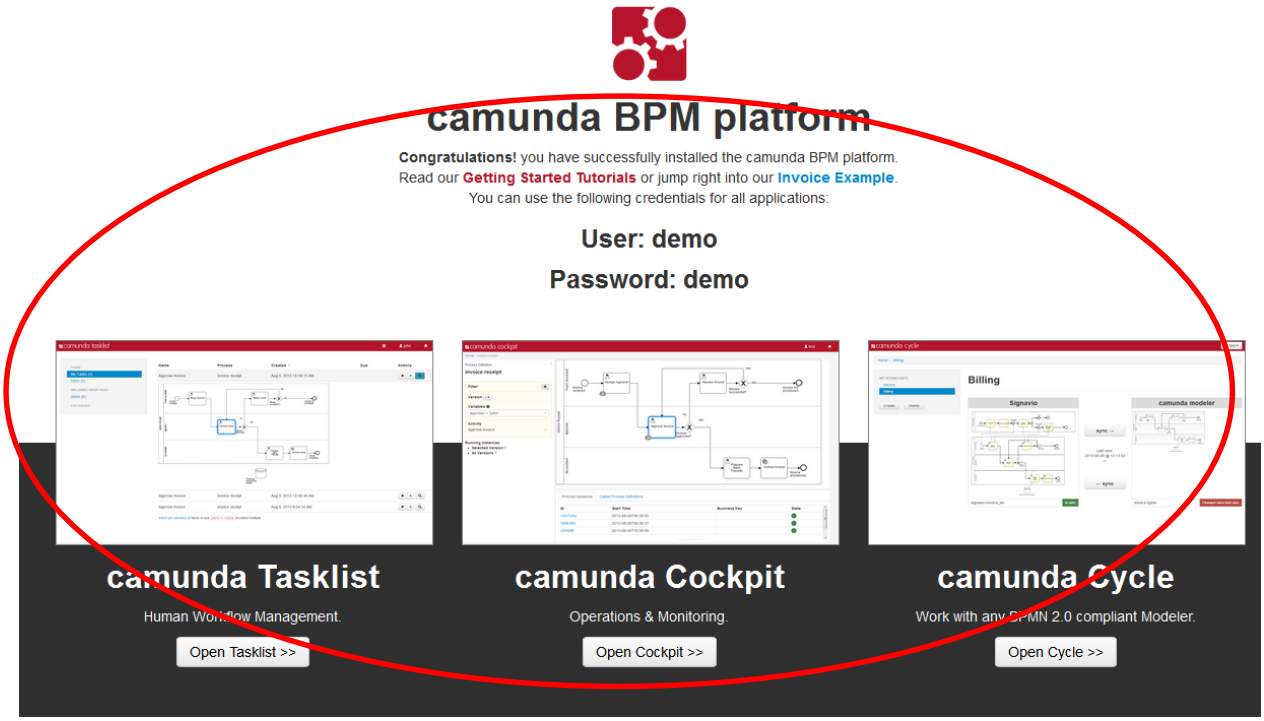
First, download a distribution of camunda BPM platform. You can chose from different application servers. In this tutorial, we will use the Apache Tomcat 7 based distribution. After having downloaded the distribution, unpack it inside a directory of your choice.



After you have successfully unpacked your distribution of camunda BPM platform, execute the script named start-camunda.bat (Windows users)



This script will start the application server and open a welcome screen in your Web browser. If the page does not open, go to http://localhost:8080/camunda-welcome/index.html.

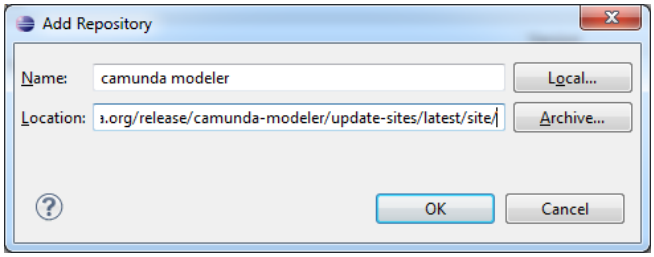


Step 2: Install Camunda Modeller

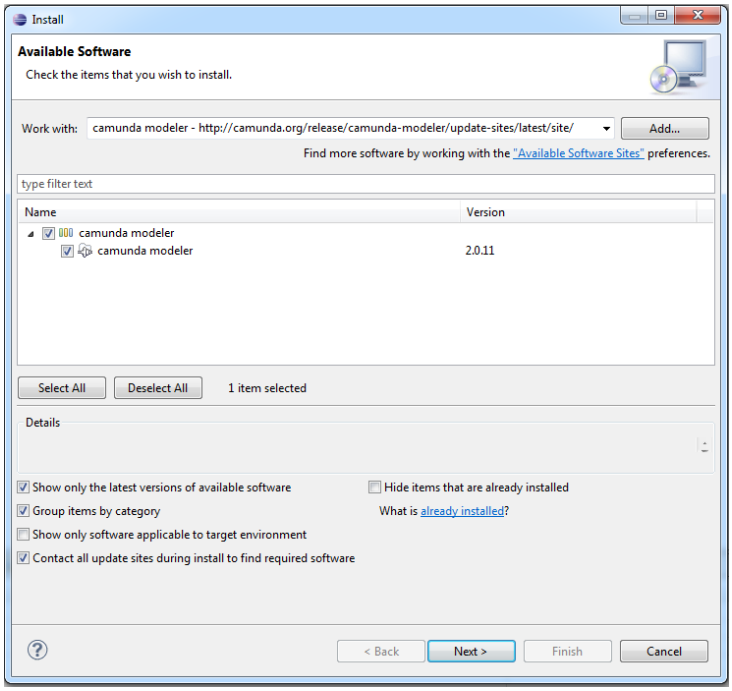
Use the following links to install the plugins:

Kepler	http://camunda.org/release/camunda-modeler/update-sites/kepler/latest/site/
Indigo and Juno	http://camunda.org/release/camunda-modeler/update-sites/latest/site/

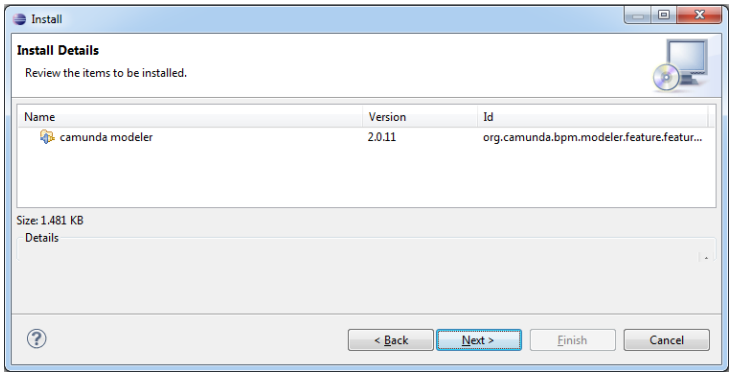
Add the update site URL (see above) as an Eclipse Repository



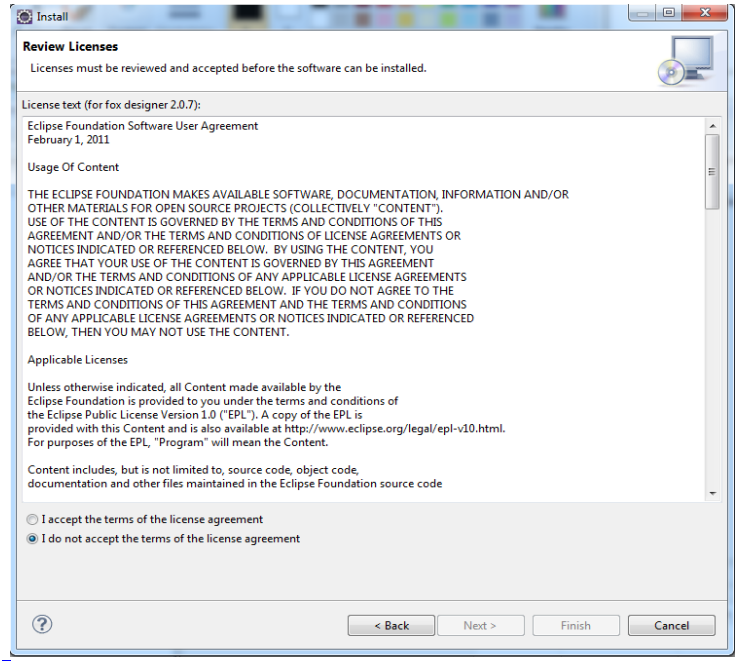
The item **camunda Modeler** appears. Select it and click *Next*



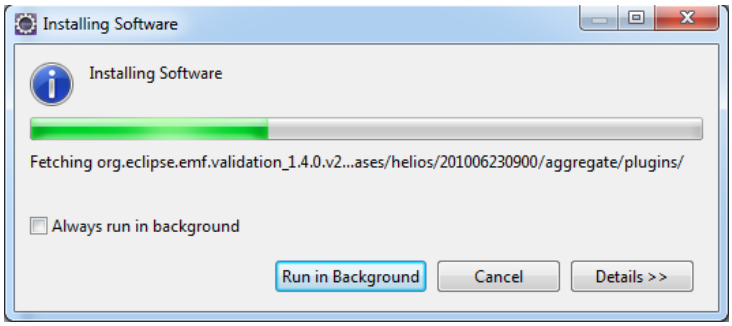
Wait for the installation details. Click *Next*



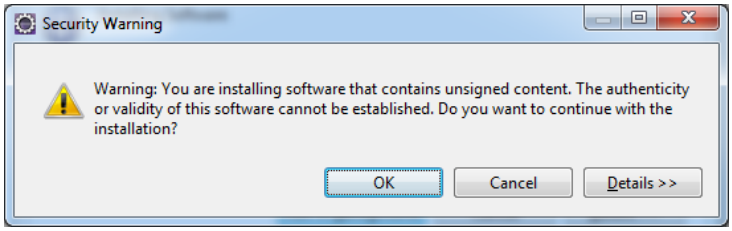
Before you can download the plugin you will be prompted to accept the User Agreement. Click *Next*



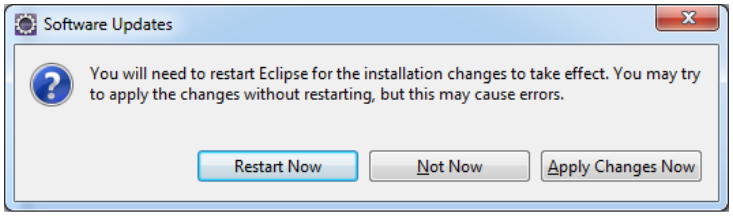
The installation progress window appears. Wait.



You will get a security warning message during the installation. Click *OK*.



After the installation has finished you will be prompted to restart your Eclipse. Click *Restart Now*. After the restart the camunda modeller is ready to use.



2.2 Creating a Maven project

Step 1: Create a new Maven Project in Eclipse

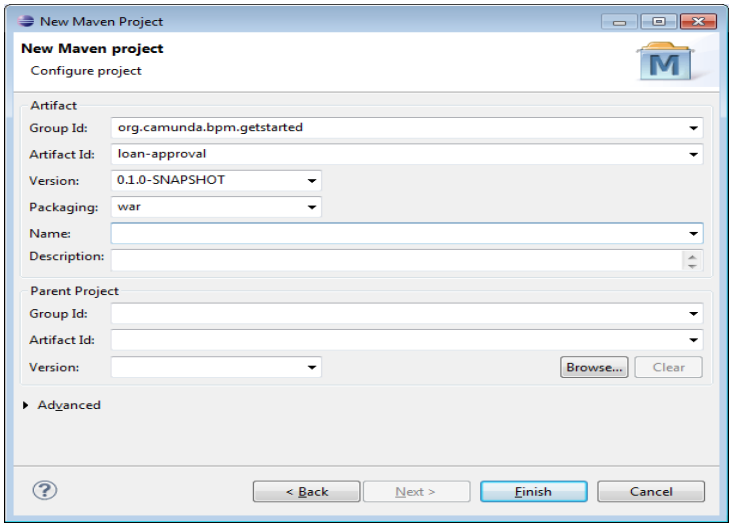
In eclipse, go to File / New / Other .... This opens the *New Project Wizard*. In the *New Project Wizard* select Maven / Maven Project. Click Next.

On the first page of the *New Maven Project Wizard* select "*Create a simple project (skip archetype selection)*". Click Next.

On the second page (see screenshot), configure the Maven coordinates for the project.

Since we are setting up a WAR Project, make sure to select Packaging: war.

When you are done, click Finish. Eclipse sets up a new Maven project. The project appears in the *Project Explorer* View.



Step 2: Add Camunda Maven Dependencies

The next step consists in setting up the Maven dependencies for your new process application. Add the following dependencies to the pom.xml file of your project:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

             xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">

      <modelVersion>4.0.0</modelVersion>

      <groupId>org.camunda.bpm.getstarted</groupId>

      <artifactId>loan-approval</artifactId>

      <version>0.1.0-SNAPSHOT</version>

      <packaging>war</packaging>


      <dependencies>

        <dependency>

          <groupId>org.camunda.bpm</groupId>

          <artifactId>camunda-engine</artifactId>

          <version>7.1.0-Final</version>

          <scope>provided</scope>

        </dependency>


        <dependency>

          <groupId>javax.servlet</groupId>

          <artifactId>javax.servlet-api</artifactId>

          <version>3.0.1</version>

          <scope>provided</scope>

        </dependency>

      </dependencies>
```

```
    <build>
      <plugins>
        <plugin>
          <groupId>org.apache.maven.plugins</groupId>
          <artifactId>maven-war-plugin</artifactId>
          <version>2.3</version>
          <configuration>
            <failOnMissingWebXml>false</failOnMissingWebXml>
          </configuration>
        </plugin>
      </plugins>
    </build>
</project>
```

Now you can perform the first build. Select the pom.xml in the Package Explorer, perform a Right Click and select Run AS / Maven Install

Step 3: Add a Process Application Class

Next, you need to create a package, e.g. org.camunda.bpm.getstarted.loanapproval and add a Process Application class to it. The Process Application class constitutes the interface between your application and the process engine.

```
package org.camunda.bpm.getstarted.loanapproval;


    import org.camunda.bpm.application.ProcessApplication;
    import org.camunda.bpm.application.impl.ServletProcessApplication;


    @ProcessApplication("Loan Approval App")
```

```
    public class LoanApprovalApplication extends
ServletProcessApplication {

      // empty implementation

    }
```

Step 4: Add a META-INF/processes.xml deployment descriptor

The last step for setting up the process application is adding the META-INF/processes.xml deployment descriptor file. This file allows us to provide a declarative configuration of the deployment(s) this process application makes to the process engine.

This file needs to be added to the src/main/resources/META-INF folder of the Maven project.

```
<?xml version="1.0" encoding="UTF-8" ?>


    <process-application

        xmlns="http://www.camunda.org/schema/1.0/ProcessApplication"

        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">


      <process-archive name="loan-approval">

        <process-engine>default</process-engine>

        <properties>

          <property name="isDeleteUponUndeploy">false</property>

          <property name="isScanForProcessDefinitions">true</property>

        </properties>

      </process-archive>

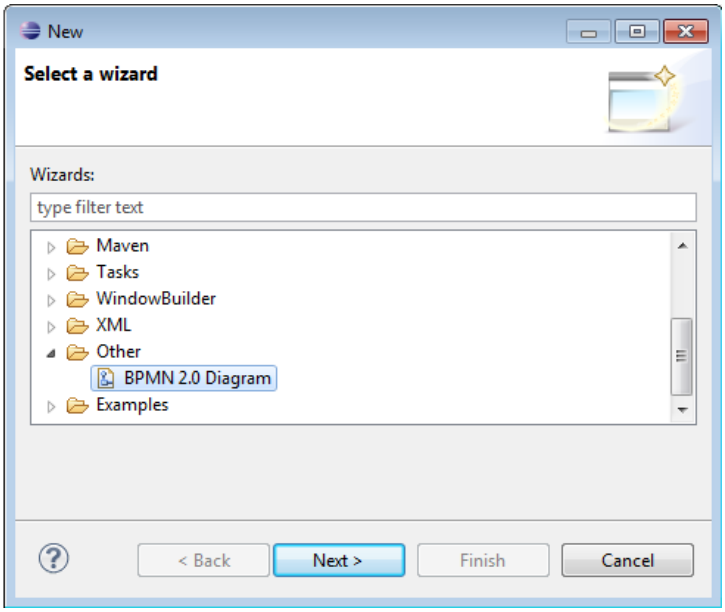
    </process-application>
```

2.3 Model a Process

Step 1: Create a new BPMN 2.0 diagram

In the eclipse *Package Explorer* select the src/main/resources folder. Right-click and select New > Other .... Go to the folder Other and select *BPMN 2.0 Diagram*. Click Next.

On the second page, you must specify the file name of the process. Put in *loan-approval.bpmn*. Click Finish



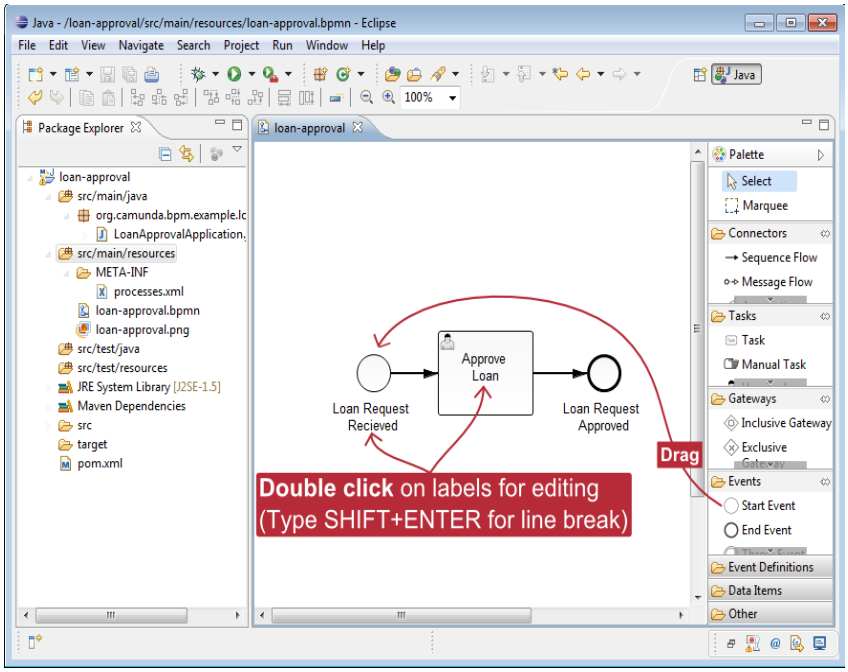
Step 2: Start with a Simple Process

Start by modeling a simple process. From the Palette on the right hand side to the canvas, drag a *Start Event*

Double Click on the Label of the Start Event. A text box opens. Type *"Loan Request Received"*.

When editing Labels, you can add line breaks by typing `Shift + Enter`.

Add a User Task to the process. Name it "*Approve Loan*". Add an End Event named "*Loan Request Approved*"

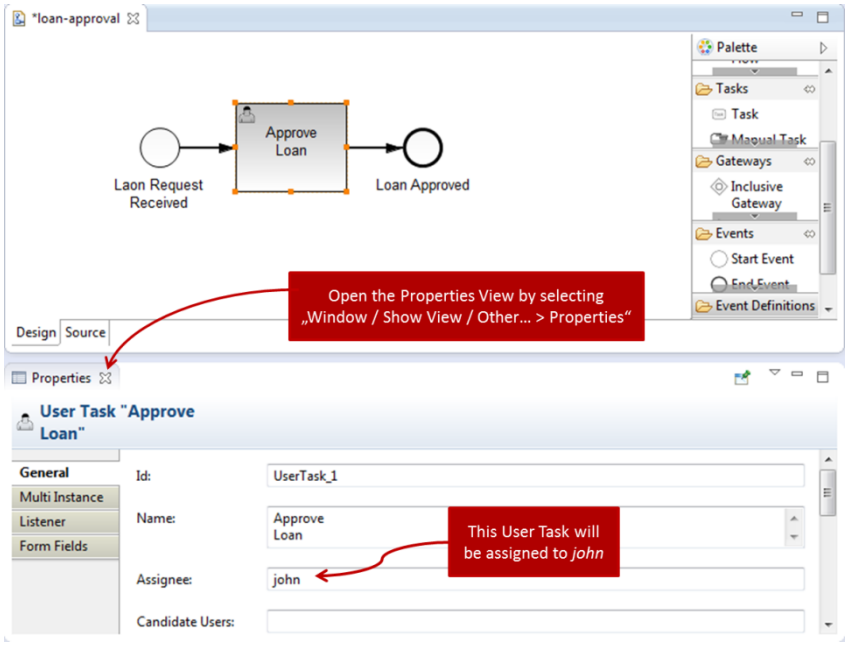


Step 3: Configure a User Task

Next, Open the properties view. If the properties view is not visible, select it from the menu: Window / Show View / Other ... This opens a dialog. From the dialog select Properties.

Select the User Task on the canvas. This updates the selection in the Properties View. Scroll to the Property named Assignee. Type *"john"*.

After you are done, save your changes.

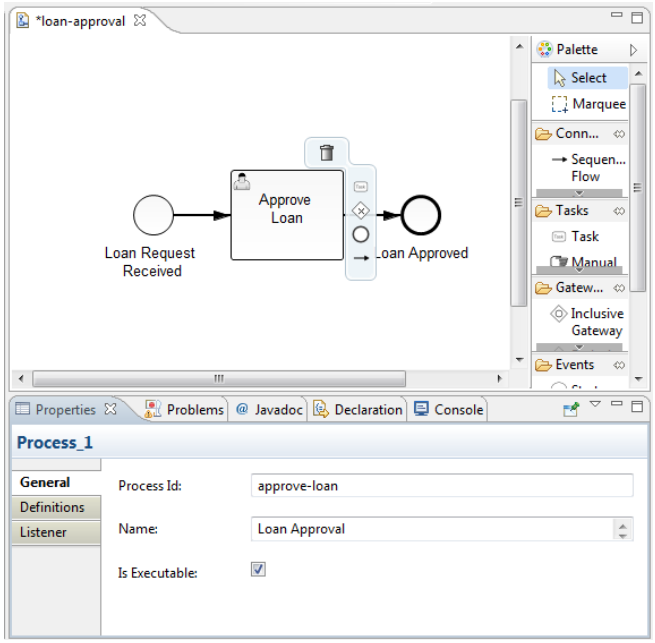


Step 4: Configure an executable Process

Since we model an executable process, we should give it an ID and set the `isExecutable` property to `true`. Open the properties view and click on a free spot of the modeling canvas. This displays the properties of the process itself.

1. First, configure an ID for the process. Type "approve-loan" in the property field *Process Id*. The ID property is used by the process engine as identifier for the executable process and it is best practice to set it to a human-readable name.
2. Second, configure the Name of the process. Type "Loan Approval" in the property field *Name*.
3. Finally, check the box of the *Is Executable* property. If you do not check this box, the process definition is ignored by the process engine.

After you are done, save your changes.

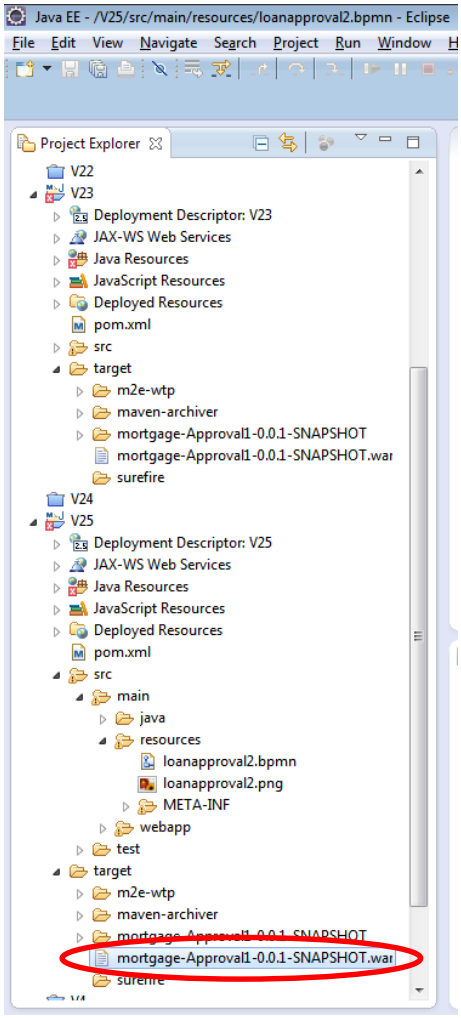


2.4 Deploy and run the Process Application

Step 1: Select the war file

A Process Application is an ordinary Java Web Application and is deployed in exactly the same way.

Select the pom.xml in the Package Explorer, perform a Right Click and select Run AS / Maven Install. This will generate a WAR file named loan-approval-0.0.1-SNAPSHOT.war In the target/ folder of your Maven project.



Step 2: Deploy to Apache Tomcat

In order to deploy the process application, copy-paste the loan-approval-0.0.1-SNAPSHOT.war from your Maven project to the $CAMUNDA_HOME/server/apache-tomcat/webapps folder.

Check the log file of the Apache Tomcat server. If you see the following log message, the deployment was successful:

```
16.03.2013 22:42:11 org.camunda.bpm.application.impl.ServletProcessApplicationDeployer onStartup
INFO: Detected @ProcessApplication class org.camunda.bpm.getstarted.loanapproval.LoanApprovalApplication
16.03.2013 22:42:11 org.camunda.bpm.container.impl.jmx.deployment.ParseProcessesXmlStep performOperationStep
INFO: Found META-INF/processes.xml file at ...webapps/loan-approval-0.0.1-SNAPSHOT/WEB-INF/classes/META-INF/proces
ses.xml
16.03.2013 22:42:11 org.camunda.bpm.container.impl.jmx.deployment.DeployProcessArchiveStep logDeploymentSummary
INFO: Deployment summary for process archive 'loan-approval':

loan-approval.bpmn

16.03.2013 22:42:11 org.camunda.bpm.engine.impl.bpmn.deployer.BpmnDeployer deploy
INFO: Processing resource loan-approval.bpmn
16.03.2013 22:42:11 org.camunda.bpm.container.impl.jmx.JmxRuntimeContainerDelegate deployProcessApplication
INFO: Process Application Loan Approval App successfully deployed.
```

Step 3: Check with camunda Cockpit and Tasklist

Now use cockpit to check whether the process is successfully deployed. Go to http://localhost:8080/camunda/app/cockpit. Your process is visible on the start screen.

Next, go to camunda Tasklist (http://localhost:8080/camunda/app/tasklist). Log in with demo / demo. After you have logged in, click on the  button to start a process instance. Select *Loan Approval* from the list. This opens a page which explains that no form has been defined for starting the process. Click on *Start process using generic form*.

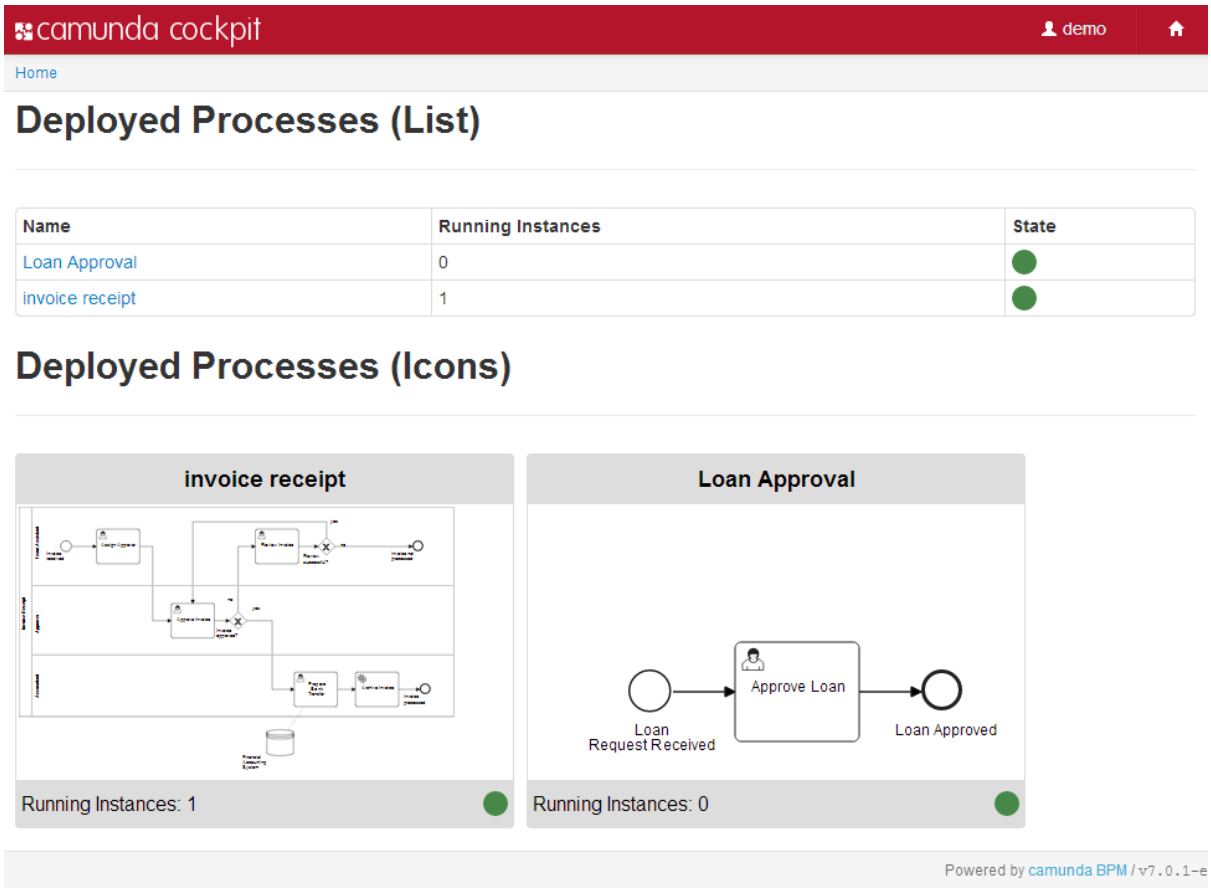
The generic form can be used whenever you have not added a dedicated form for a User Task or a Start Event. Click on the  button to get a new row. Fill in the form as shown in the screenshot. When you are done, click 

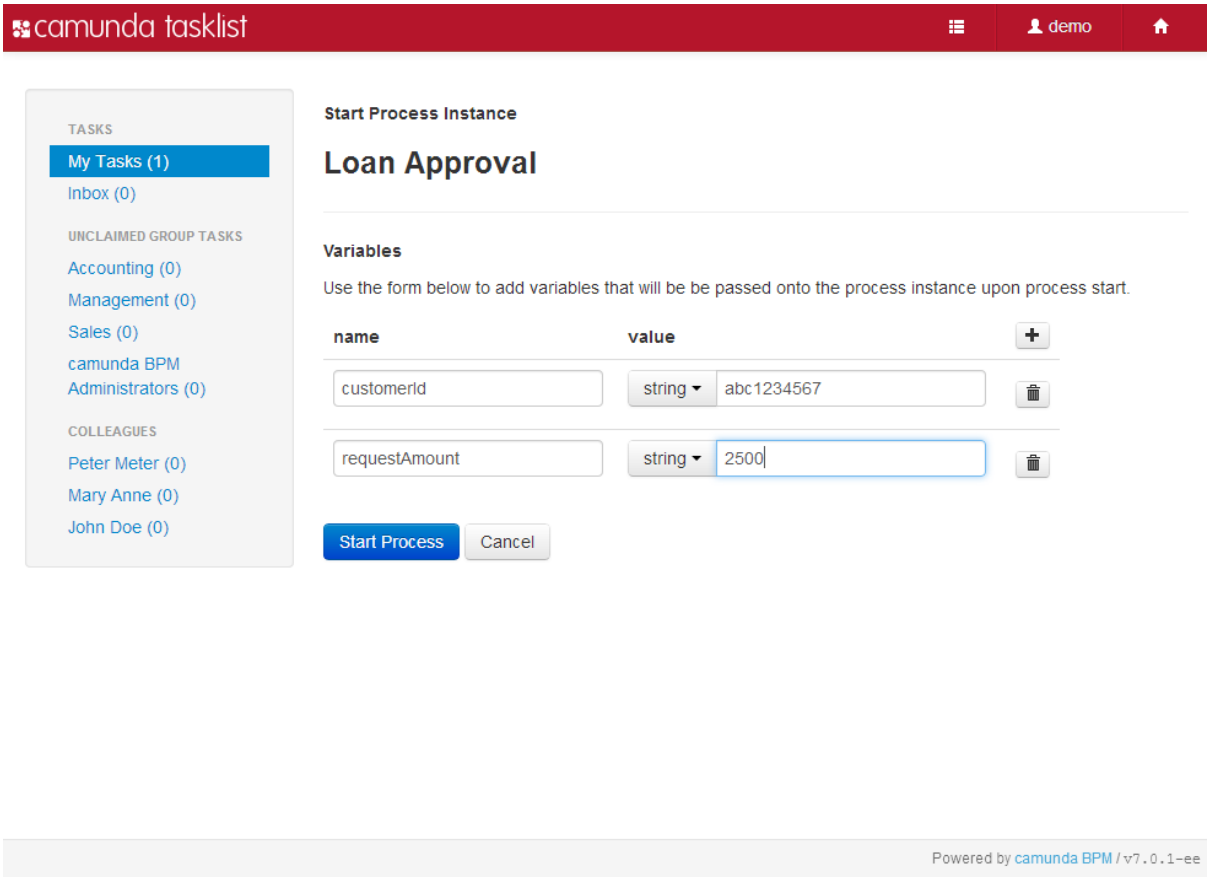
.

If you now go back to camunda Cockpit, you see the newly created process instance that is waiting in the User Task.

Log out of the Tasklist and log back in with the user credentials "john / john". Now you see the *Approve Loan* task in your inbox. Click on the  button. This displays the process diagram highlighting the User Task that is waiting for you to work on it.

To work on the task, click the ▶ button. Again, there is no task form associated with the process. Click on *Work on task using generic form*. This displays the variables you have put in in the first step.



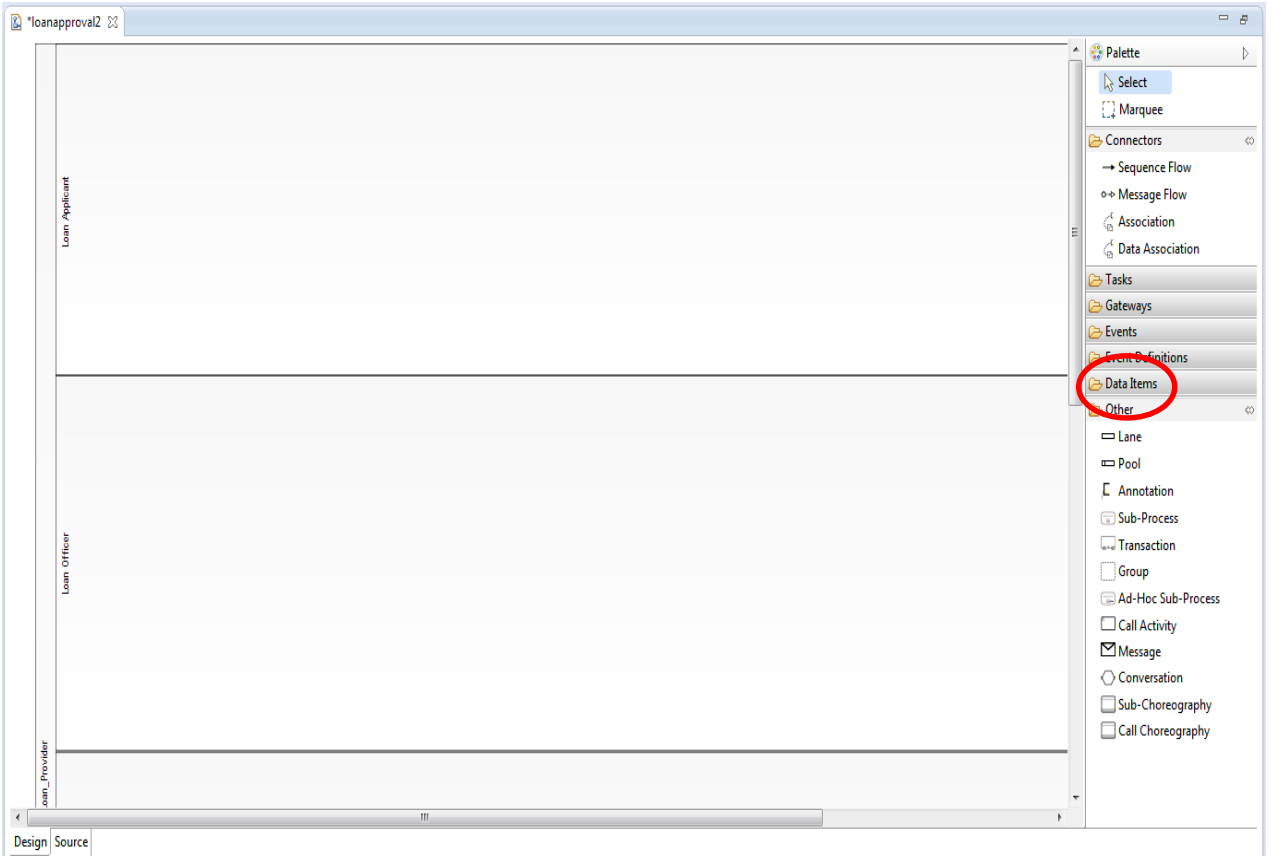

camunda tasklist
demo
TASKS
My Tasks (1)
Inbox (0)
UNCLAIMED GROUP TASKS
Accounting (0)
Management (0)
Sales (0)
camunda BPM Administrators (0)
COLLEAGUES
Peter Meter (0)
Mary Anne (0)
John Doe (0)
Start Process Instance
Loan Approval
Variables
Use the form below to add variables that will be be passed onto the process instance upon process start.
name
value
customerId
string
abc1234567
requestAmount
string
2500
Start Process
Cancel
Powered by camunda BPM / v7.0.1-ee

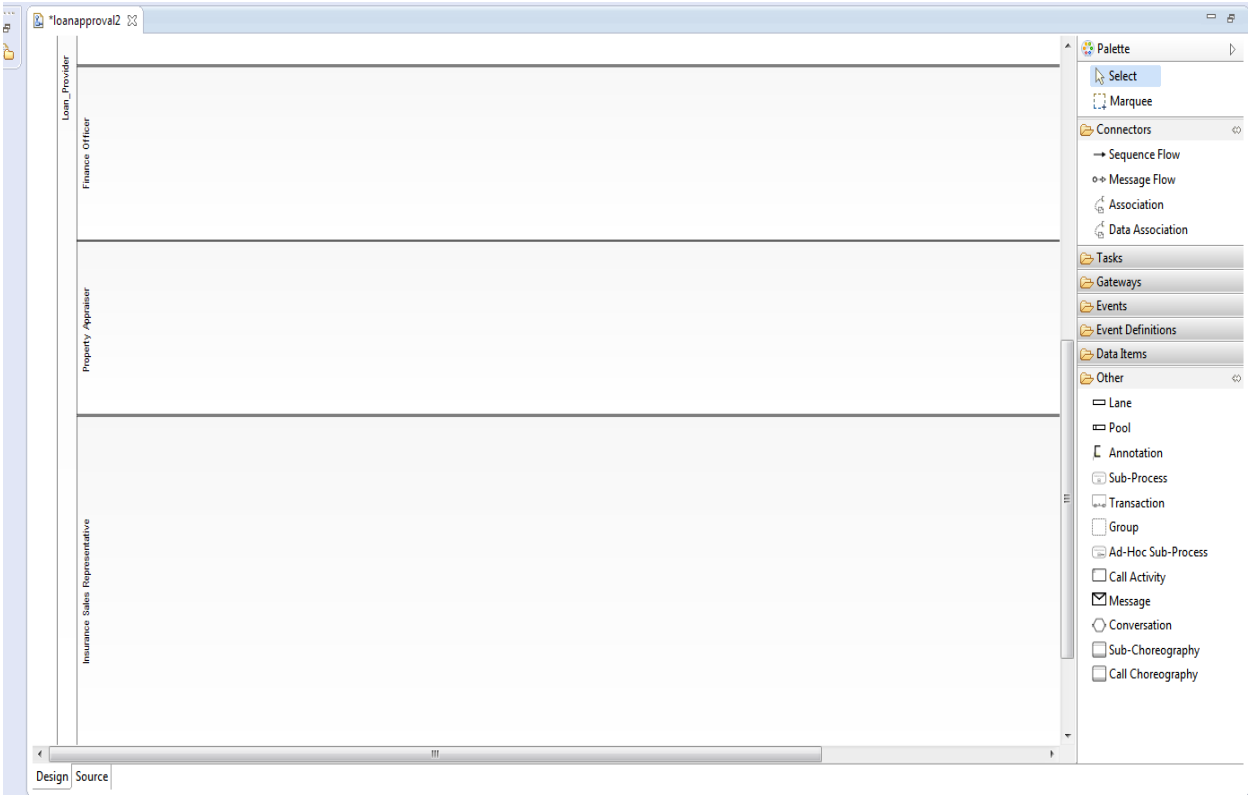
3.0 Implementation of Loan Approval Process

Step 1

Create a pool: Loan Provider

Create lanes: Loan Applicant, Loan Officer, Finance Officer, Property Appraiser and Insurance Sales representative





Please Note:* *There is a bug with respect to using* ***two or more pools* *and has been noted under constraints 1 of the Appendix in report*

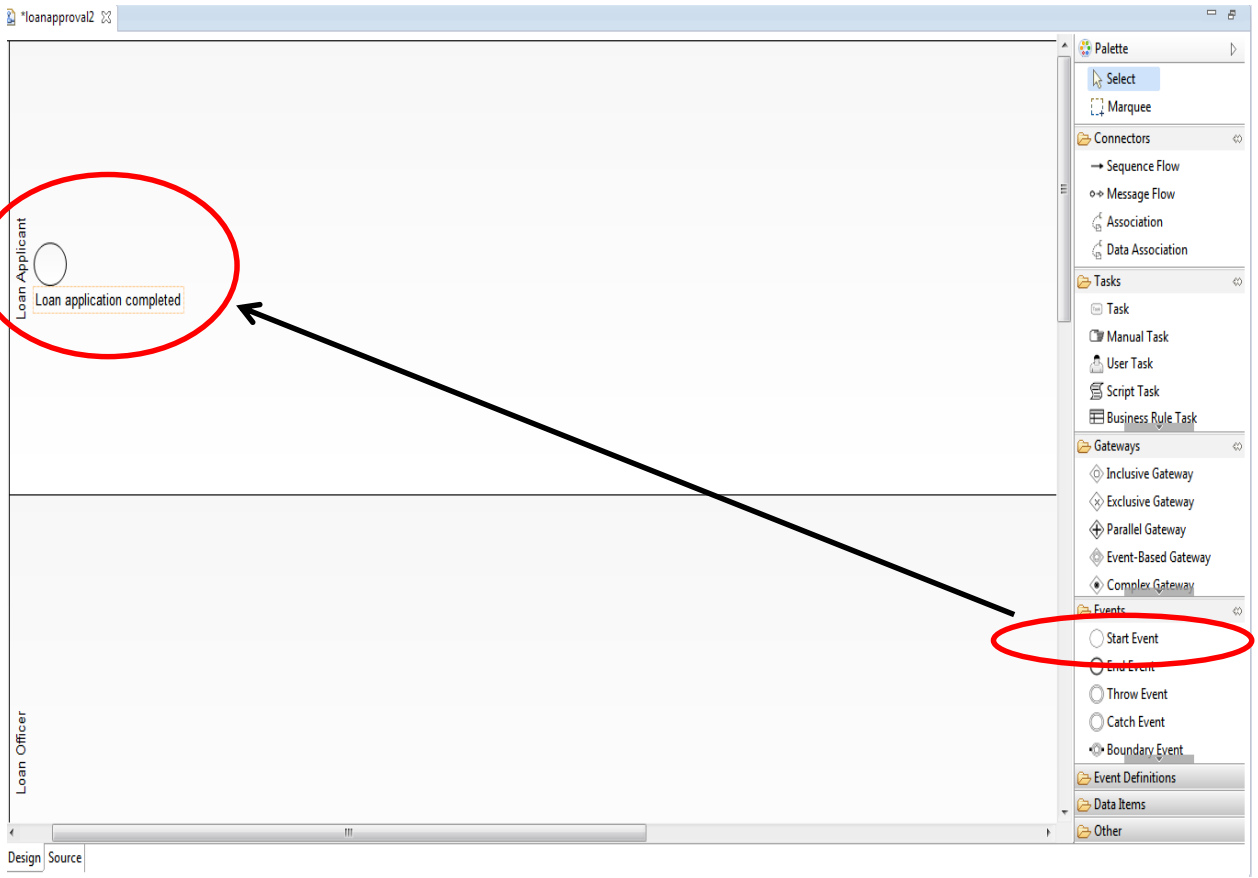
Step 2

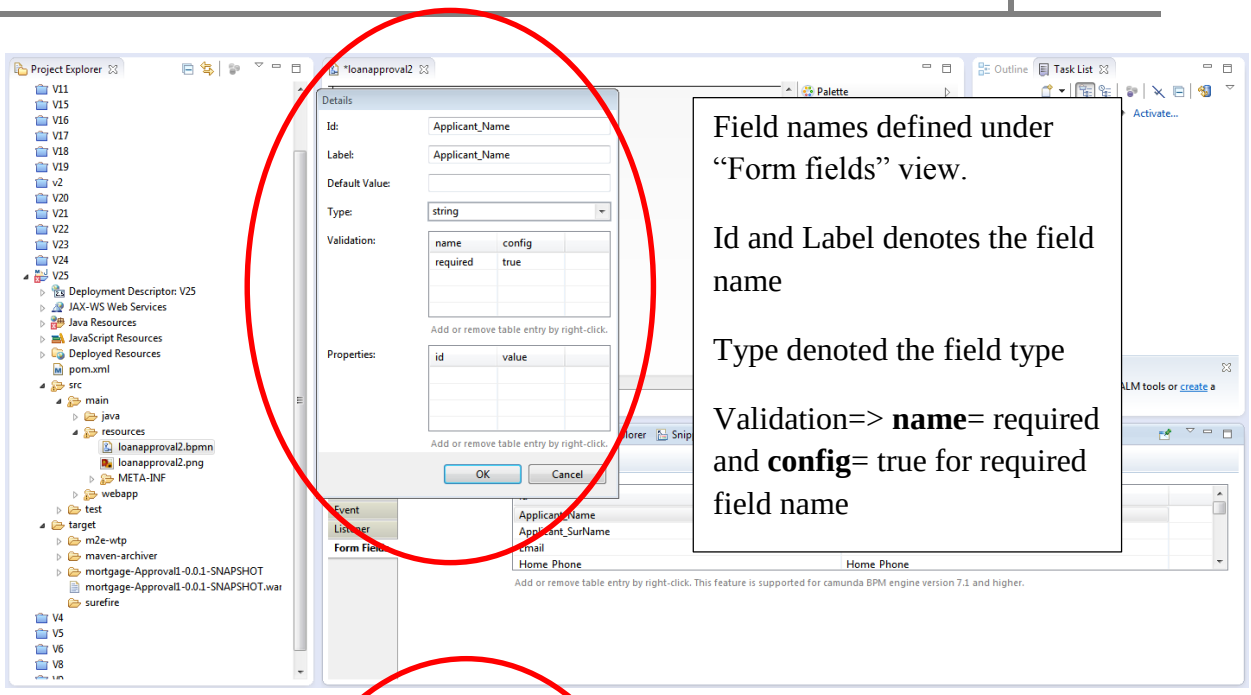
Create a Start event in Loan Applicant lane and name it 'Loan application completed'.

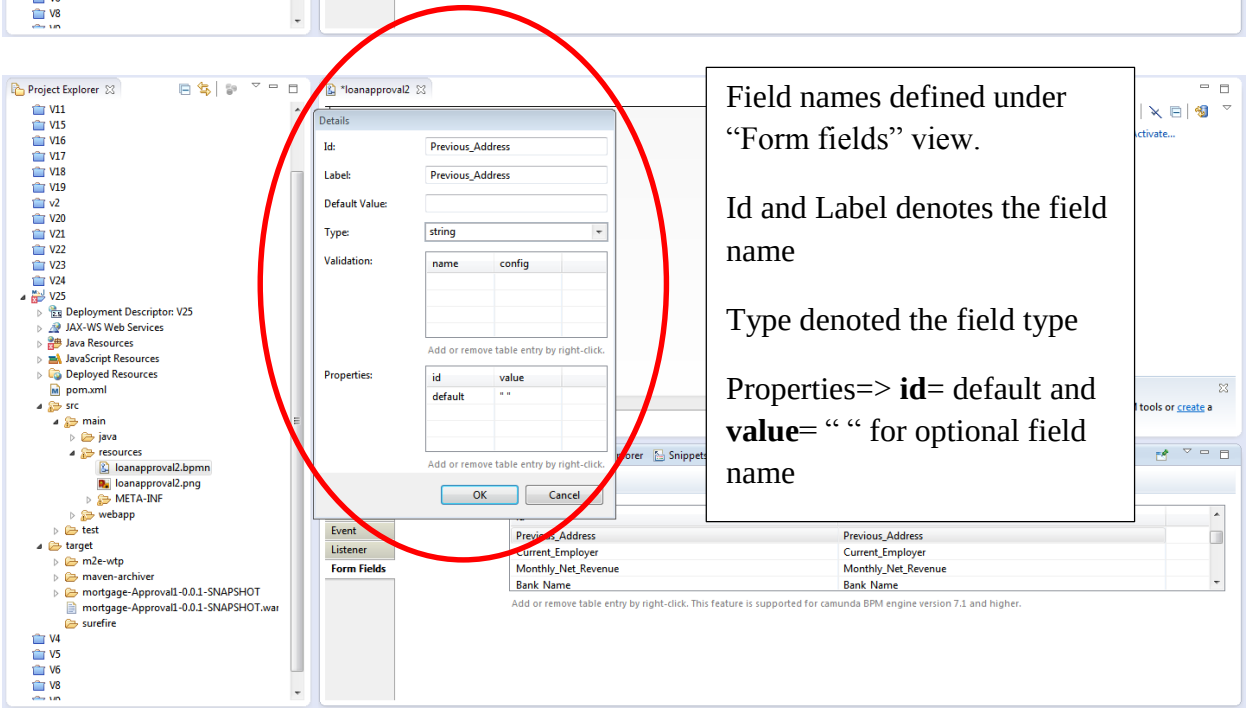
Create a form under start event with the following field name and types:

Loan Application		
Field Name	**Field Type**	**Required(R)/Optional(O)**
Applicant_Name	String	R
Applicant_Surname	String	R
Email	String	R
Home_Phone	Long	R
Mobile_Phone	Long	R
Current_Address	String	R
Previous_Address	String	O
Current_Employer	String	R
Monthly_Net_Revenue	Long	R
Bank_Name	String	R
Account_Type	String	R
Account_Number	String	R
Account_Balance	Long	R
Property_Type	String	R
Property_Address	String	R

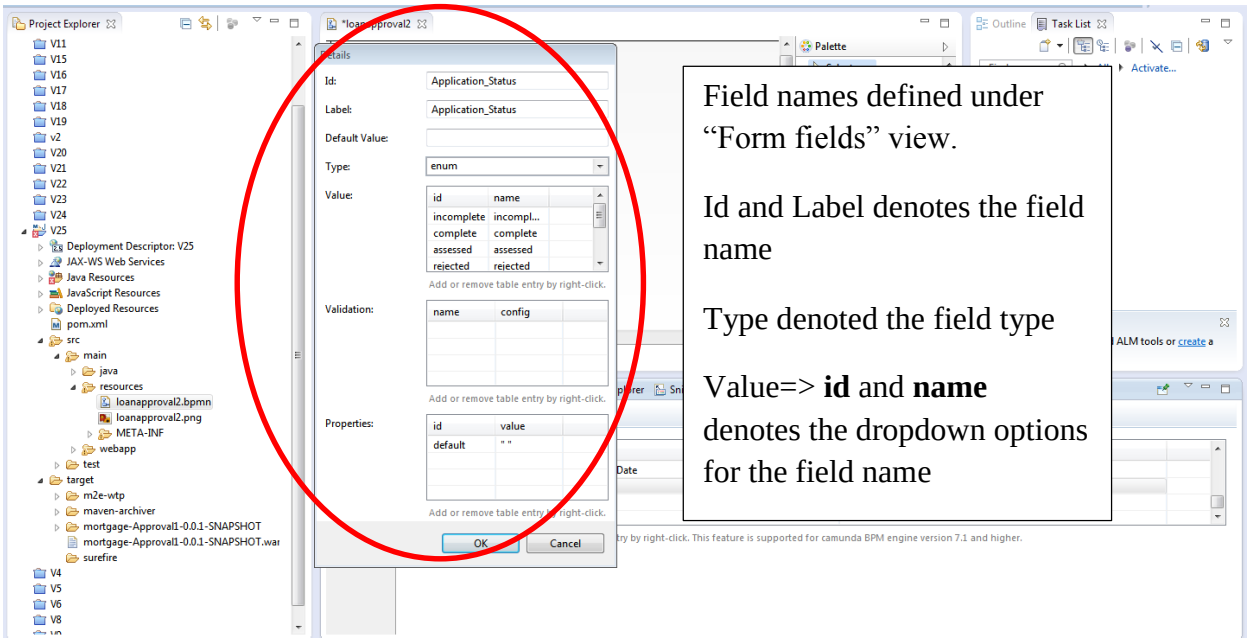
Purchasing_Price	Long	R
Loan_Type	String	R
Loan_Provider	String	R
Loan_Amount	Long	R
Loan_Duration	String	R
Loan_Start_Date	Date	R
Loan_End_Date	Date	R
Interest_Type	Enum (Variable/Fixed)	R
Interest_Rate	Long	R
Insurance_Quote_Required	Boolean	R
Administration Information		
Application_Identifier	String	O
Application_Submission_Date	String	O
Application_Revision_Date	String	O
Application_Status	Enum (Complete/ Incomplete/ Assessed/ Rejected/ Cancelled/ Approved)	O
Comment_on_Status	String	O







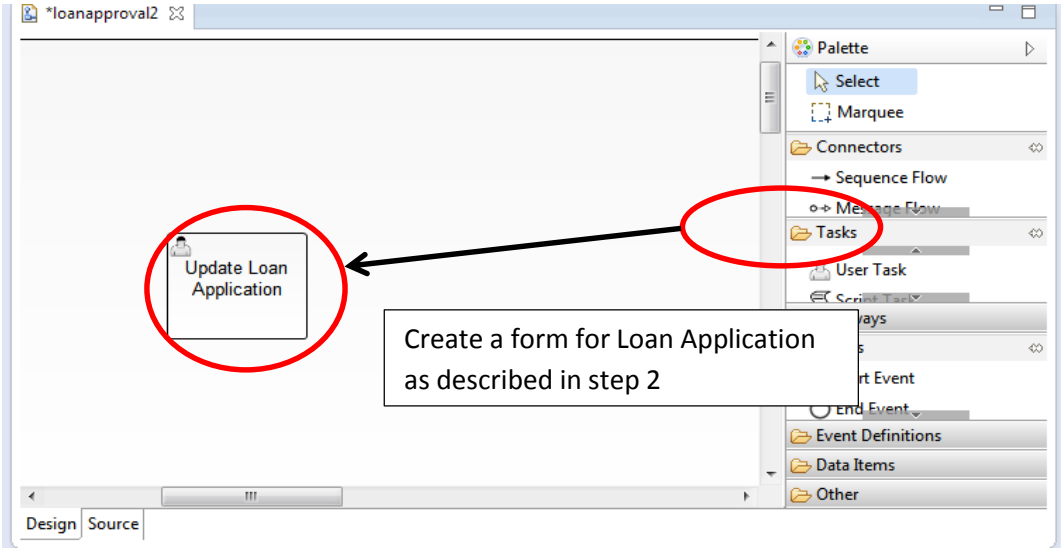
Please Note:* *There is a bug with respect to **default and **initial user assignment** validation and has been noted in the constraints 2 and 4 of the Appendix.*

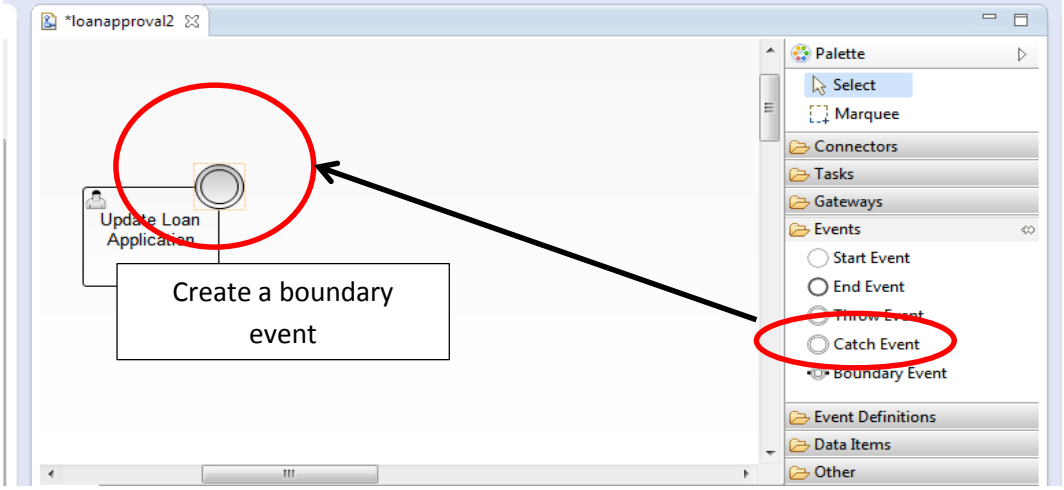


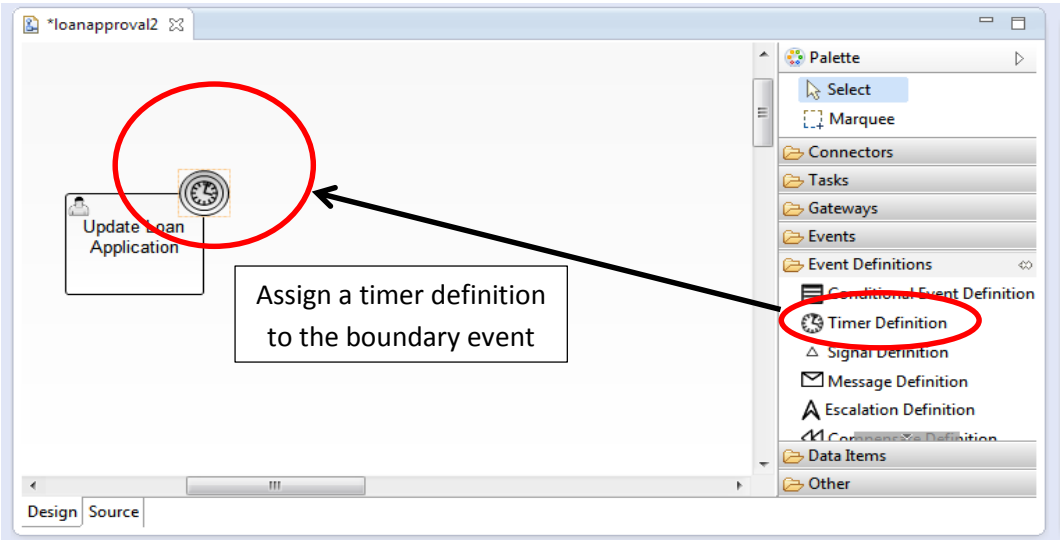
Step 3

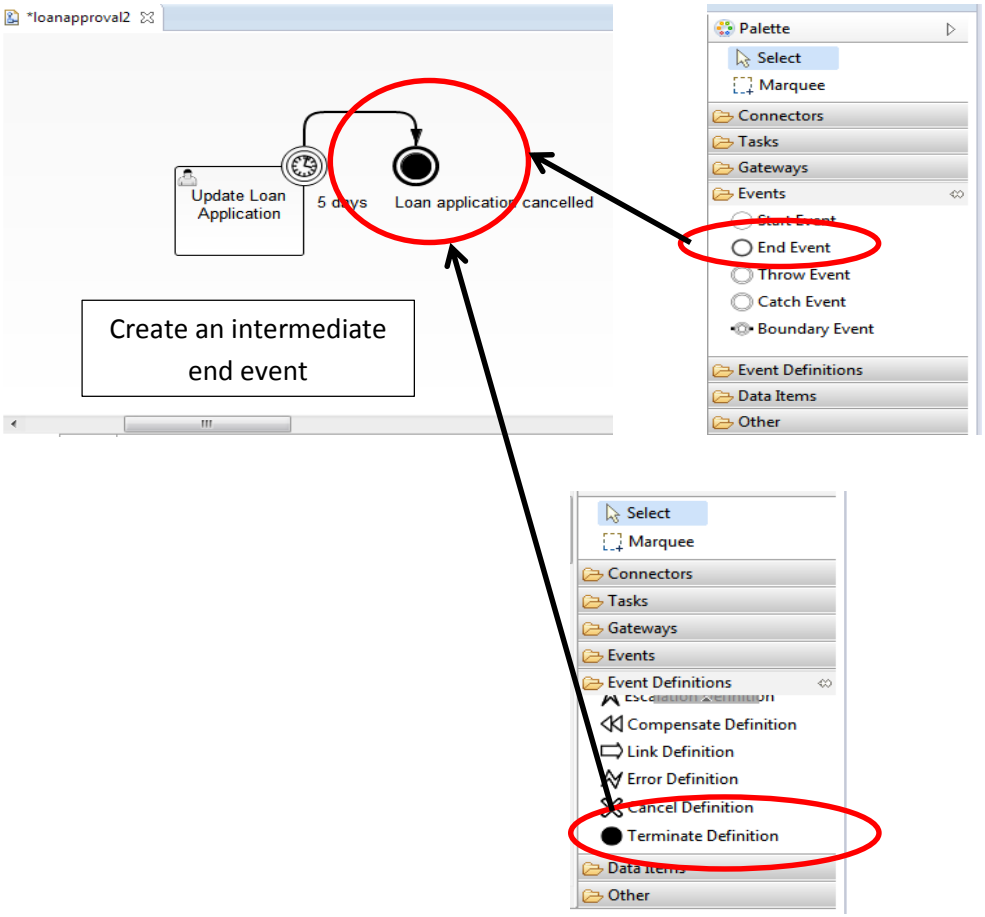
Create a User Task: Update Loan Application and assign user 'Mary' (General>>Assignee) as Loan Applicant

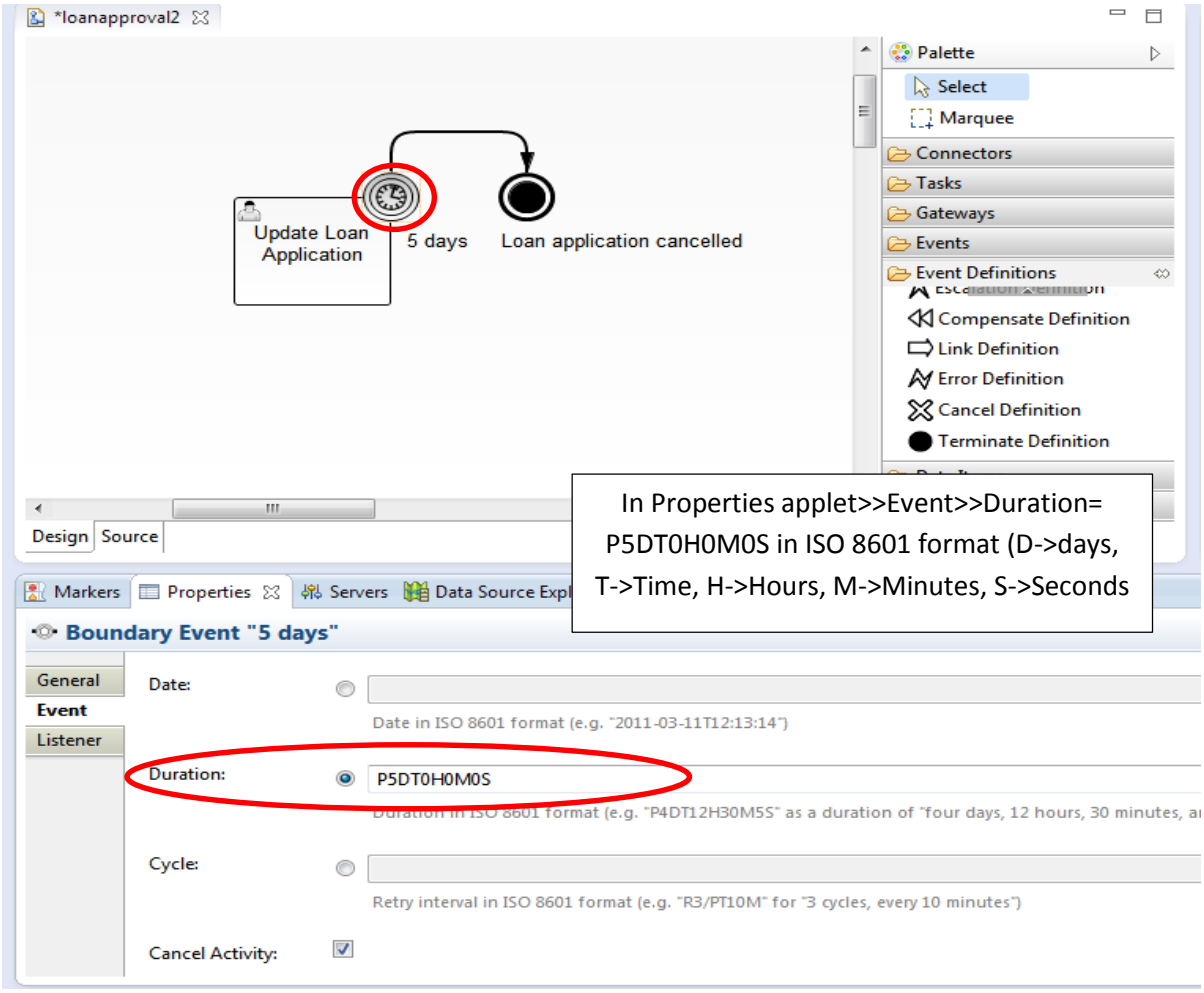
Create a timer for the user task so that the loan application is cancelled after a time interval of 5 days




*loanapproval2
Palette
Select
Marquee
Connectors
Tasks
Gateways
Events
Start Event
End Event
Catch Event
Boundary Event
Event Definitions
Data Items
Other
Update Loan Application
Create a boundary event

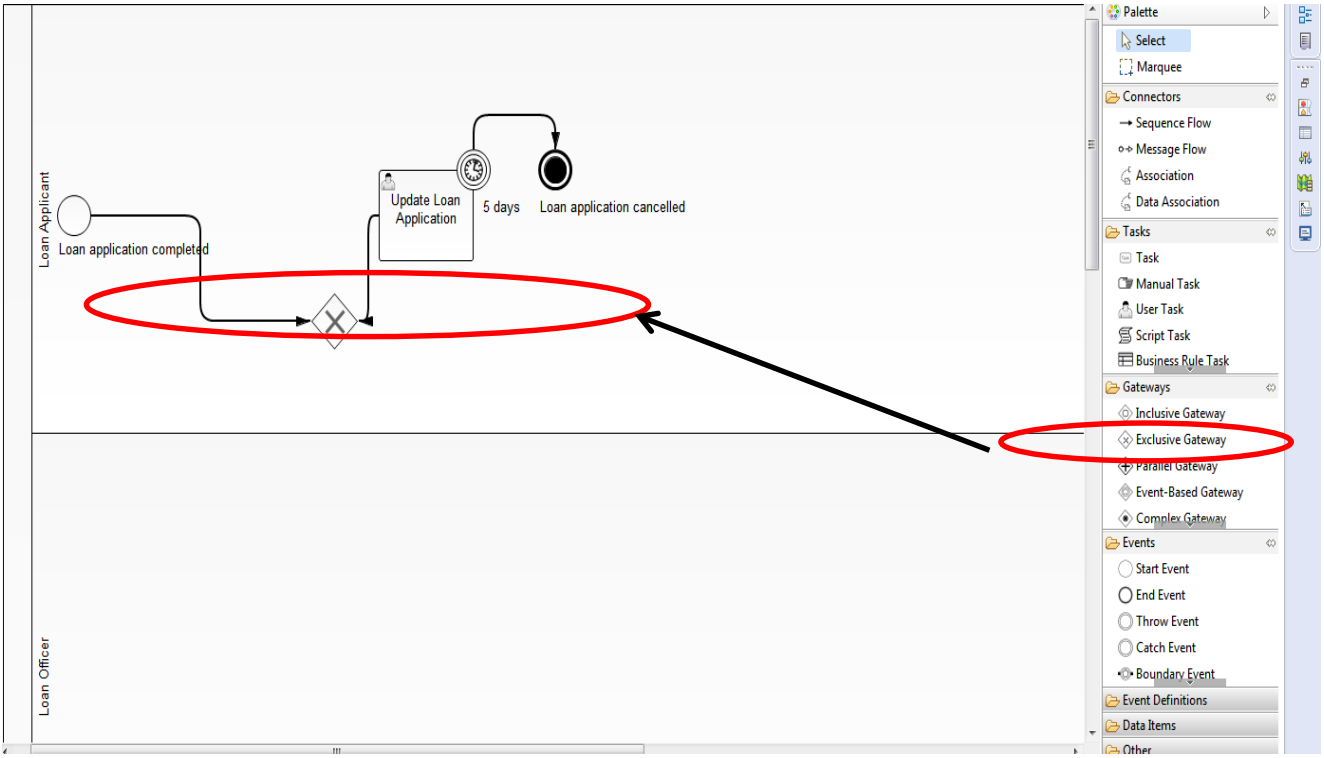

*loanapproval2
Palette
Select
Marquee
Connectors
Tasks
Gateways
Events
Event Definitions
Timer Definition
Message Definition
Escalation Definition
Data Items
Other
Design
Source
Update Loan Application
Assign a timer definition to the boundary event


*loanapproval2
Update Loan Application
5 days
Loan application cancelled
Create an intermediate end event
Palette
Select
Marquee
Connectors
Tasks
Gateways
Events
End Event
Throw Event
Catch Event
Boundary Event
Event Definitions
Data Items
Other
Select
Marquee
Connectors
Tasks
Gateways
Events
Event Definitions
Compensate Definition
Link Definition
Error Definition
Cancel Definition
Terminate Definition
Other



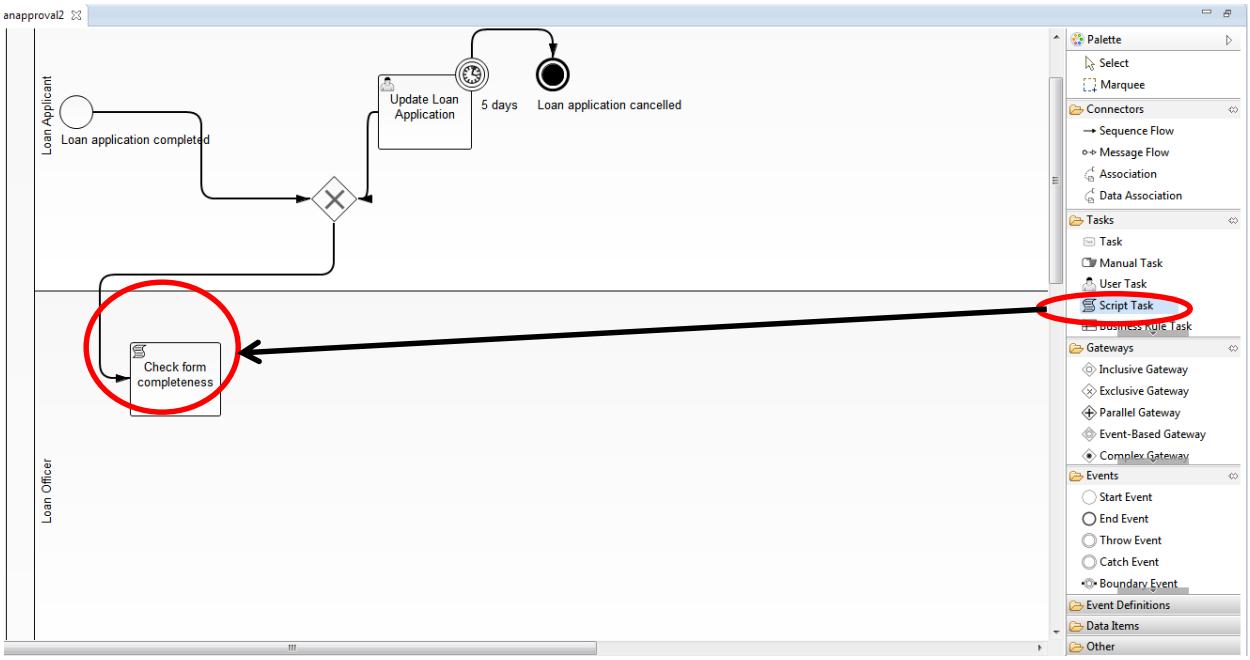
Step 4

Create an Event-Based join gateway to join the two tasks



Step 5

Create a script task to check if Email, Home_Phone and Mobile_Phone fields are in the right format and add Application_Identifier, Application_Submission_Date and Application_Revision_Date to the loan application form along with the appropriate comments in Comments_on_Status field (if format is wrong). Furthermore if the application is in right format, the flow should proceed to 'Credit History' task else it should revert to 'Update Loan Application' task



Script Language: Groovy

Script:

```
if(Email ==~ /[a-z_][a-z0-9_]+@[a-z0-9_.?]+.[a-z]{1,3}/) {
   Application_Status= 'complete';
} else {
   Application_Status = 'incomplete';
   execution.setVariable ("Email", "Wrong Format");
   execution.setVariable ("Comment_on_Status", "Revision Needed");
}

if(Home_Phone ==~ /\d{10}$/) {
   Application_Status = 'complete';
} else  {
   Application_Status = 'incomplete';
   execution.setVariable ("Home_Phone", "Wrong Format");
   execution.setVariable ("Comment_on_Status", "Revision Needed");
}

if(Mobile_Phone ==~ /\d{10}$/) {
   Application_Status = 'complete';
} else  {
   Application_Status = 'incomplete';
   execution.setVariable ("Mobile_Phone", "Wrong Format");
```

```
    execution.setVariable ("Comment_on_Status", "Revision Needed");
}

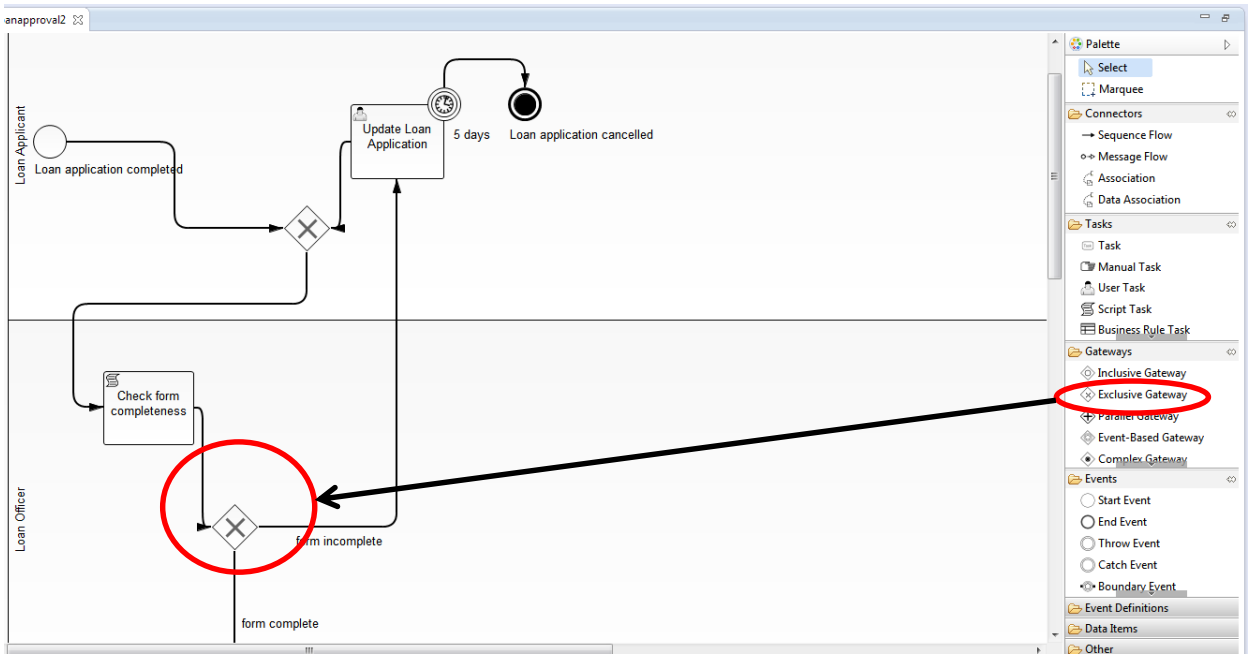
Application_Identifier = execution.getVariable ("ProcessInstanceId");
execution.setVariable ("Application_Identifier",
Application_Identifier);
execution.setVariable ("Application_Submission_Date", (new Date
()).format("dd/MM/yyyy"));
execution.setVariable ("Application_Revision_Date", (new Date
()).format("dd/MM/yyyy"));
```

Step 6

Add an Exor split gateway with the following Listener expression: ${Application_Status=='complete'} or ${Application_Status=='incomplete'} (for routing appropriately).

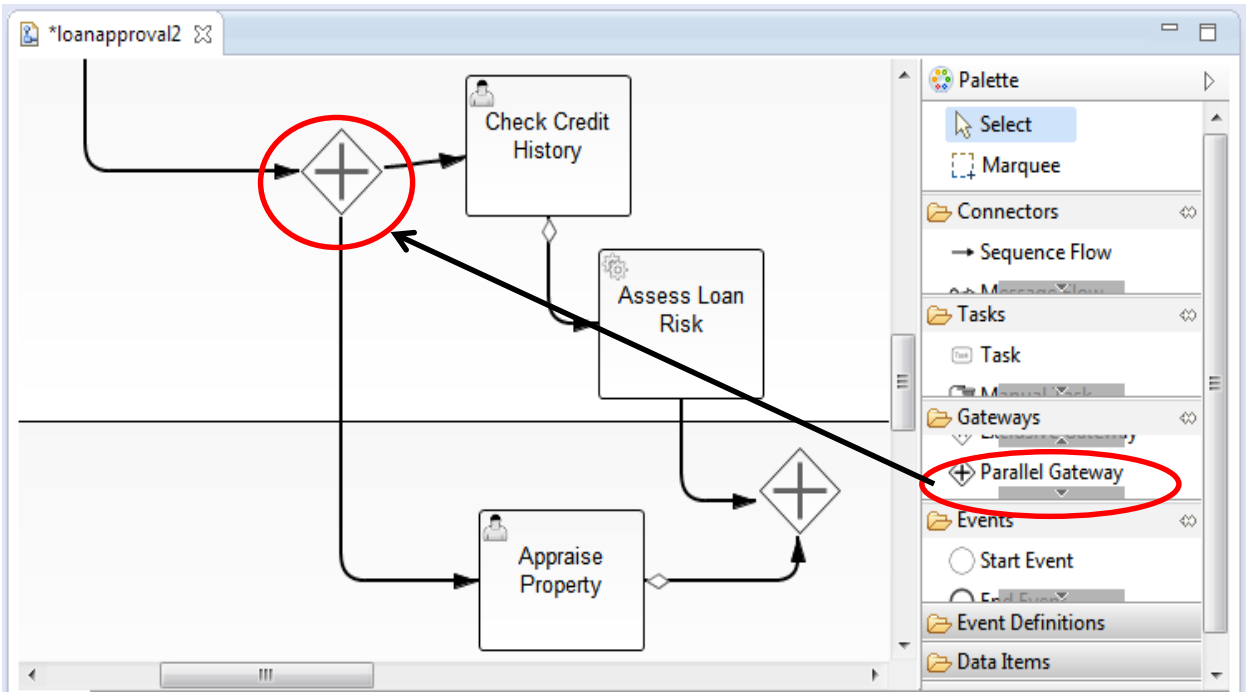
For 'Update Loan Application' user task sequence flow; assign the flow condition: ${Application_Status=='complete'}

For 'Check Credit History' user task sequence flow; assign the flow condition: ${Application_Status=='incomplete'}



Step 7

Add a parallel split gateway to execute user tasks 'Check Credit History' and 'Appraise Property' simultaneously.

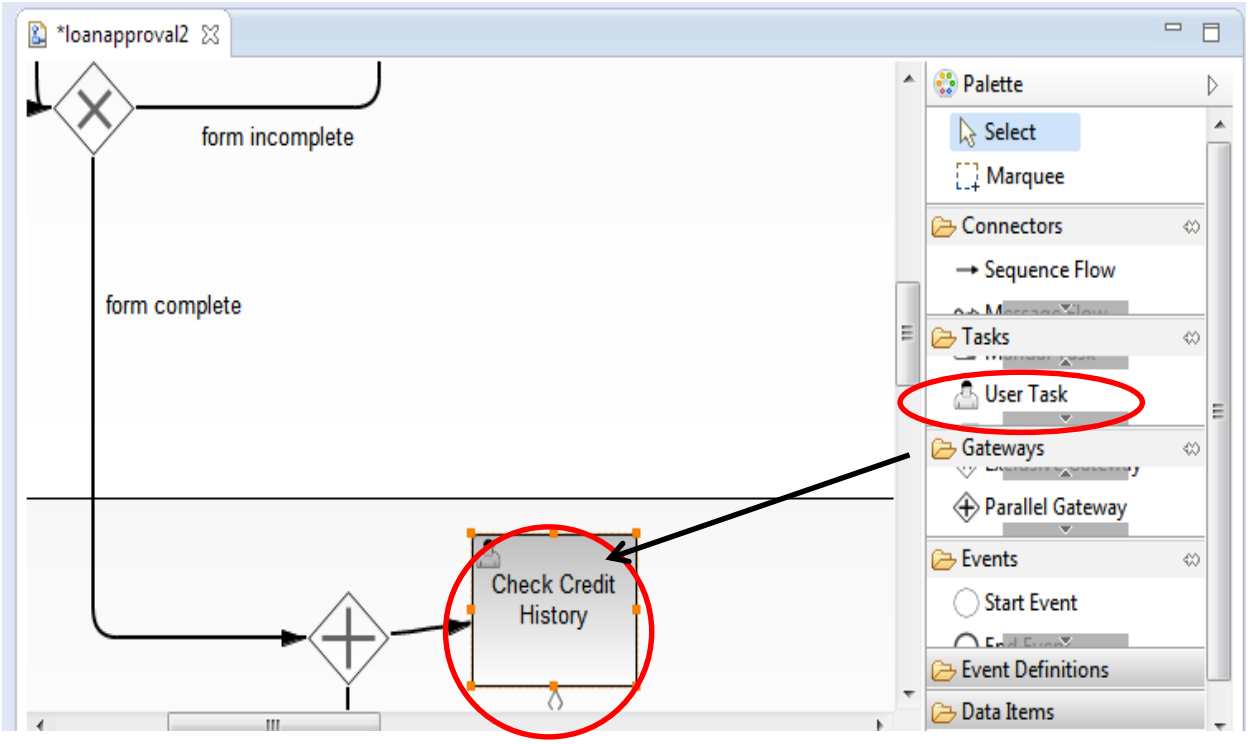


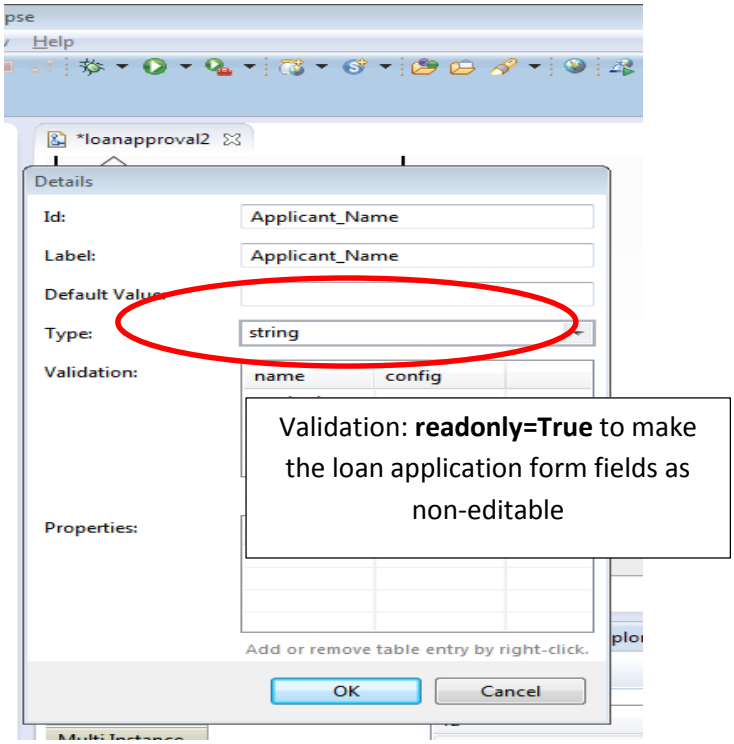
Step 8

Create a User Task: Check Credit History and assign user 'Peter' (General>>Assignee) as Finance Officer. Create a Credit History Report form as per Step 2. The loan application form should be visible to the Finance Officer, but in read-only format.

Credit History Report		
Field Name	**Field Type**	**Required(R)/Optional(O)/Not Applicable (N/A)**
Finance Officer Identifier	String	N/A
Court Judgement Information	String	O
Bankruptcy Information	String	O
Credit Assessment	Enum (B/BB/BBB/A/AA/AAA)	N/A
Loan Application History: Amount	Long	N/A
Loan Application History: Duration	String	N/A

Loan Application History: Start Date	Date	N/A
Loan Application History: End Date	Date	N/A
Loan Application History: Interest	Long	N/A
Overdue Credit Accounts: Amount	Long	N/A
Overdue Credit Accounts: Duration	String	N/A
Overdue Credit Accounts: Start Date	Date	N/A
Overdue Credit Accounts: End Date	Date	N/A
Overdue Credit Accounts: Interest	Long	N/A
Current Credit Card Provider: Amount	Long	N/A
Current Credit Card Provider: Duration	String	N/A
Current Credit Card Provider: Start Date	Date	N/A
Current Credit Card Provider: End Date	Date	N/A
Current Credit Card Provider: Interest	Long	N/A

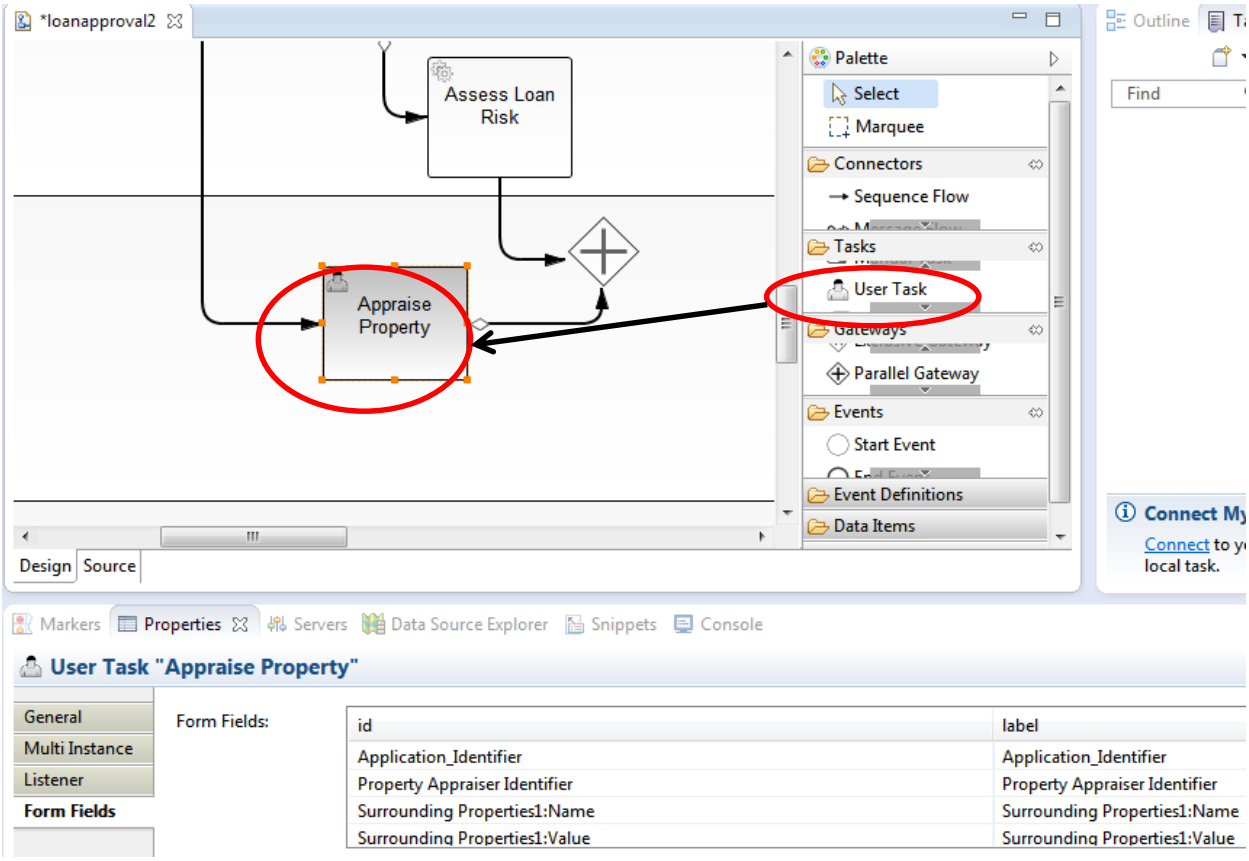




Step 9

Create a User Task: Appraise Property and assign user 'Mary' (General>>Assignee) as Property Appraiser. Create a Property Appraisal Report form as per Step 2.

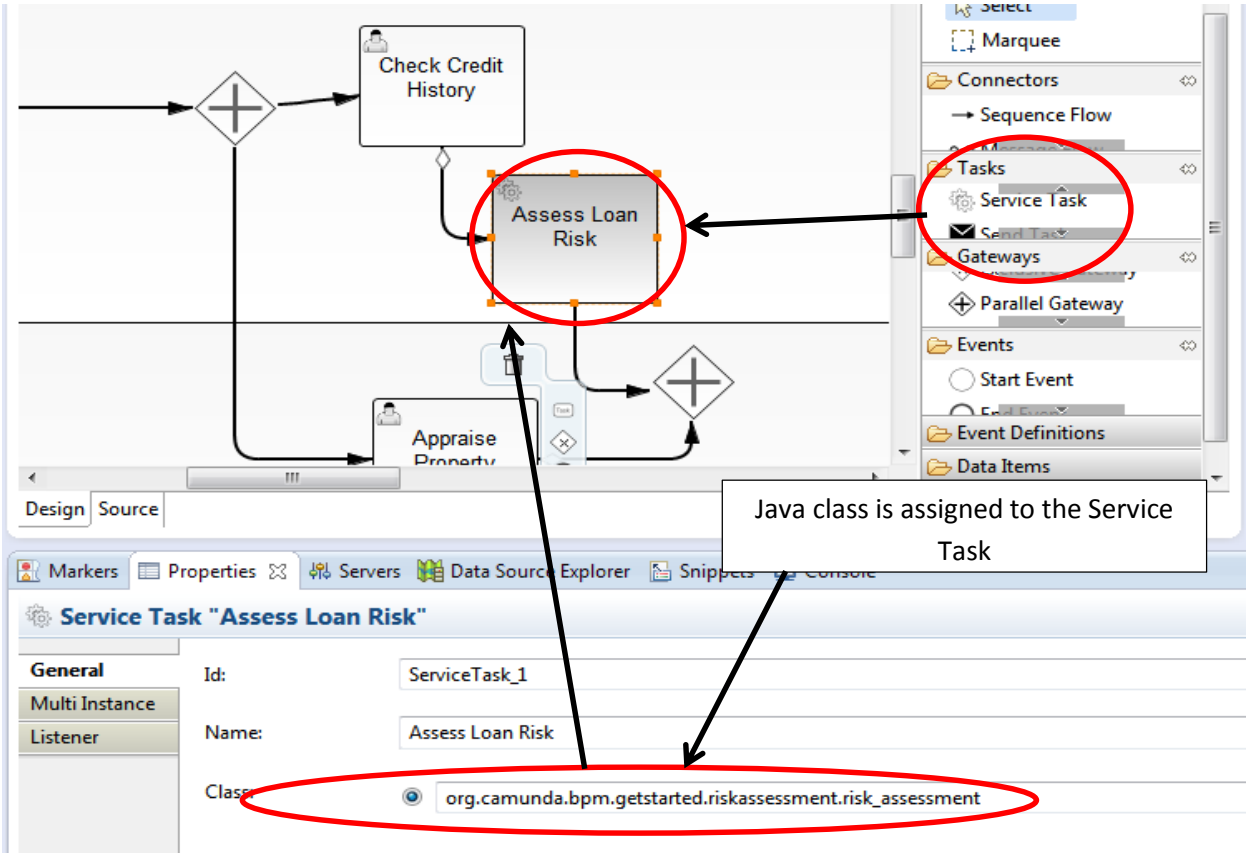
Property Appraisal Report		
Field Name	**Field Type**	**Required(R)/Optional(O)/Not Applicable (N/A)/ Read Only (RO)**
Application_Identifier	String	RO
Property Appraiser Identifier	String	N/A
Surrounding Properties1: Name	String	N/A
Surrounding Properties1: Value	Long	N/A
Surrounding Properties2: Name	String	N/A
Surrounding Properties2: Value	Long	N/A
Surrounding Properties3: Name	String	N/A
Surrounding Properties3: Value	Long	N/A
Estimate Value	Long	N/A
Comments_on_property	String	N/A

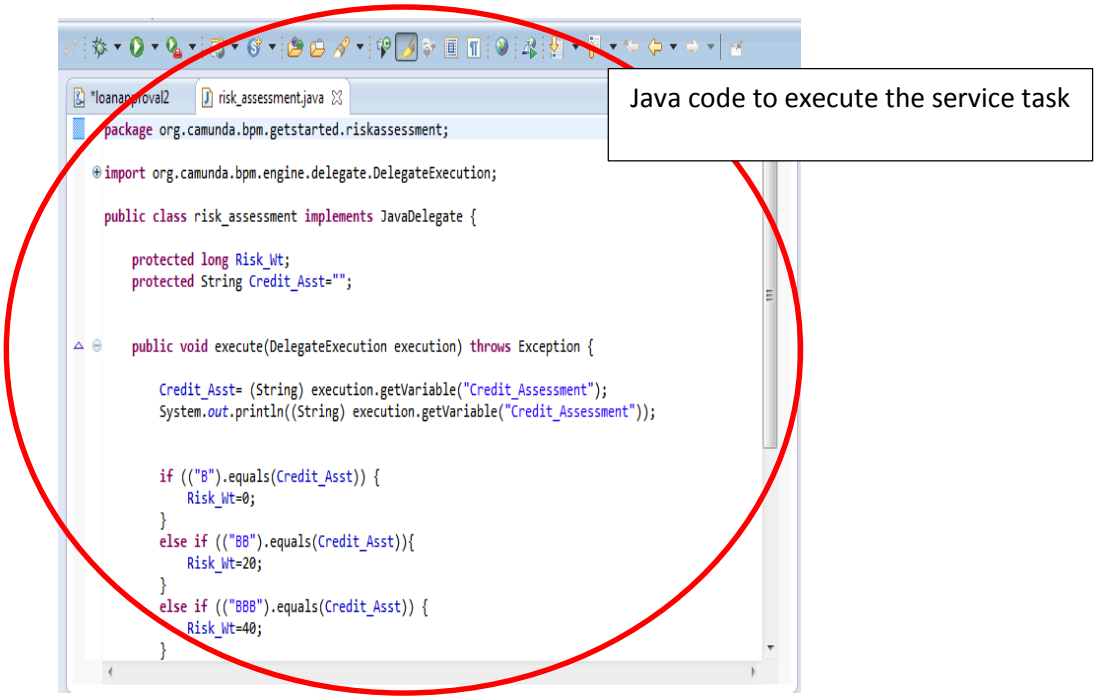


Step 10

Create a Service Task: Assess Loan Risk to automatically assign Risk Weights as per Credit Assessment selected. Create a Java class to execute this task.

Credit Assessment	Risk Weight
B	0
BB	20
BBB	40
A	60
AA	80
AAA	100


Check Credit History
Assess Loan Risk
Appraise
Marquee
Connectors
Sequence Flow
Tasks
Service Task
Gateways
Parallel Gateway
Events
Start Event
Event Definitions
Data Items
Design
Source
Java class is assigned to the Service Task
Markers
Properties
Servers
Data Source Explorer
Service Task "Assess Loan Risk"
General
Multi Instance
Listener
Id:
ServiceTask_1
Name:
Assess Loan Risk
org.camunda.bpm.getstarted.riskassessment.risk_assessment


risk_assessment.java
Java code to execute the service task
package org.camunda.bpm.getstarted.riskassessment;
import org.camunda.bpm.engine.delegate.DelegateExecution;
public class risk_assessment implements JavaDelegate {
protected long Risk_Wt;
protected String Credit_Asst="";
public void execute(DelegateExecution execution) throws Exception {
Credit_Asst= (String) execution.getVariable("Credit_Assessment");
System.out.println((String) execution.getVariable("Credit_Assessment"));
if (("B").equals(Credit_Asst)) {
Risk_Wt=0;
}
else if (("BB").equals(Credit_Asst)){
Risk_Wt=20;
}
else if (("BBB").equals(Credit_Asst)) {
Risk_Wt=40;
}

Java code to execute the Service Task:

```
package org.camunda.bpm.getstarted.riskassessment;

import org.camunda.bpm.engine.delegate.DelegateExecution;
import org.camunda.bpm.engine.delegate.JavaDelegate;

public class risk_assessment implements JavaDelegate {

            protected long Risk_Wt;
            protected String Credit_Asst="";

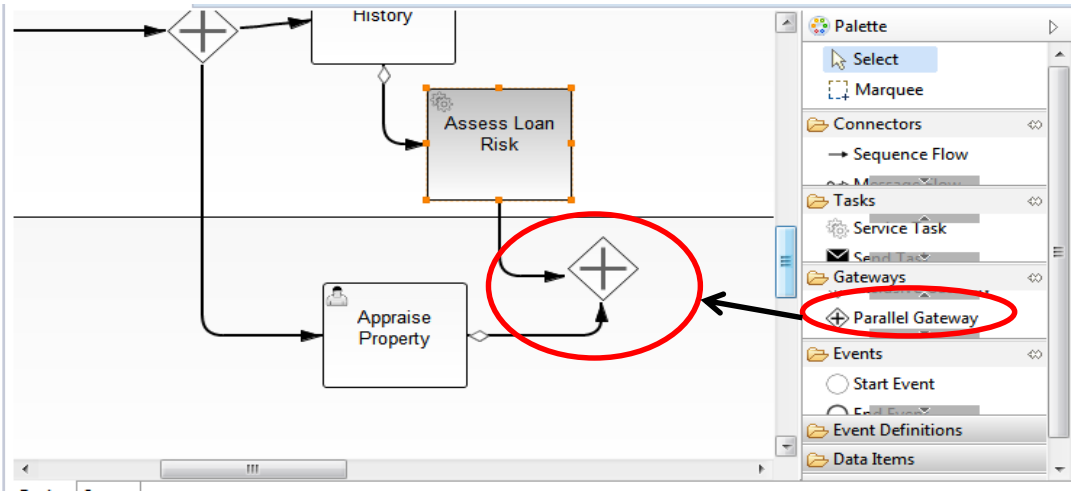

            public void execute(DelegateExecution execution) throws
Exception {

                  Credit_Asst= (String)
execution.getVariable("Credit_Assessment");
                  System.out.println((String)
execution.getVariable("Credit_Assessment"));


                  if (("B").equals(Credit_Asst)) {
                        Risk_Wt=0;
                  }
                  else if (("BB").equals(Credit_Asst)){
                        Risk_Wt=20;
                  }
                  else if (("BBB").equals(Credit_Asst)) {
                        Risk_Wt=40;
                  }
                  else if (("A").equals(Credit_Asst)) {
                        Risk_Wt=60;
                  }
                  else if (("AA").equals(Credit_Asst)) {
                        Risk_Wt=80;
                  }
                  else if (("AAA").equals(Credit_Asst)) {
                        Risk_Wt=100;
                  }
                  execution.setVariable("Risk_Weight", Risk_Wt);
                  System.out.println("Risk Weight" + Risk_Wt);
                  System.out.println("Credit_Assessment = " + Credit_Asst);
            }
}
```

Step 11

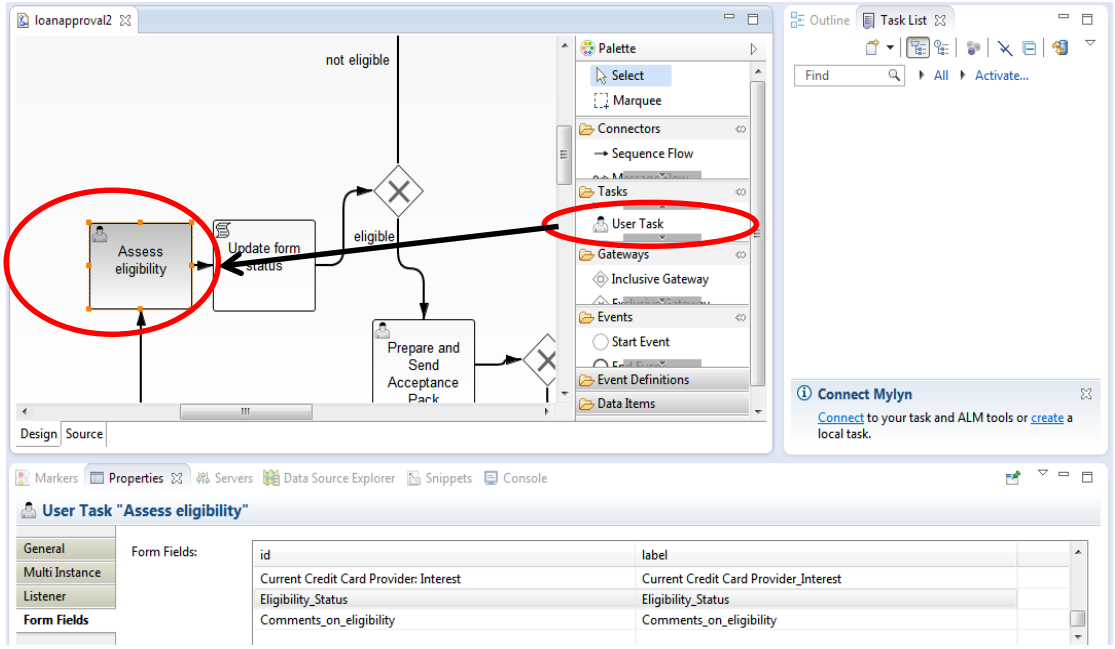
Add a parallel join gateway to execute the outputs of 'Assess Loan Risk' and 'Appraise Property' tasks simultaneously.



Step 12

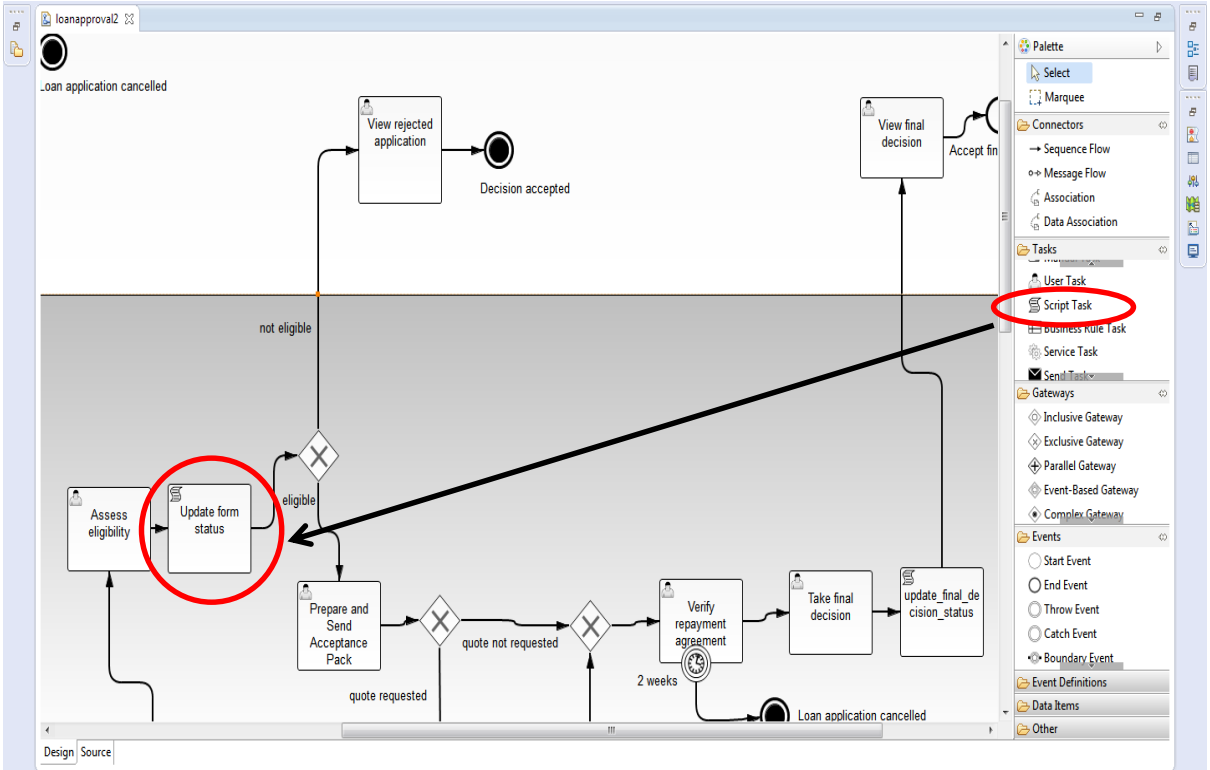
Create a User Task: Assess eligibility and assign user 'John' (General>>Assignee) as Loan Officer. Loan Application form, Credit History report and Property Appraisal report must be visible to the Loan Officer but in read-only format. Create the following fields in the user form as per step2

Field Name	**Field Type**	**Required(R)/Optional(O)/Not Applicable (N/A)/ Read Only (RO)**
Loan Officer Identifier	String	N/A
Eligibility_Status	Enum (eligible/ineligible)	N/A
Comments_on_eligibility	String	N/A



Step 13

Create a script task-Update form status to check for eligibility status of loan application. If eligibility status=’eligible’, change Application_Status=’assessed’ and assign sequence flow to ‘Prepare and Acceptance Task’. If eligibility status=’ineligible’, Application_Status=’rejected’ change assign sequence flow to ‘View Rejected Application’ task.



Script Language: Groovy

Script:

```
if ((Eligibility_Status=='ineligible')) {
Application_Status='rejected';
}
else {
Application_Status='assessed';
}
out:println "Application_Status" + Application_Status;
out:println "Applicant_Name" + Applicant_Name;
out:println "Applicant_SurName" + Applicant_SurName;
out:println "Email" + Email;
out:println "Home_Phone" + Home_Phone;
out:println "Mobile_Phone" + Mobile_Phone;
out:println "Current_Address" + Current_Address;
out:println "Previous_Address" + Previous_Address;
out:println "Current_Employer" + Current_Employer;
out:println "Monthly_Net_Revenue" + Monthly_Net_Revenue;
out:println "Bank_Name" + Bank_Name;
out:println "Account_Type" + Account_Type;
out:println "Account_Number" + Account_Number;
out:println "Account_Balance" + Account_Balance;
out:println "Property_Type" + Property_Type;
out:println "Property_Address" + Property_Address;
out:println "Purchasing_Price" + Purchasing_Price;
out:println "Loan_Type" + Loan_Type;
out:println "Loan_Provider" + Loan_Provider;
out:println "Loan_Amount" + Loan_Amount;
out:println "Loan_Duration" + Loan_Duration;
out:println "Loan_Start_Date" + Loan_Start_Date;
out:println "Loan_End_Date" + Loan_End_Date;
```

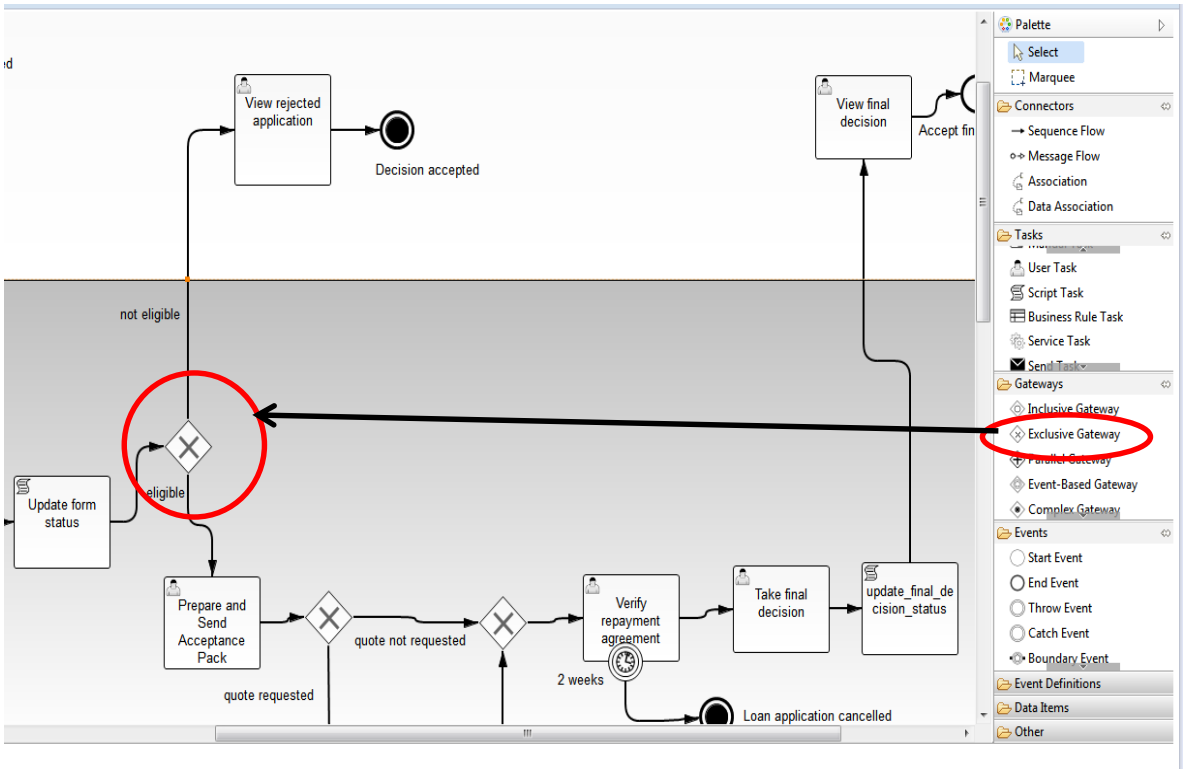
```
out:println "Interest_Rate" + Interest_Rate;
out:println "Interest_Type" + Interest_Type;
out:println "Application_Submission_Date" + Application_Submission_Date;
out:println "Application_Revision_Date" + Application_Revision_Date;
out:println "Comment_on_Status" + Comment_on_Status;
```

Step 14

Add an Exor split gateway with the following Listener expression: ${Application_Status=='assessed'} or ${Application_Status=='rejected'} (for routing appropriately).

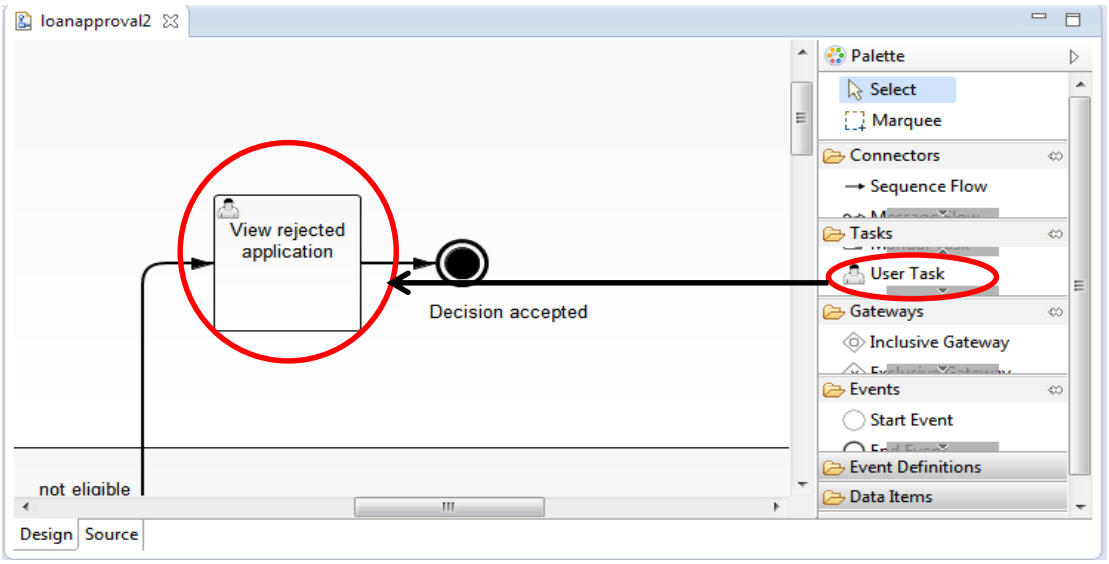
For 'View rejected Application' user task sequence flow; assign the flow condition: ${Application_Status=='rejected'}

For 'Prepare and Send Acceptance Pack' user task sequence flow; assign the flow condition: ${Application_Status=='assessed'}



Step 15

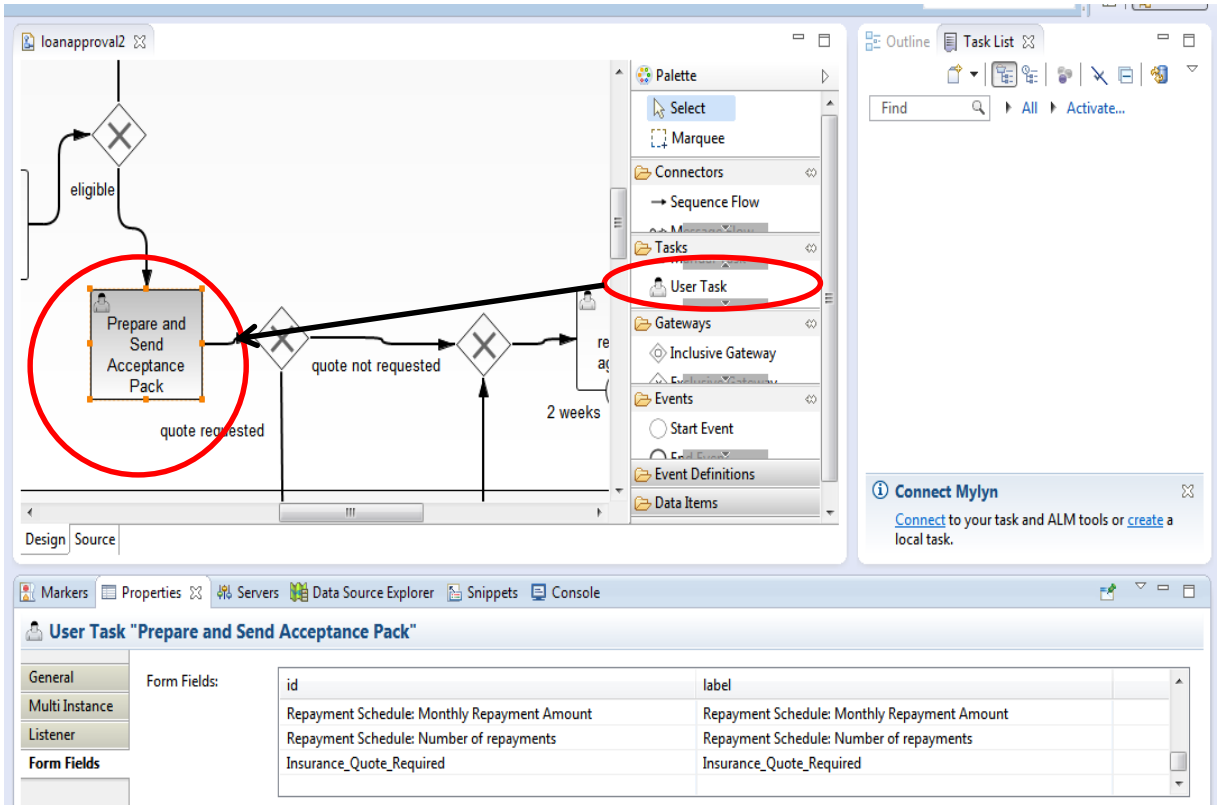
Create a User Task: View Rejected Application and assign user ‘Mary’ (General>>Assignee) as Loan Applicant. Create an intermediate end event ‘Decision Accepted’ as per step3. The user must be able to view the loan application in read-only format.



Step 16

Create a User Task: Prepare and Send Acceptance Pack and assign user ‘John’ (General>>Assignee) as Loan Officer. Loan Application form, Credit History report, Property Appraisal report and fields added in ‘Assess Eligibility’ user task must be viewed in read-only format. The following fields must be added.

Repayment Agreement		
Field Name	**Field Type**	**Required(R)/Optional(O)/Not Applicable (N/A)/ Read Only (RO)**
Application_Identifier	String	RO
Repayment Schedule: Monthly Repayment Amount	Long	N/A
Repayment Schedule: Number of repayments	Long	N/A
Insurance_Quote_Required	Enum (complete/incomplete)	N/A



Step 17

Add an Xor split gateway with the following Listener expression:

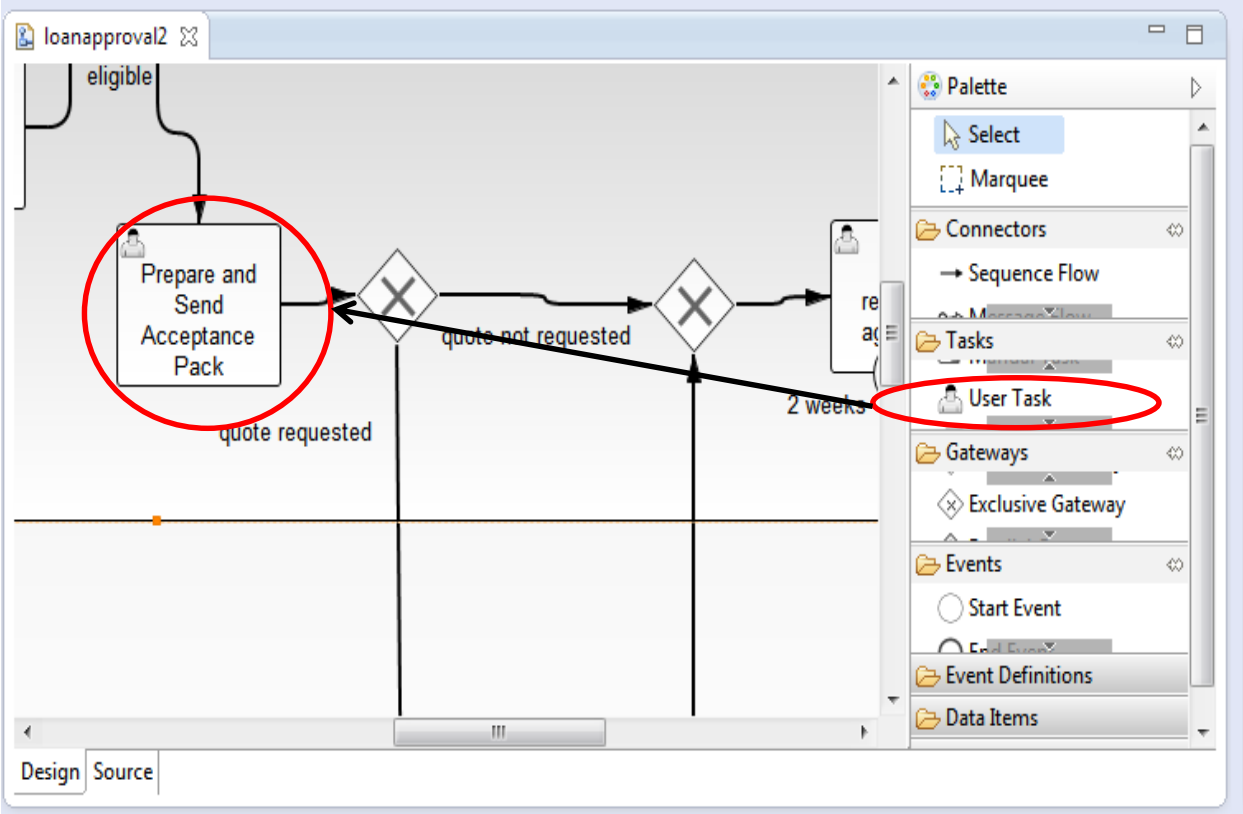
${Insurance_Quote_Required=='complete'} or

${Insurance_Quote_Required=='incomplete'} (for routing appropriately).

For ‘Prepare and send home insurance quote’ user task sequence flow; assign the flow condition: ${ Insurance_Quote_Required=='incomplete'}

For the Exor join gateway sequence flow; assign the flow condition:

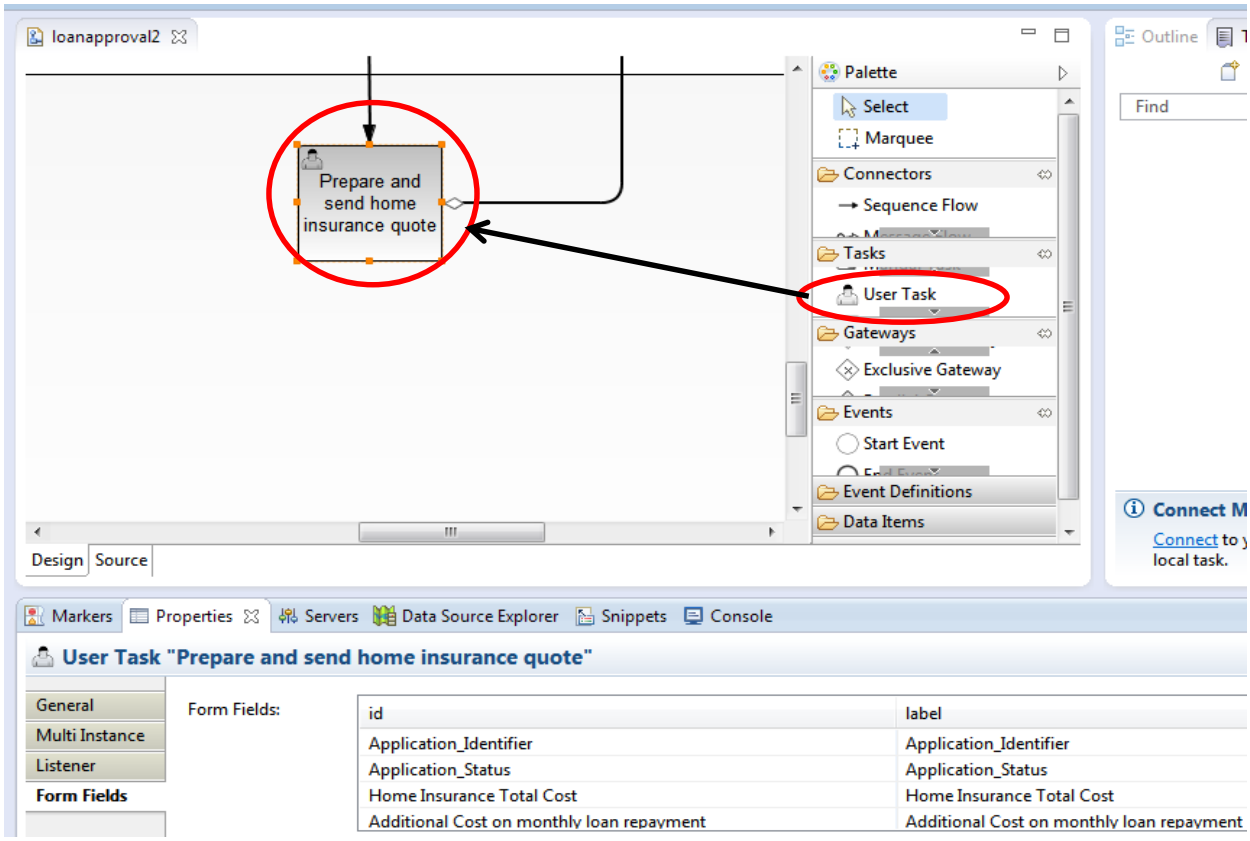
${Insurance_Quote_Required=='complete'}



Step 18

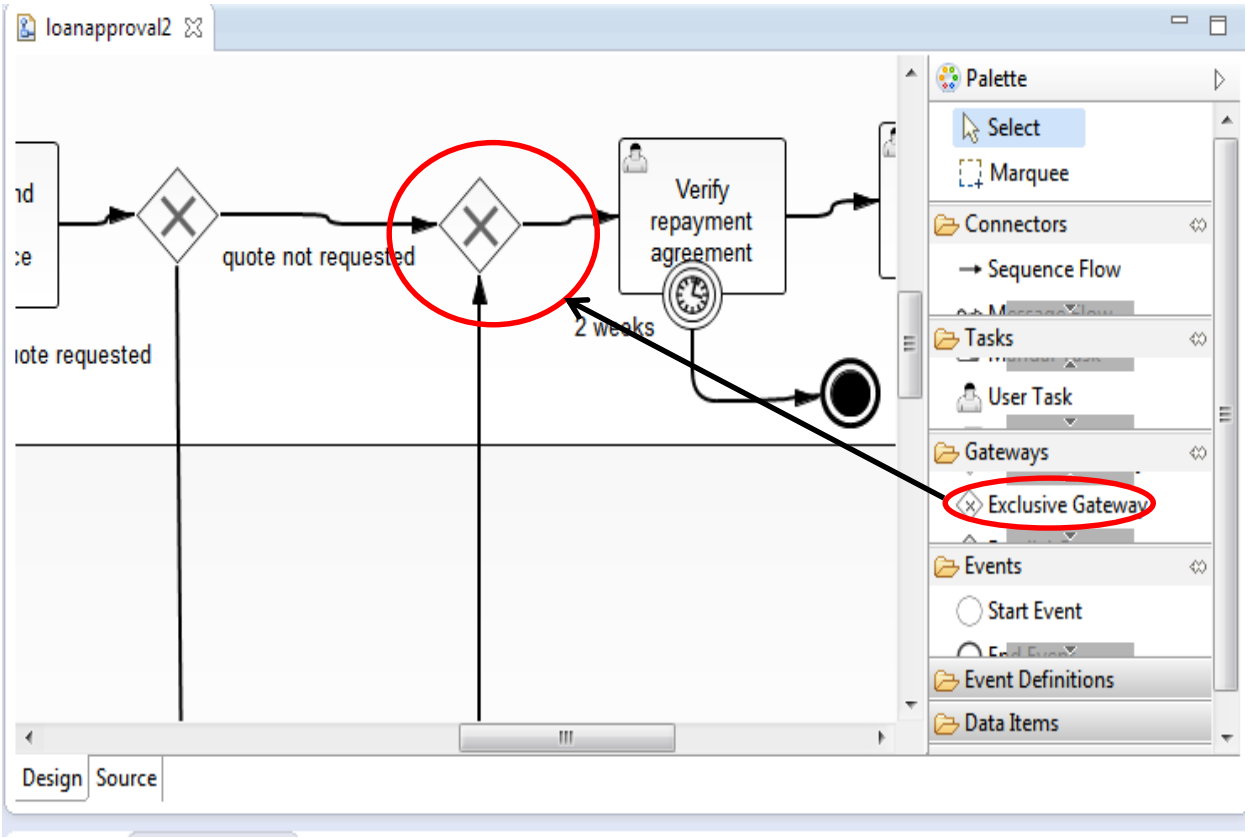
Create a User Task: Prepare and Send Acceptance Pack and assign user 'Mary' (General>>Assignee) as Insurance Sales Representative. The following fields must be added:

Home Insurance Quote		
Field Name	**Field Type**	**Required(R)/Optional(O)/Not Applicable (N/A)/ Read Only (RO)**
Application_Identifier	String	RO
Home Insurance Total Cost	Long	N/A
Additional Cost on monthly loan repayment	Long	N/A
Insurance Terms and Conditions	Boolean	N/A
Insurance Sales Representative Identifier	String	N/A



Step 19

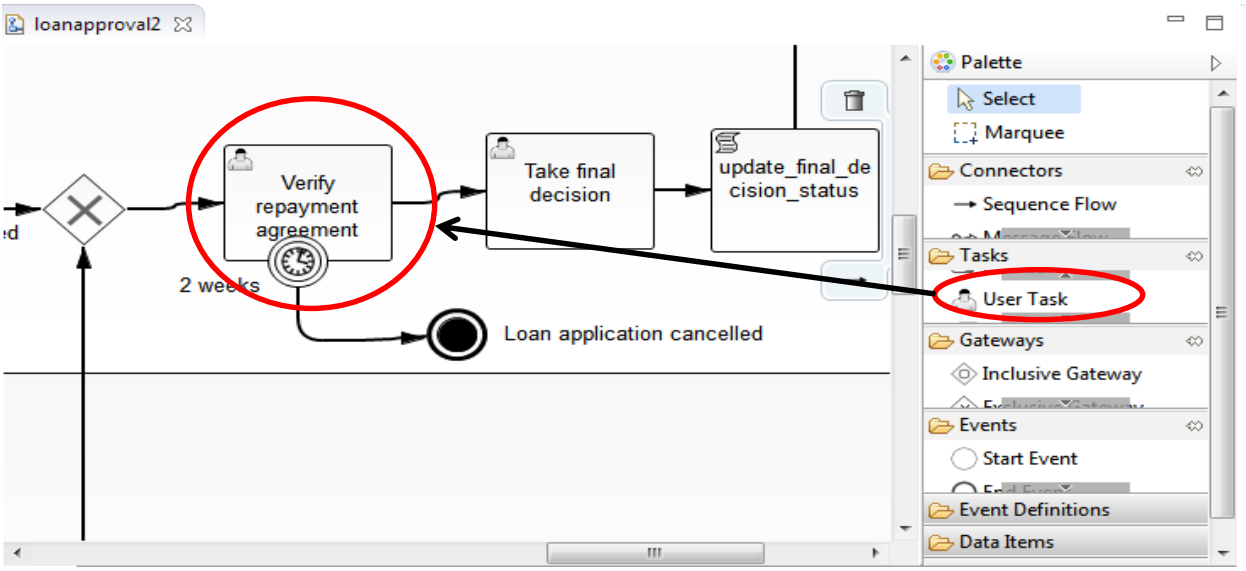
An Exor-join gateway is used to alternatively route the outputs of 'Prepare and send home insurance quote' user task and 'quote not requested' sequence flow

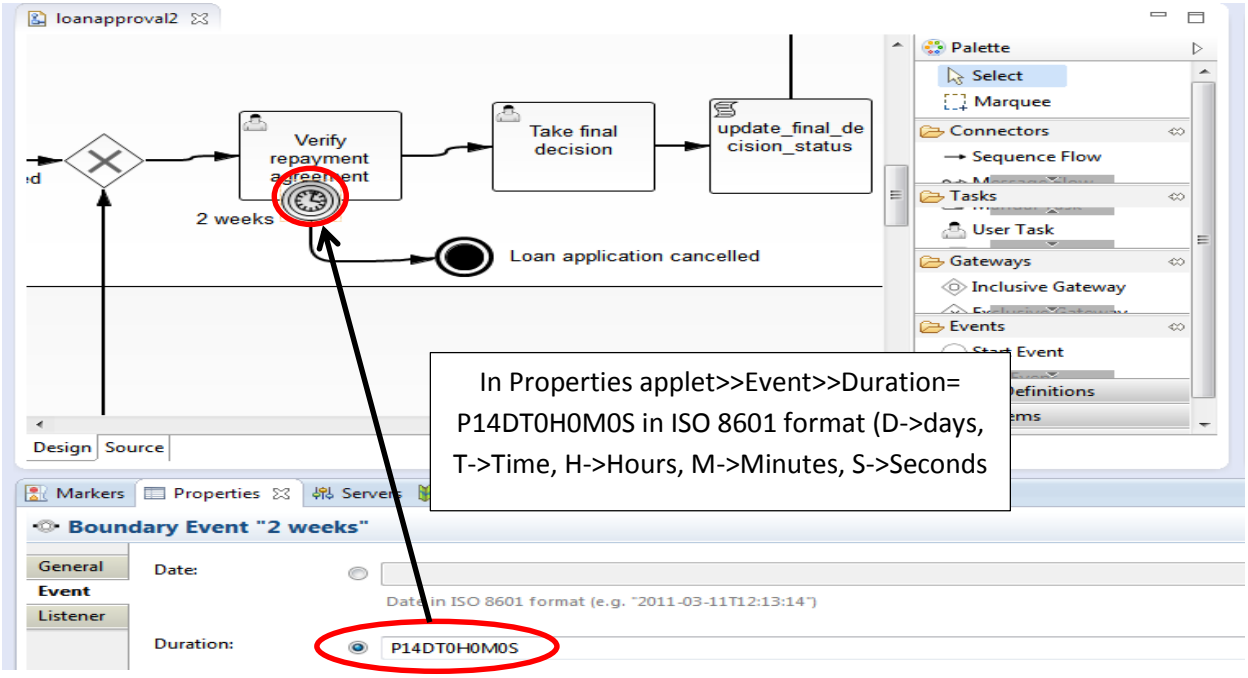


Step 20

Create a User Task: Verify repayment agreement and assign user 'John' (General>>Assignee) as the Loan Officer. Loan Application form, Credit History report, Property Appraisal report, Repayment agreement and home insurance quote must be viewed in read-only format. A timer event with a wait period of 2 weeks (14 days) is added as per step 3, at the end of which the loan application is cancelled. The following fields are added under agreement summary.

Agreement Summary		
Field Name	**Field Type**	**Required(R)/Optional(O)/Not Applicable (N/A)/ Read Only (RO)**
Application_Identifier	String	RO
Conditions agreed	Boolean	N/A
Repayment agreed	Boolean	N/A

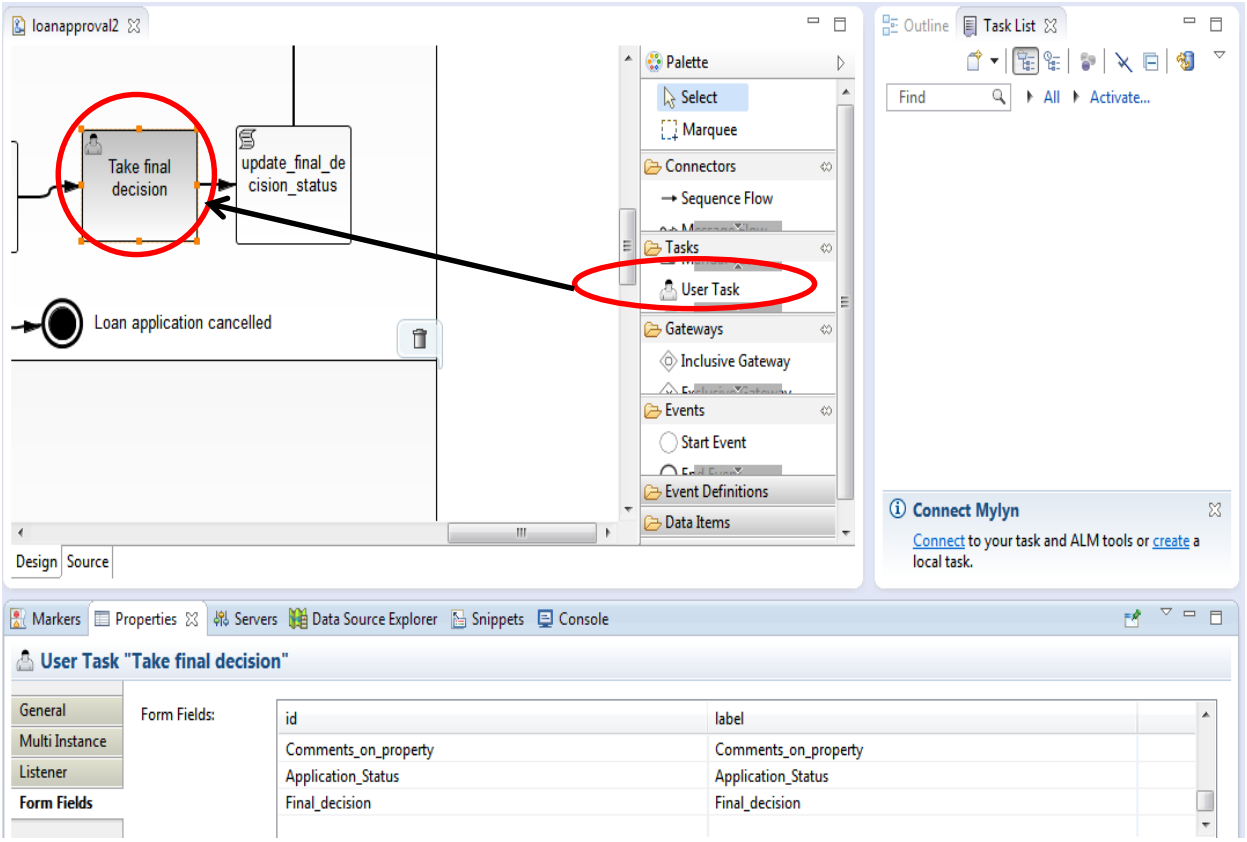




Step 21

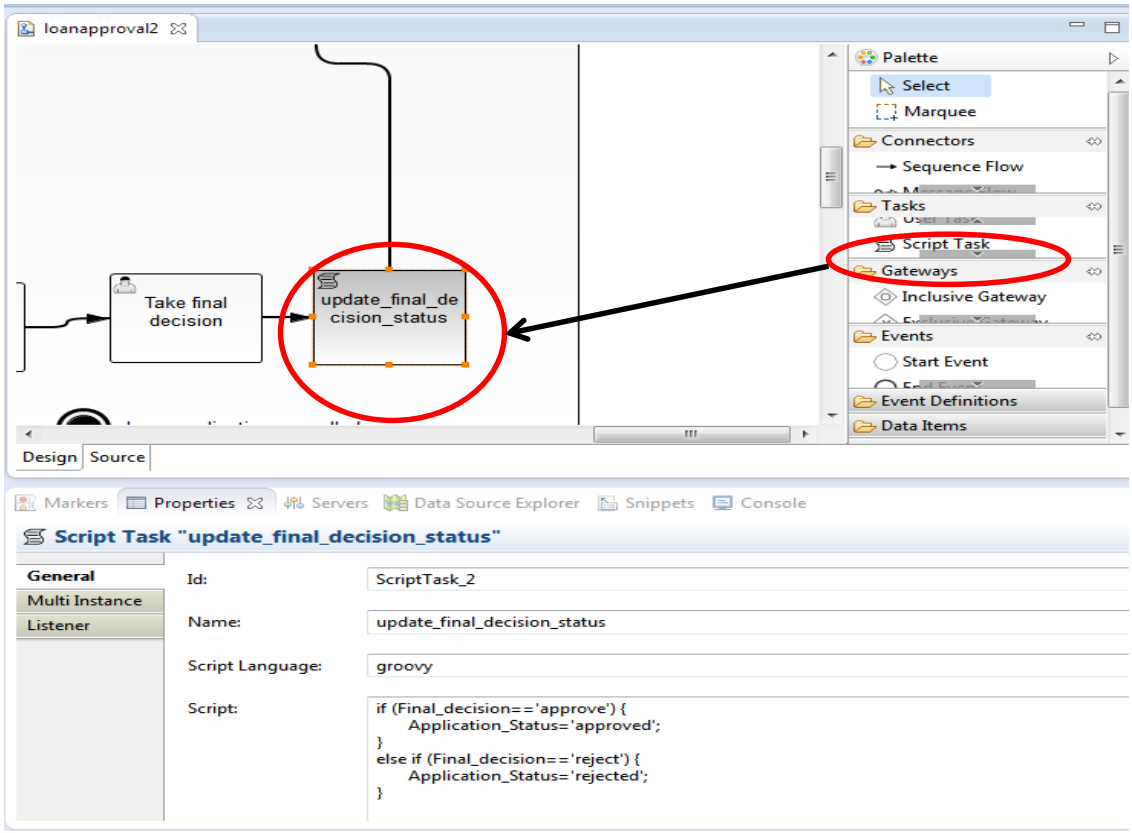
Create a User Task: Take final decision and assign user ‘Peter’ (General>>Assignee) as the Loan Officer. Loan Application form, Credit History report, Property Appraisal report, Repayment agreement, home insurance quote and Agreement Summary must be viewed in read-only format. The following fields are added.

Field Name	Field Type	Required(R)/Optional(O)/Not Applicable (N/A)/ Read Only (RO)
Final_decision	Enum (approve/reject)	N/A



Step 22

Create a script task-Update final decision status to notify the final decision to the loan applicant. Depending on the Final_decision field output from Take final decision user task, Application_Status is changed either to approved or rejected.



Script Language: Groovy

Script:

```
if (Final_decision=='approve') {
            Application_Status='approved';
}
else if (Final_decision=='reject') {
            Application_Status='rejected';
}

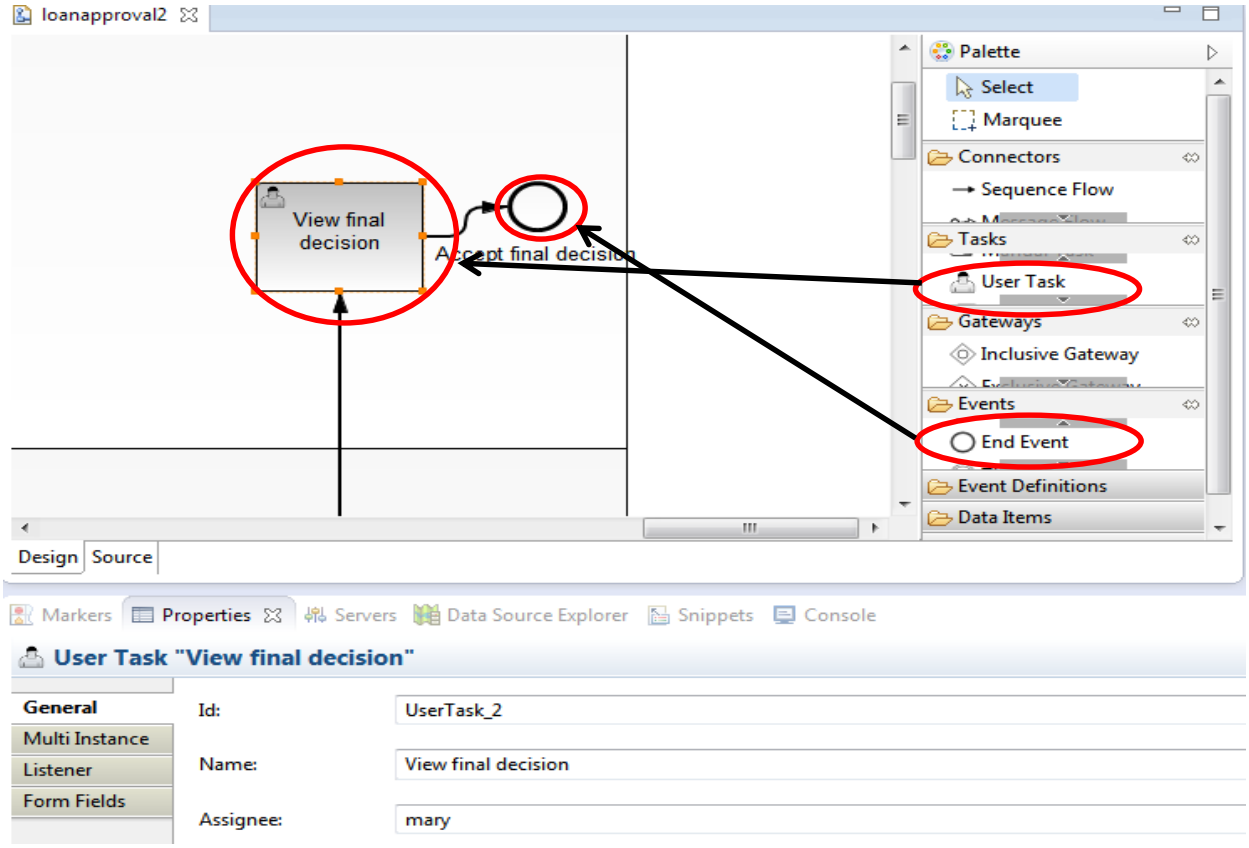
else {
            Application_Status='unknown';
}

out:println "Application_Status" + Application_Status;
out:println "Applicant_Name" + Applicant_Name;
out:println "Applicant_SurName" + Applicant_SurName;
out:println "Email" + Email;
out:println "Home_Phone" + Home_Phone;
out:println "Mobile_Phone" + Mobile_Phone;
out:println "Current_Address" + Current_Address;
out:println "Previous_Address" + Previous_Address;
out:println "Current_Employer" + Current_Employer;
out:println "Monthly_Net_Revenue" + Monthly_Net_Revenue;
out:println "Bank_Name" + Bank_Name;
out:println "Account_Type" + Account_Type;
out:println "Account_Number" + Account_Number;
out:println "Account_Balance" + Account_Balance;
```

```
out:println "Property_Type" + Property_Type;
out:println "Property_Address" + Property_Address;
out:println "Purchasing_Price" + Purchasing_Price;
out:println "Loan_Type" + Loan_Type;
out:println "Loan_Provider" + Loan_Provider;
out:println "Loan_Amount" + Loan_Amount;
out:println "Loan_Duration" + Loan_Duration;
out:println "Loan_Start_Date" + Loan_Start_Date;
out:println "Loan_End_Date" + Loan_End_Date;
out:println "Interest_Rate" + Interest_Rate;
out:println "Interest_Type" + Interest_Type;
out:println "Application_Submission_Date" + Application_Submission_Date;
out:println "Application_Revision_Date" + Application_Revision_Date;
out:println "Comment_on_Status" + Comment_on_Status;
```

Step 23

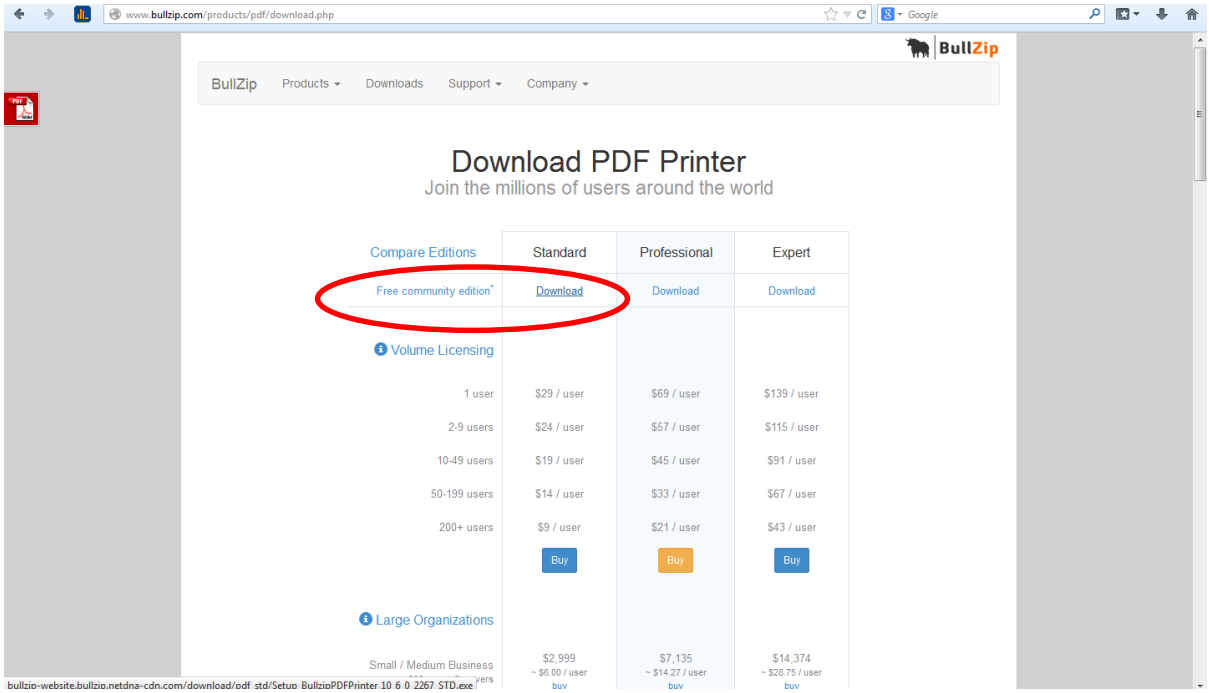
Create a User Task: View final decision and assign user ‘Mary’ (General>>Assignee) as Loan Applicant. Create an end event ‘Accept Final decision’ to complete the process. The user must be able to view the loan application in read-only format.

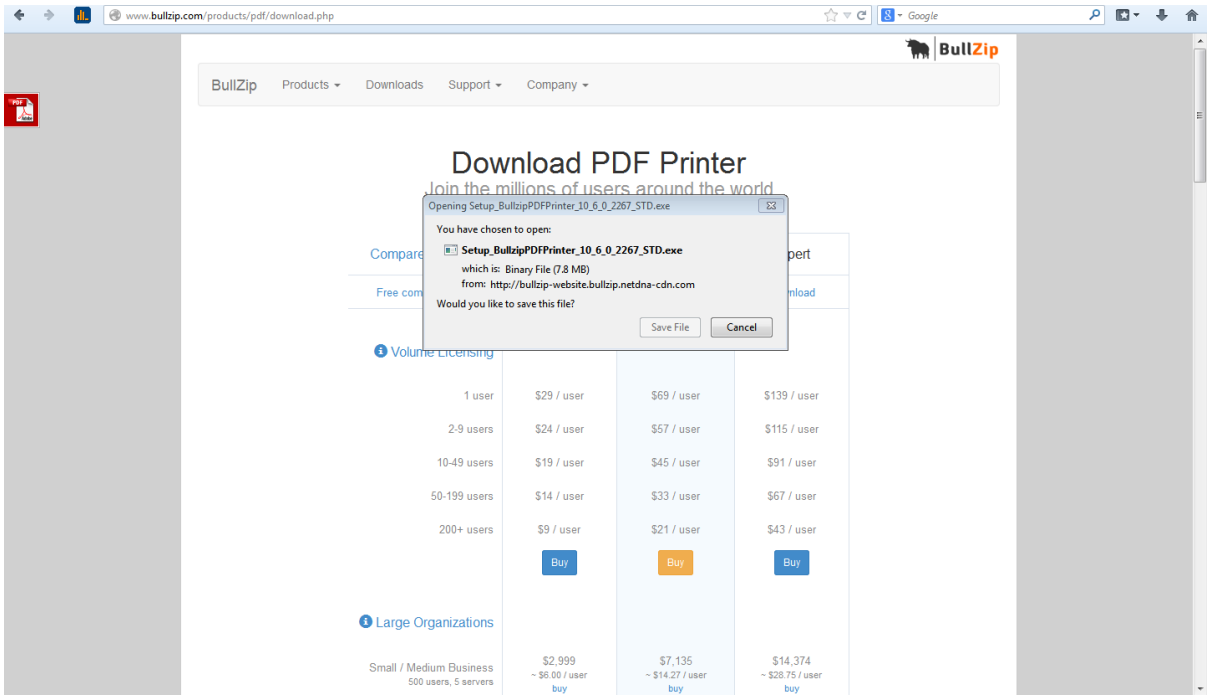


Step 24

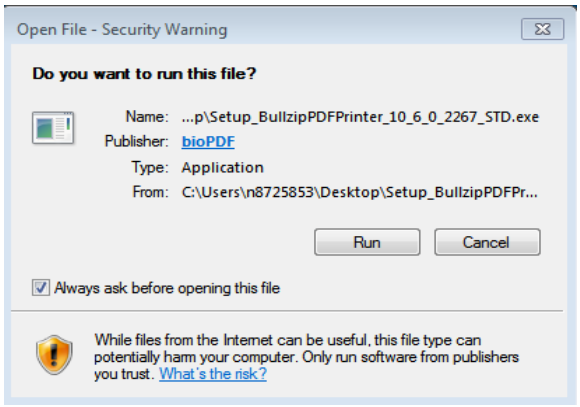
Download the bullzip pdf printer from the following link:

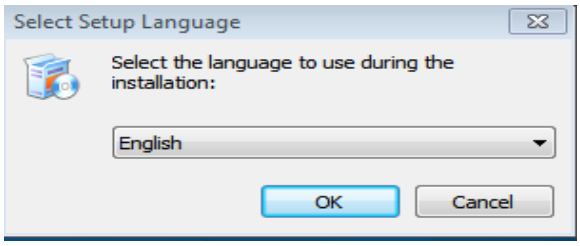
http://www.bullzip.com/products/pdf/download.php

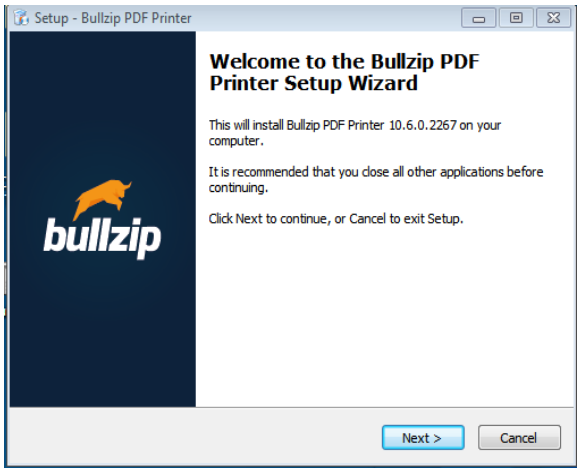


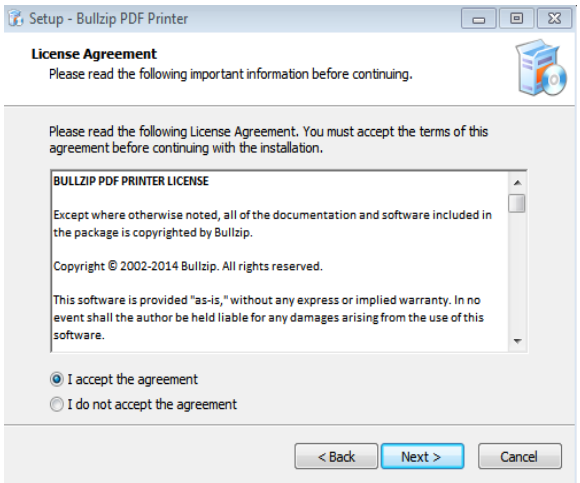


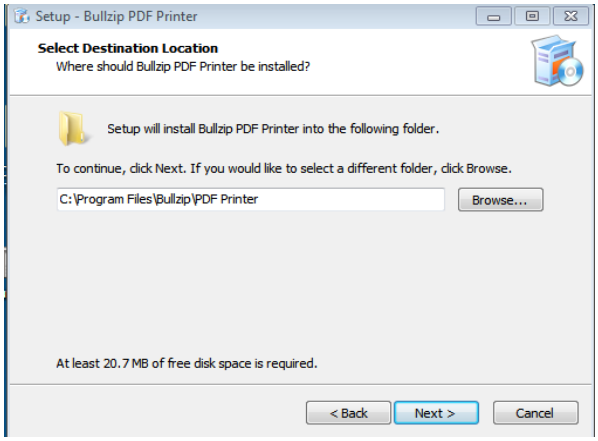
Install Bullzip printer

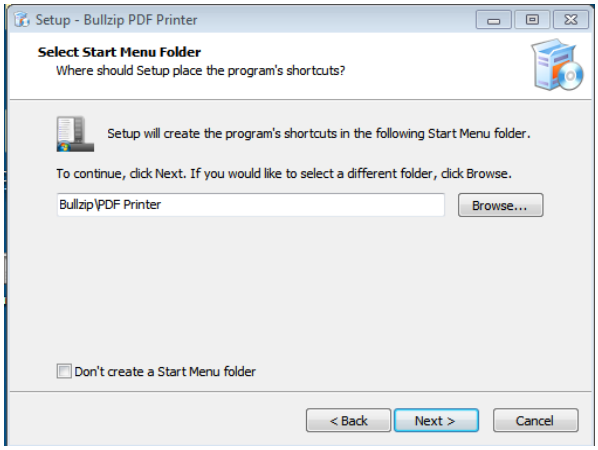


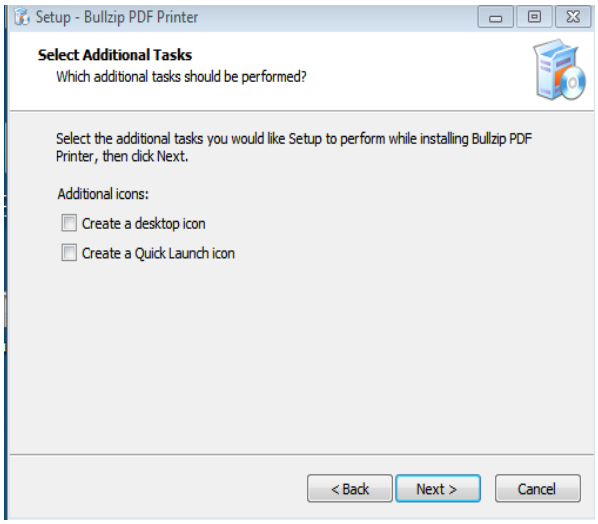


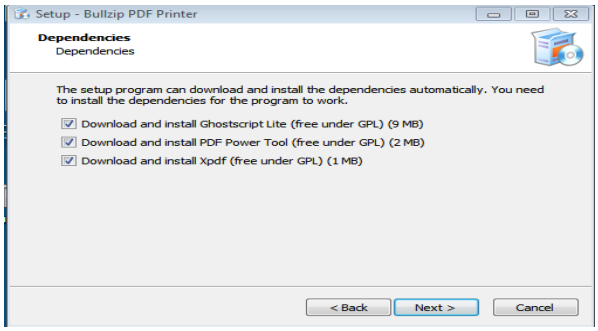


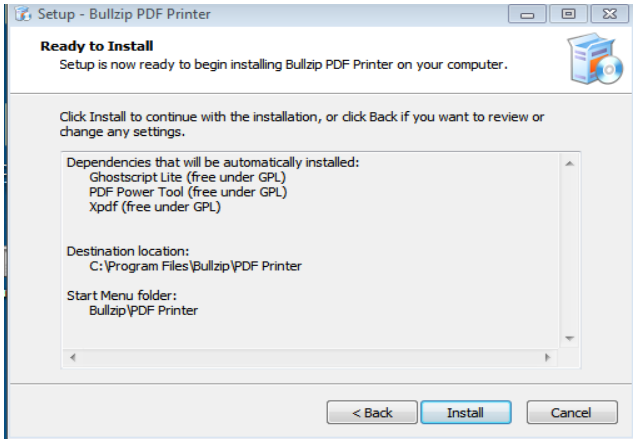


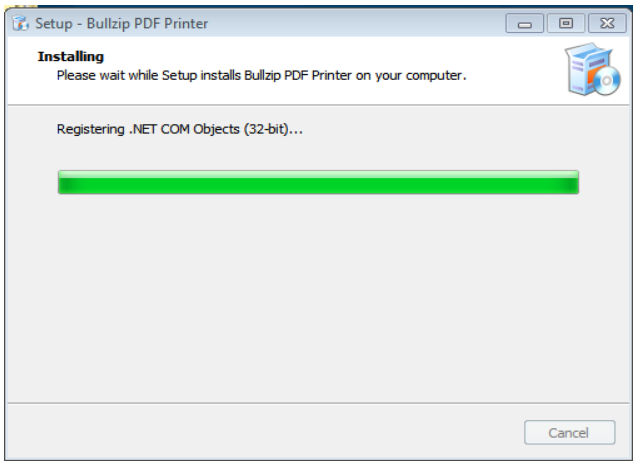

Setup - Bullzip PDF Printer
Select Destination Location
Where should Bullzip PDF Printer be installed?
Setup will install Bullzip PDF Printer into the following folder.
To continue, click Next. If you would like to select a different folder, click Browse.
C:\Program Files\Bullzip\PDF Printer
Browse...
At least 20.7 MB of free disk space is required.
< Back
Next >
Cancel

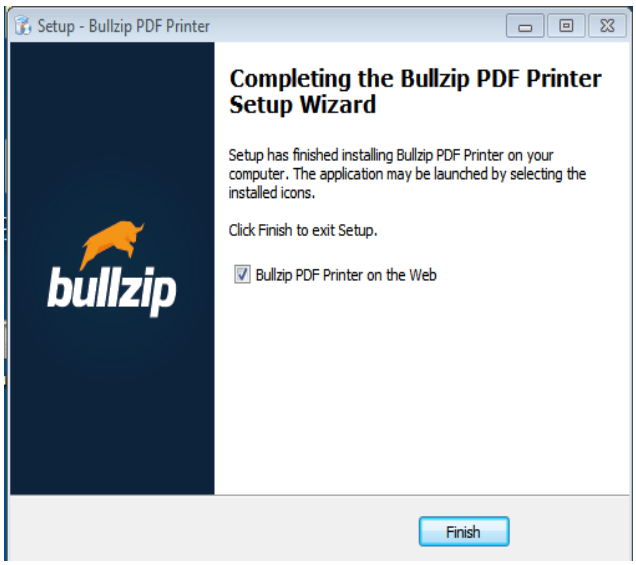

Setup - Bullzip PDF Printer
Select Start Menu Folder
Where should Setup place the program's shortcuts?
Setup will create the program's shortcuts in the following Start Menu folder.
To continue, click Next. If you would like to select a different folder, click Browse.
Bullzip\PDF Printer
Browse...
Don't create a Start Menu folder
< Back
Next >
Cancel


Setup - Bullzip PDF Printer
Select Additional Tasks
Which additional tasks should be performed?
Select the additional tasks you would like Setup to perform while installing Bullzip PDF Printer, then click Next.
Additional icons:
Create a desktop icon
Create a Quick Launch icon
< Back
Next >
Cancel


Setup - Bullzip PDF Printer
Dependencies
Dependencies
The setup program can download and install the dependencies automatically. You need to install the dependencies for the program to work.
Download and install Ghostscript Lite (free under GPL) (9 MB)
Download and install PDF Power Tool (free under GPL) (2 MB)
Download and install Xpdf (free under GPL) (1 MB)
< Back
Next >
Cancel

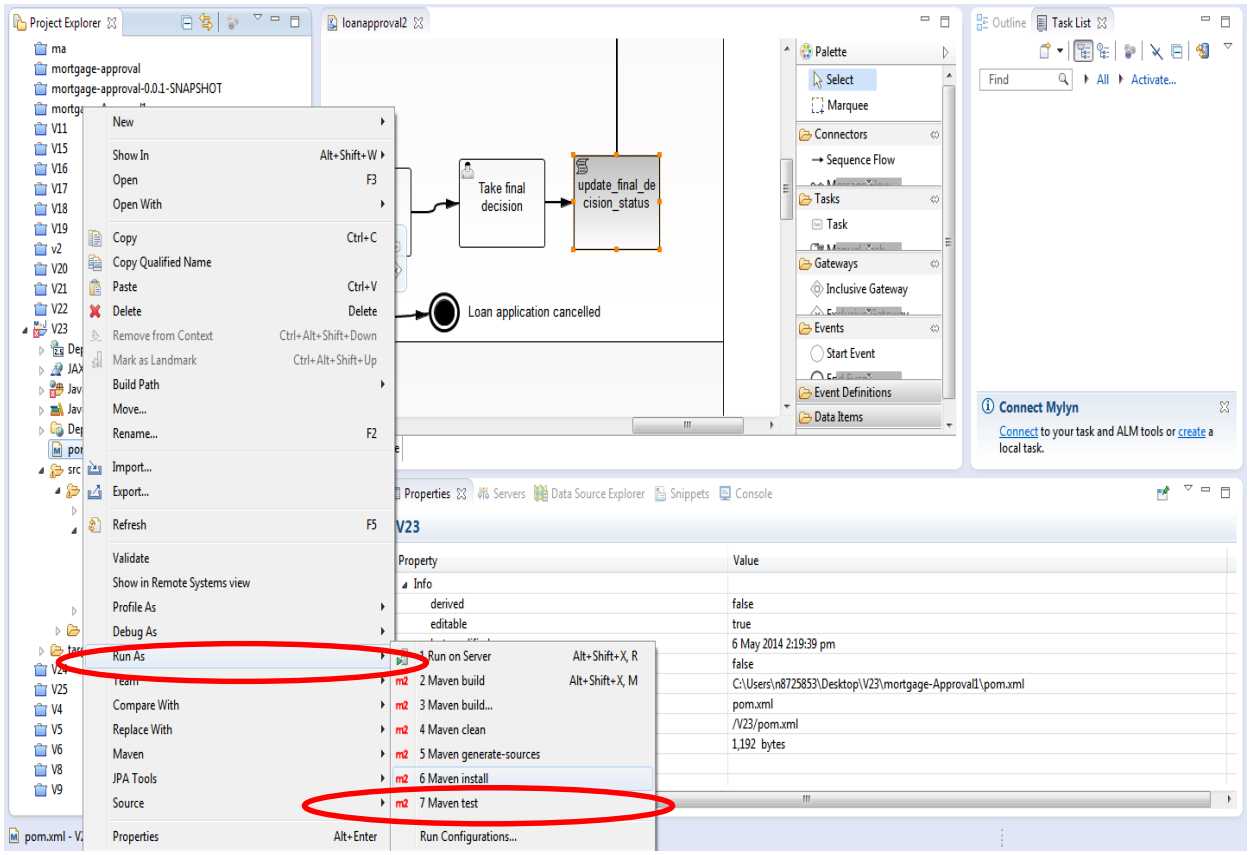

Setup - Bullzip PDF Printer
Ready to Install
Setup is now ready to begin installing Bullzip PDF Printer on your computer.
Click Install to continue with the installation, or click Back if you want to review or change any settings.
Dependencies that will be automatically installed:
Ghostscript Lite (free under GPL)
PDF Power Tool (free under GPL)
Xpdf (free under GPL)
Destination location:
C:\Program Files\Bullzip\PDF Printer
Start Menu folder:
Bullzip\PDF Printer
< Back
Install
Cancel


Setup - Bullzip PDF Printer
Installing
Please wait while Setup installs Bullzip PDF Printer on your computer.
Registering .NET COM Objects (32-bit)...
Cancel

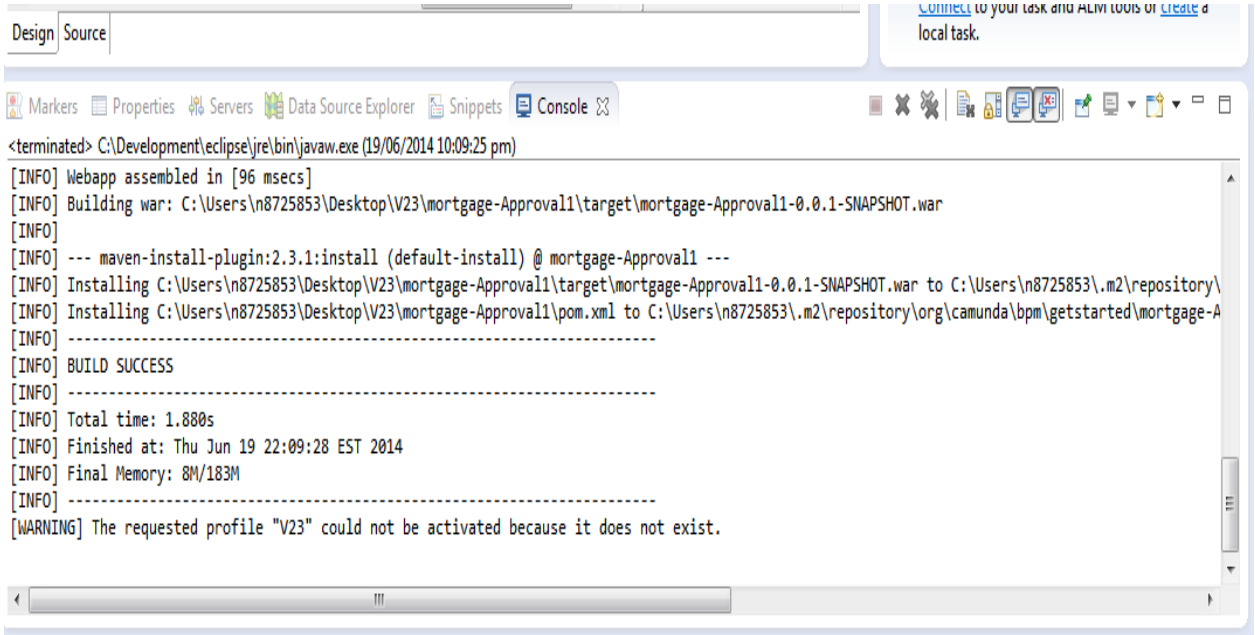

Setup - Bullzip PDF Printer
bullzip
Completing the Bullzip PDF Printer Setup Wizard
Setup has finished installing Bullzip PDF Printer on your computer. The application may be launched by selecting the installed icons.
Click Finish to exit Setup.
Bullzip PDF Printer on the Web
Finish

4.0 Validation and Deployment

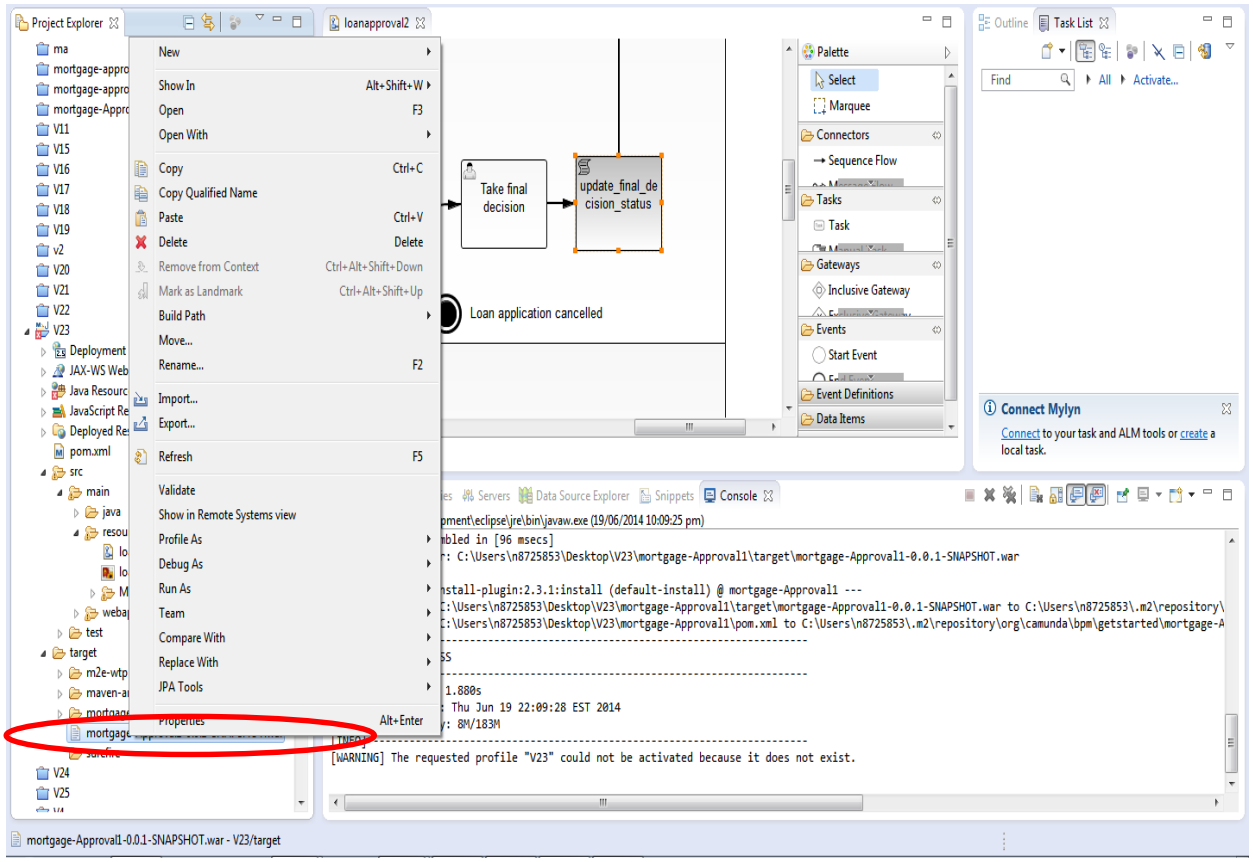
Step1: Right click on pom.xml>>Run as>>Maven Install



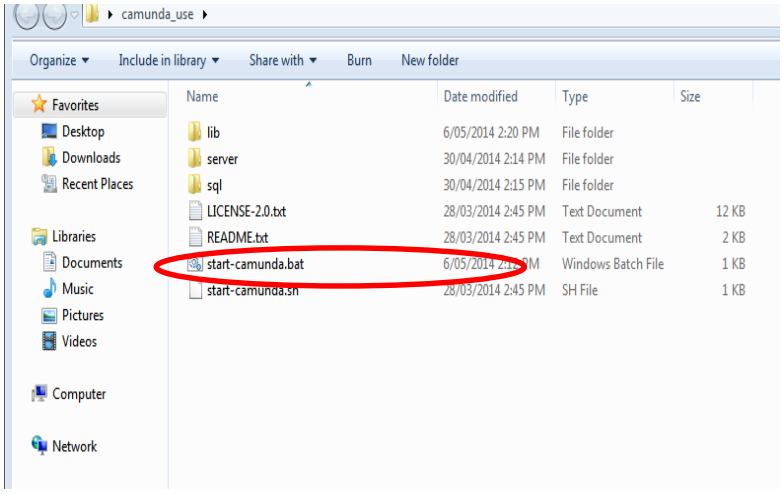
The following results must be displayed after compilation



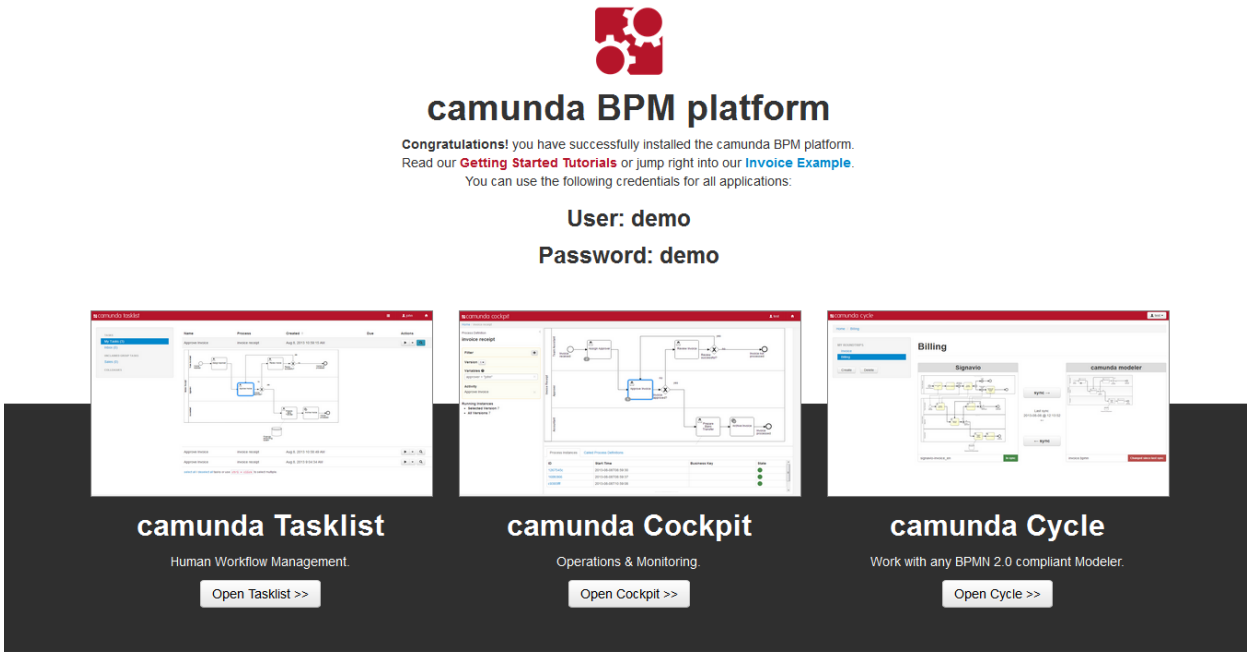
Step2: Navigate to target>>mortgage-Approval-0.0.1-SNAPSHOT.war and copy the .war file



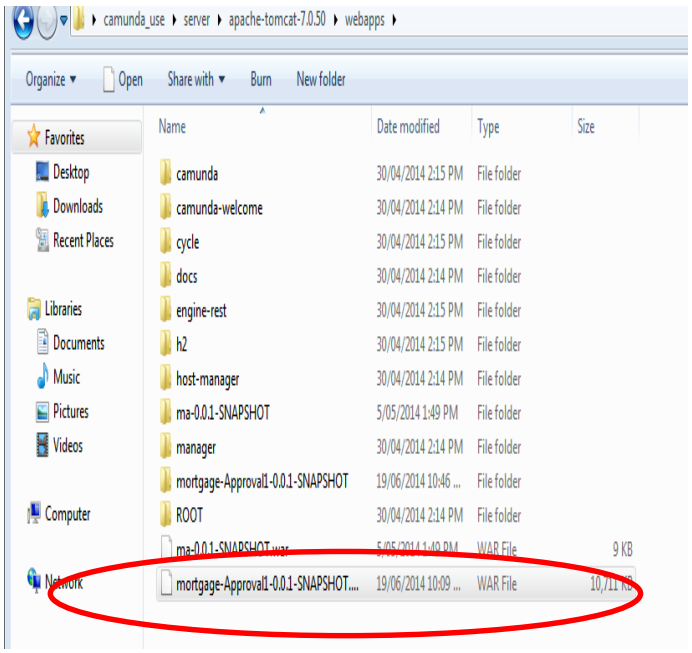
Step3: Launch start-camunda.bat



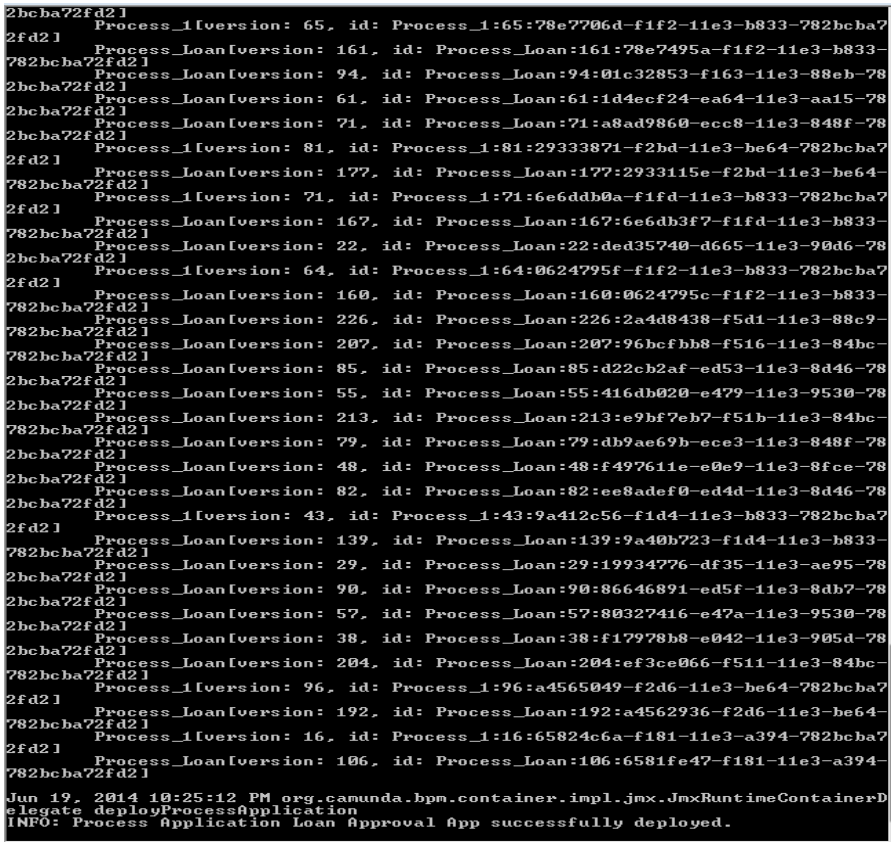
Camunda BPM Platform is displayed



Step4: Navigate to webapps folder in the downloaded Camunda BPM platform (The path used in the tutorial is: camunda_use>>server>>apache-tomcat-7.0.50>>webapps) and paste the .war file



If deployment is successful, the following message is displayed:



5.0 User Designations

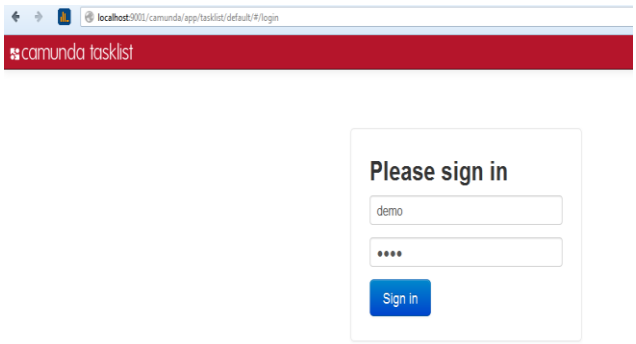
User Name	Authority Level	Specific task in Loan Process
Demo	Loan Applicant	Loan application completed start event (*Reason for separate login for specific task is to overcome constraint 4 mentioned in Appendix)*
Mary	Loan Applicant	All tasks in loan applicant lane
John	Loan Officer	All tasks in loan officer lane
Peter	Finance Officer	All tasks in finance officer lane
Mary	Property Appraiser	All tasks in property appraiser lane
Mary	Insurance Sales Representative	All tasks in insurance sales representative lane
Peter	Second Loan Officer	Take final decision task

6.0 Execution

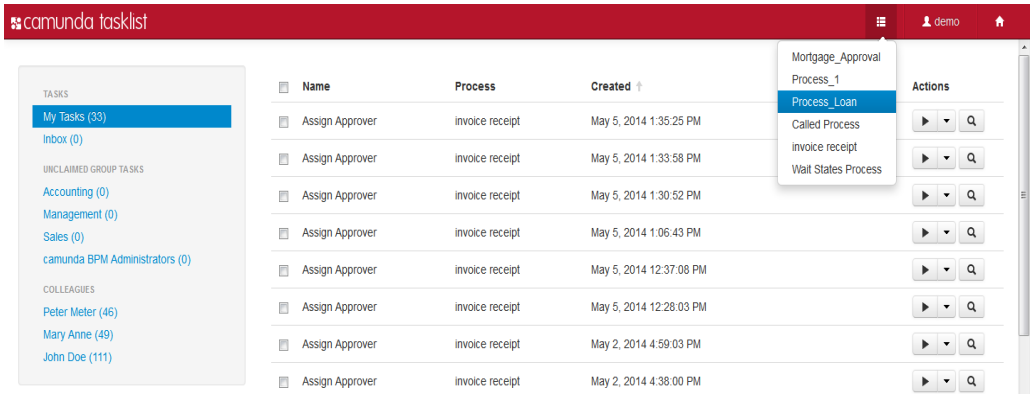
6.1 Scenario 1:

Loan application information is accurate and insurance quote is not required

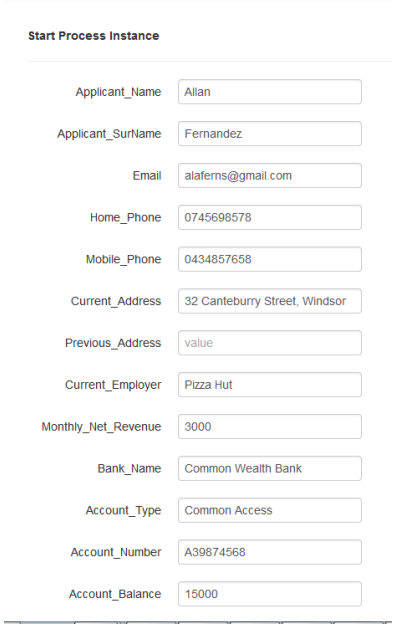
Click on Camunda Tasklist and login with the user credentials demo/demo (for loan applicant)

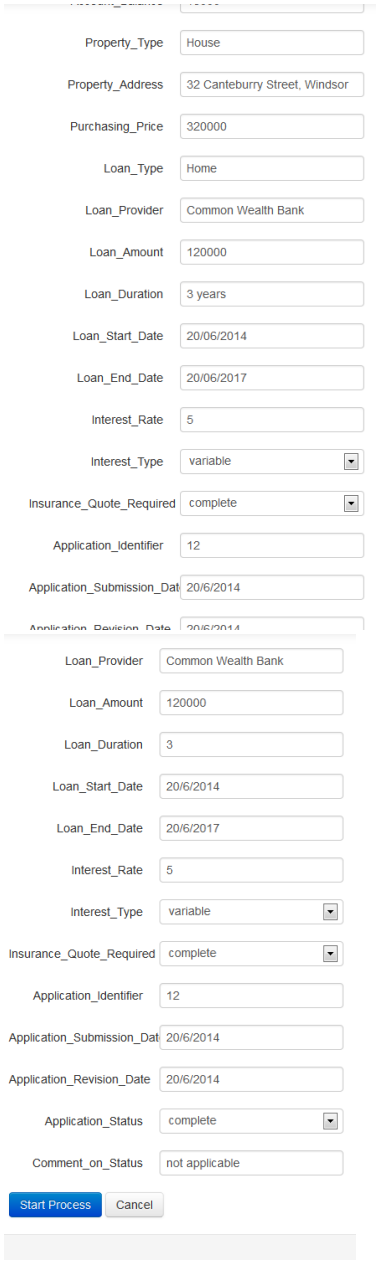


Choose the running instance as 'Process_Loan'



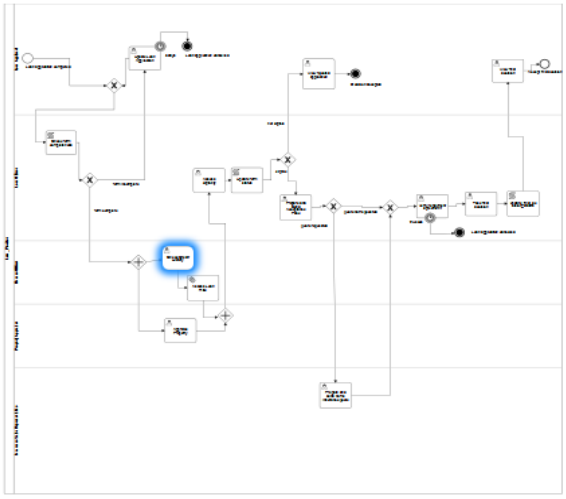
Fill in all the fields of the loan application form except for 'Previous-Address'




Property_Type
House
Property_Address
32 Canteburry Street, Windsor
Purchasing_Price
320000
Loan_Type
Home
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
20/06/2014
Loan_End_Date
20/06/2017
Interest_Rate
5
Interest_Type
variable
Insurance_Quote_Required
complete
Application_Identifier
12
Application_Submission_Dat
20/6/2014
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3
Loan_Start_Date
20/6/2014
Loan_End_Date
20/6/2017
Interest_Rate
5
Interest_Type
variable
Insurance_Quote_Required
complete
Application_Identifier
12
Application_Submission_Dat
20/6/2014
Application_Revision_Date
20/6/2014
Application_Status
complete
Comment_on_Status
not applicable
Start Process
Cancel

Login as peter/peter (Finance Officer) to check Credit History Report

Name	Process	Created
Check Credit History		Jun 19, 2014 11:03:16 PM

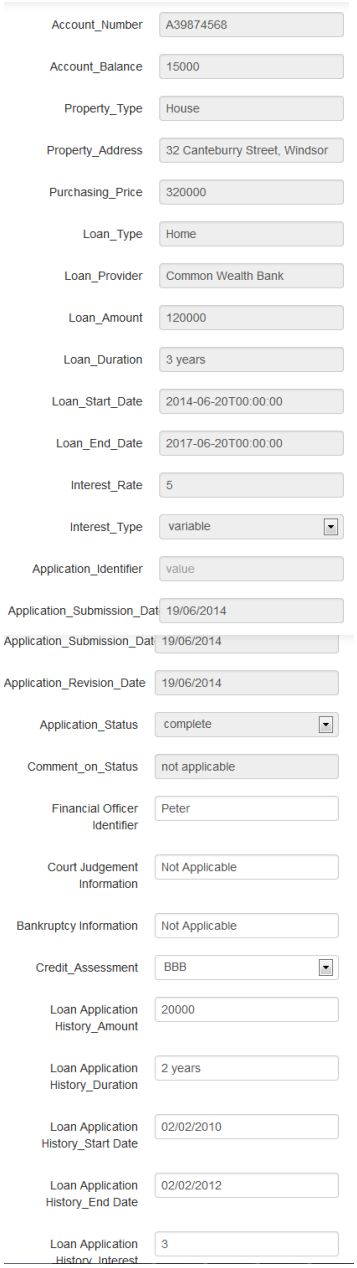


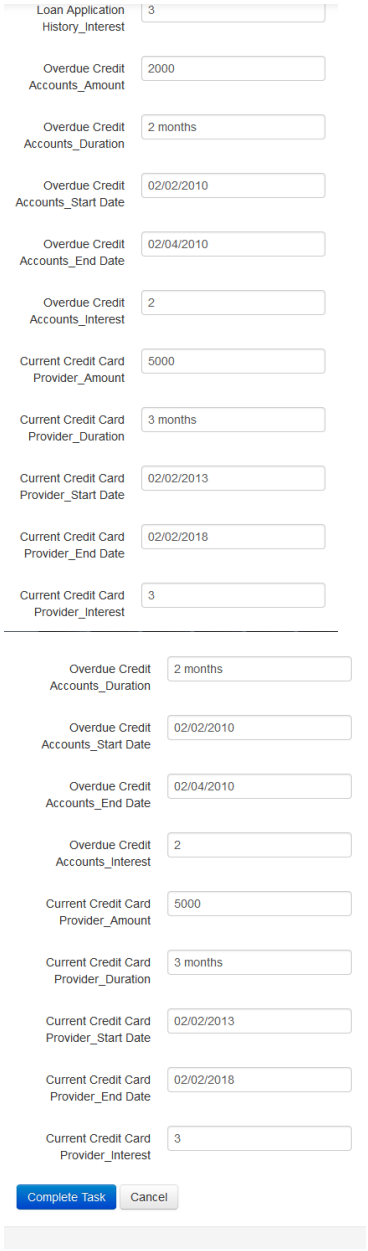
Complete Task

Check Credit History

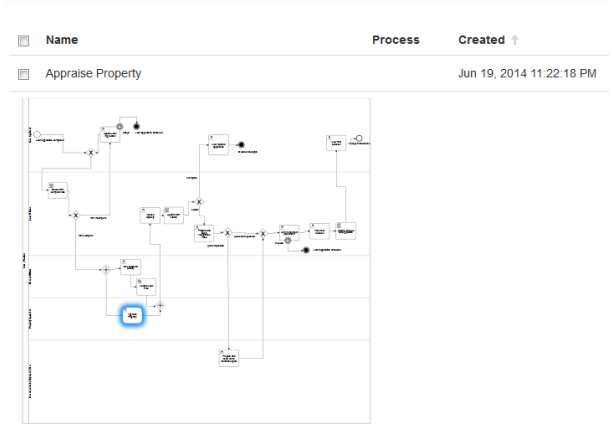
Created: Jun 19, 2014 11:03:16 PM

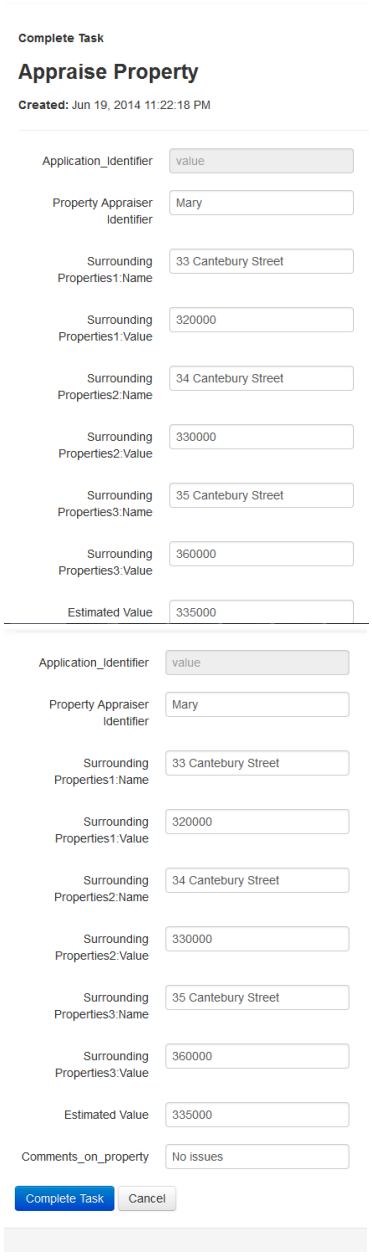
Applicant_Name	Allan
Applicant_SurName	Fernandez
Email	alaferns@gmail.com
Home_Phone	0745698578
Mobile_Phone	0434857658
Current_Address	32 Canteburry Street, Windsor
Previous_Address	32 Canteburry Street, Windsor
Current_Employer	Pizza Hut
Monthly_Net_Revenue	3000
Bank_Name	Common Wealth Bank
Account_Type	Common Access
Account_Number	A39874568


Account_Number
A39874568
Account_Balance
15000
Property_Type
House
Property_Address
32 Canteburry Street, Windsor
Purchasing_Price
320000
Loan_Type
Home
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
2014-06-20T00:00:00
Loan_End_Date
2017-06-20T00:00:00
Interest_Rate
5
Interest_Type
variable
Application_Identifier
value
Application_Submission_Dat
19/06/2014
Application_Submission_Dat
19/06/2014
Application_Revision_Date
19/06/2014
Application_Status
complete
Comment_on_Status
not applicable
Financial Officer Identifier
Peter
Court Judgement Information
Not Applicable
Bankruptcy Information
Not Applicable
Credit_Assessment
BBB
Loan Application History_Amount
20000
Loan Application History_Duration
2 years
Loan Application History_Start Date
02/02/2010
Loan Application History_End Date
02/02/2012
Loan Application
3

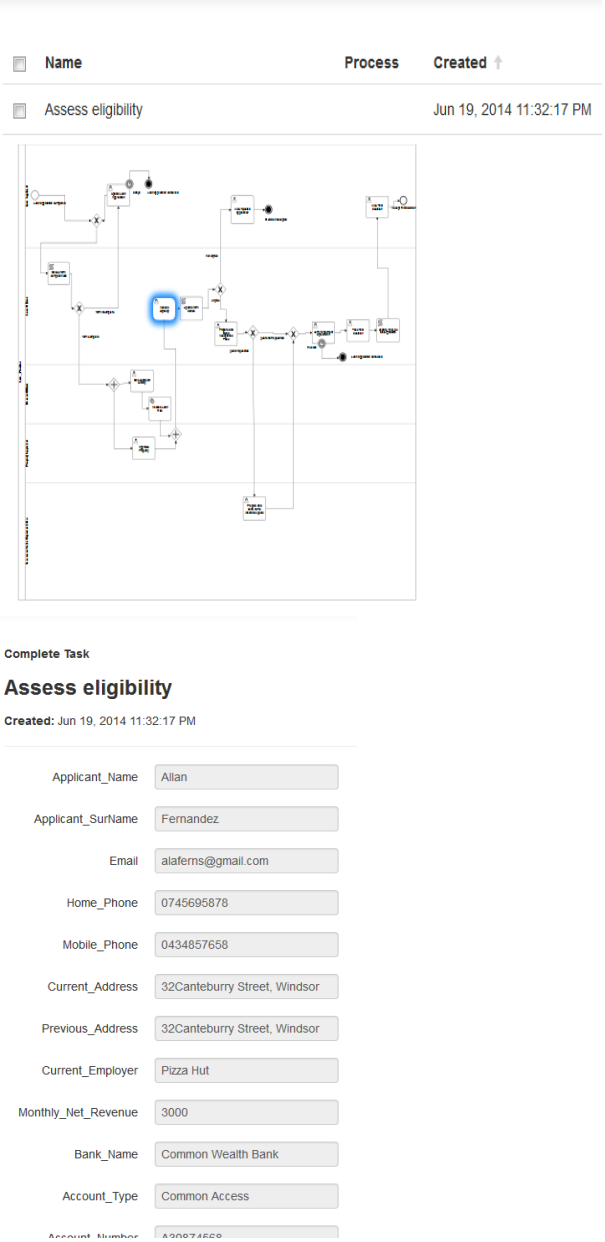


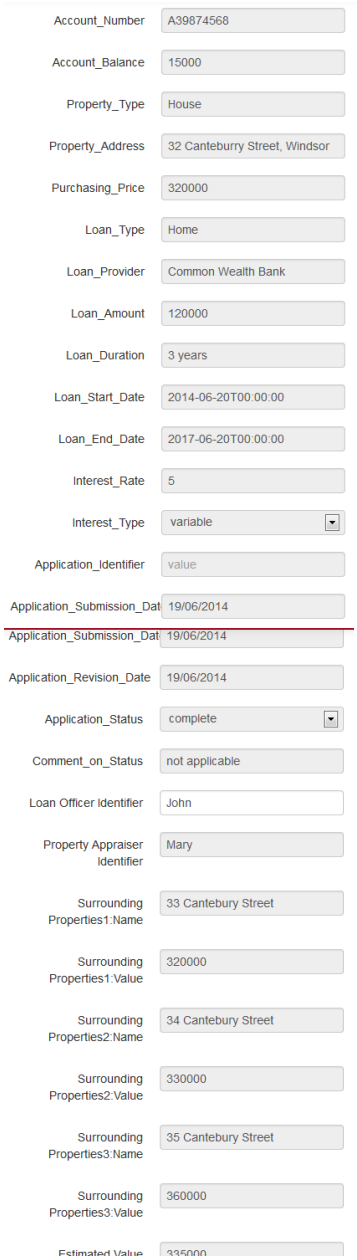
Login as mary/mary (Property Appraiser) to check property appraisal

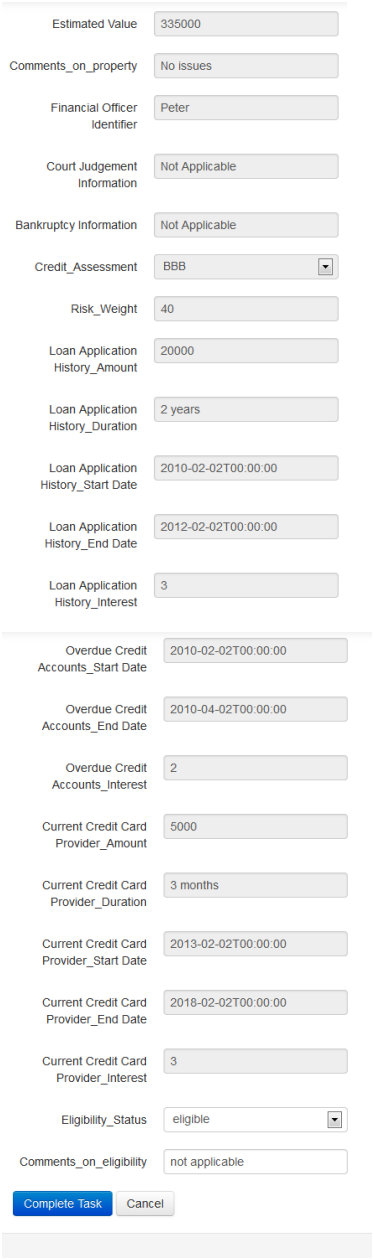



Complete Task
Appraise Property
Created: Jun 19, 2014 11:22:18 PM
Application_Identifier
value
Property Appraiser Identifier
Mary
Surrounding Properties1:Name
33 Cantebury Street
Surrounding Properties1:Value
320000
Surrounding Properties2:Name
34 Cantebury Street
Surrounding Properties2:Value
330000
Surrounding Properties3:Name
35 Cantebury Street
Surrounding Properties3:Value
360000
Estimated Value
335000
Application_Identifier
value
Property Appraiser Identifier
Mary
Surrounding Properties1:Name
33 Cantebury Street
Surrounding Properties1:Value
320000
Surrounding Properties2:Name
34 Cantebury Street
Surrounding Properties2:Value
330000
Surrounding Properties3:Name
35 Cantebury Street
Surrounding Properties3:Value
360000
Estimated Value
335000
Comments_on_property
No issues
Complete Task
Cancel

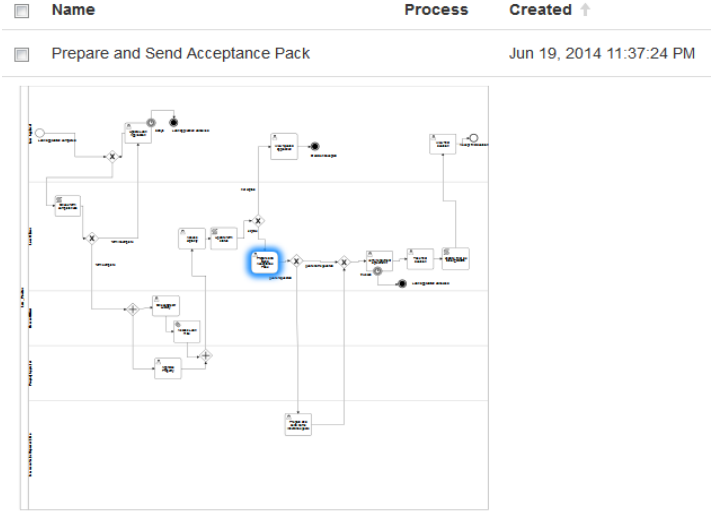
Login as john/john (Loan Officer) to check eligibility




Account_Number
A39874568
Account_Balance
15000
Property_Type
House
Property_Address
32 Canteburry Street, Windsor
Purchasing_Price
320000
Loan_Type
Home
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
2014-06-20T00:00:00
Loan_End_Date
2017-06-20T00:00:00
Interest_Rate
5
Interest_Type
variable
Application_Identifier
value
Application_Submission_Dat
19/06/2014
Application_Submission_Dat
19/06/2014
Application_Revision_Date
19/06/2014
Application_Status
complete
Comment_on_Status
not applicable
Loan Officer Identifier
John
Property Appraiser Identifier
Mary
Surrounding Properties1:Name
33 Cantebury Street
Surrounding Properties1:Value
320000
Surrounding Properties2:Name
34 Cantebury Street
Surrounding Properties2:Value
330000
Surrounding Properties3:Name
35 Cantebury Street
Surrounding Properties3:Value
360000
Estimated Value
335000



Login as john/john (Loan Officer) to prepare and send acceptance pack

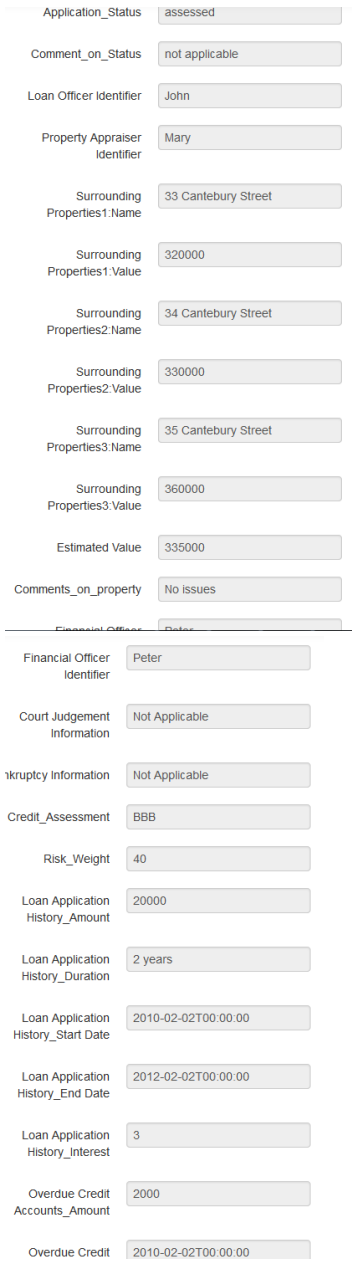


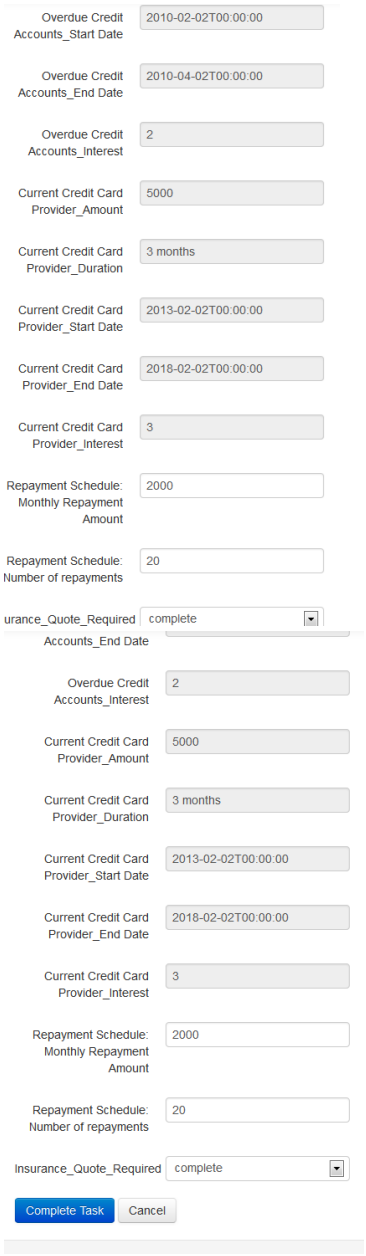
Complete Task

Prepare and Send Acceptance Pack

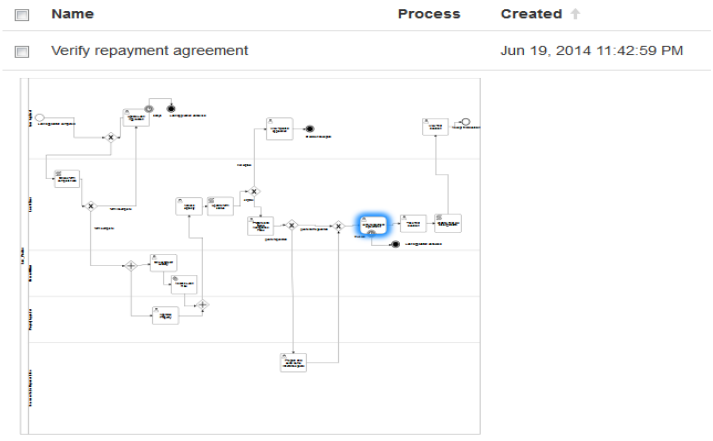
Created: Jun 19, 2014 11:37:24 PM

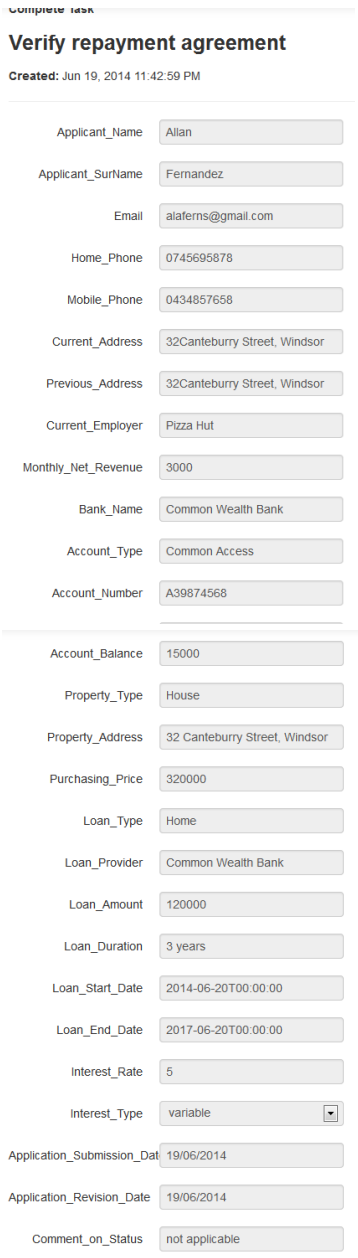
Applicant_Name	Allan
Applicant_SurName	Fernandez
Email	alaferns@gmail.com
Home_Phone	0745695878
Mobile_Phone	0434857658
Current_Address	32Canteburry Street, Windsor
Previous_Address	32Canteburry Street, Windsor
Current_Employer	Pizza Hut
Monthly_Net_Revenue	3000
Bank_Name	Common Wealth Bank
Account_Type	Common Access
Account_Number	A39874568
Account_Balance	15000
Property_Type	House
Property_Address	32 Canteburry Street, Windsor
Purchasing_Price	320000
Loan_Type	Home
Loan_Provider	Common Wealth Bank
Loan_Amount	120000
Loan_Duration	3 years
Loan_Start_Date	2014-06-20T00:00:00
Loan_End_Date	2017-06-20T00:00:00
Interest_Rate	5
Interest_Type	variable
Application_Identifier	value
Application_Submission_Dat	19/06/2014
Application_Revision_Date	19/06/2014

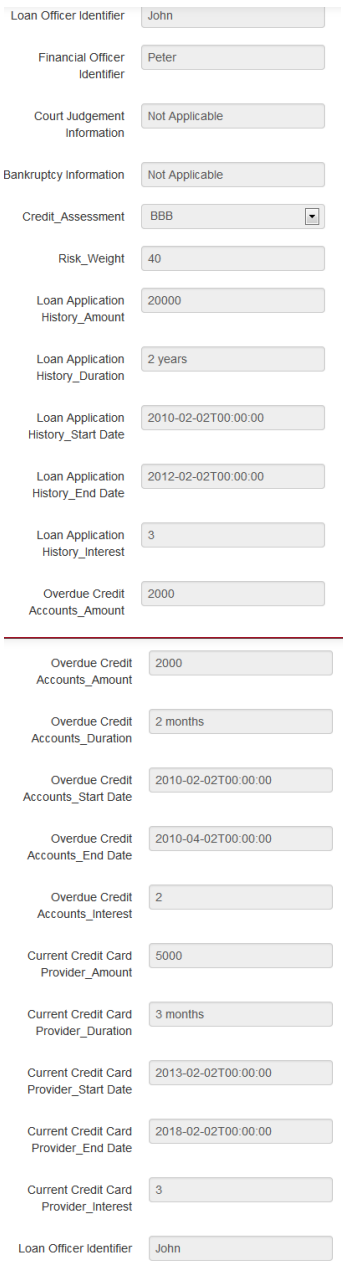

Application_Status
assessed
Comment_on_Status
not applicable
Loan Officer Identifier
John
Property Appraiser Identifier
Mary
Surrounding Properties1:Name
33 Cantebury Street
Surrounding Properties1:Value
320000
Surrounding Properties2:Name
34 Cantebury Street
Surrounding Properties2:Value
330000
Surrounding Properties3:Name
35 Cantebury Street
Surrounding Properties3:Value
360000
Estimated Value
335000
Comments_on_property
No issues
Financial Officer Identifier
Peter
Court Judgement Information
Not Applicable
ıkruptcy Information
Not Applicable
Credit_Assessment
BBB
Risk_Weight
40
Loan Application History_Amount
20000
Loan Application History_Duration
2 years
Loan Application History_Start Date
2010-02-02T00:00:00
Loan Application History_End Date
2012-02-02T00:00:00
Loan Application History_Interest
3
Overdue Credit Accounts_Amount
2000
Overdue Credit
2010-02-02T00:00:00

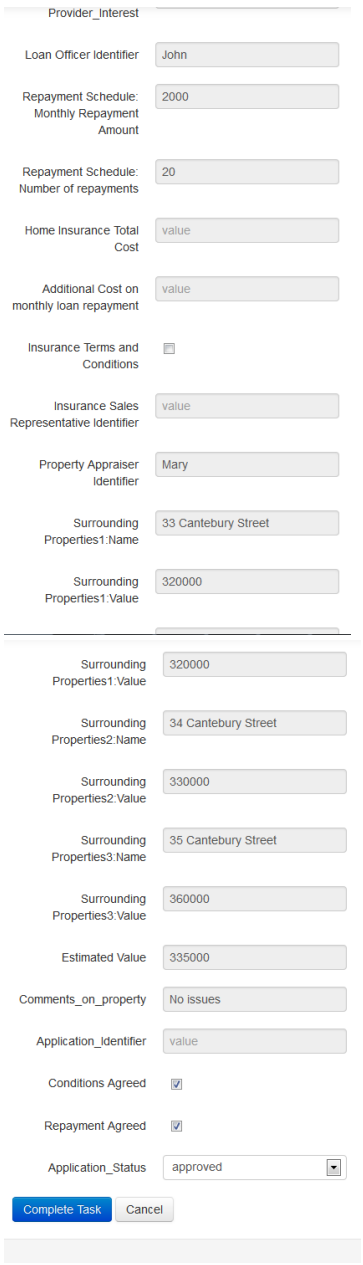


Login as john/john (Loan Officer) to verify repayment agreement

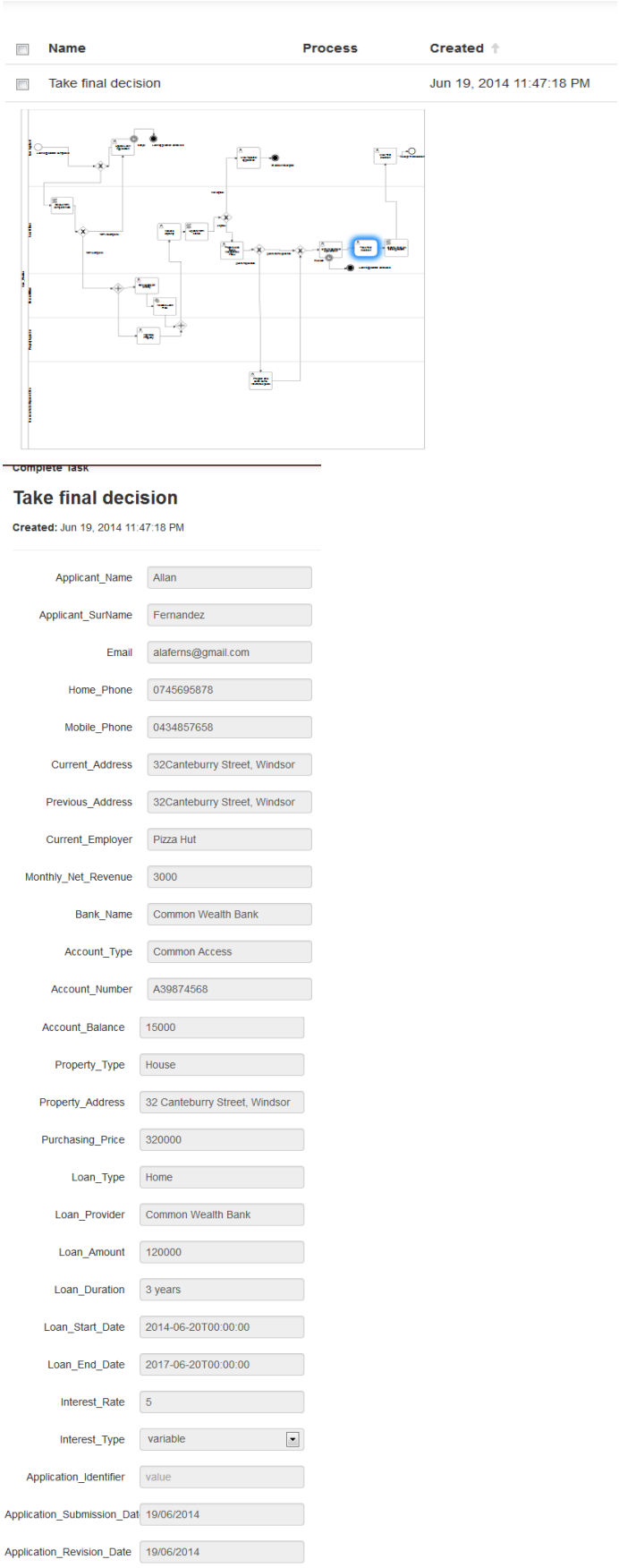


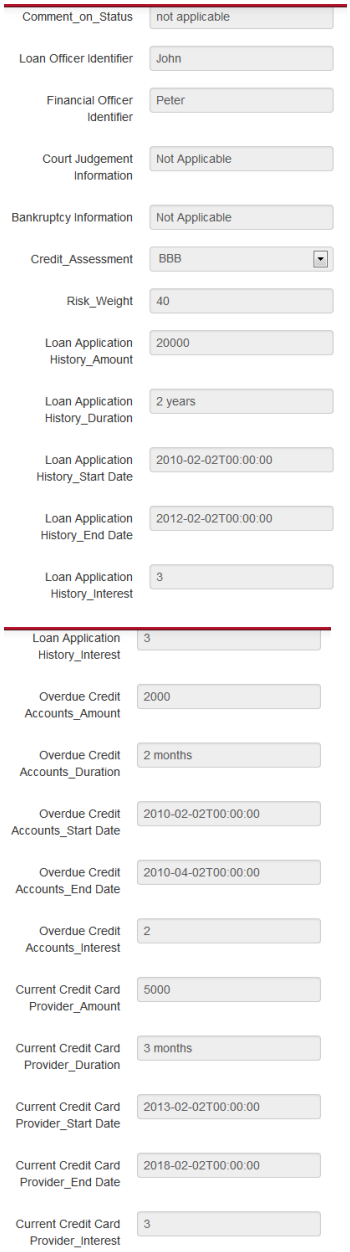

Complete Task
Verify repayment agreement
Created: Jun 19, 2014 11:42:59 PM
Applicant_Name
Allan
Applicant_SurName
Fernandez
Email
alaferns@gmail.com
Home_Phone
0745695878
Mobile_Phone
0434857658
Current_Address
32Canteburry Street, Windsor
Previous_Address
32Canteburry Street, Windsor
Current_Employer
Pizza Hut
Monthly_Net_Revenue
3000
Bank_Name
Common Wealth Bank
Account_Type
Common Access
Account_Number
A39874568
Account_Balance
15000
Property_Type
House
Property_Address
32 Canteburry Street, Windsor
Purchasing_Price
320000
Loan_Type
Home
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
2014-06-20T00:00:00
Loan_End_Date
2017-06-20T00:00:00
Interest_Rate
5
Interest_Type
variable
Application_Submission_Dat
19/06/2014
Application_Revision_Date
19/06/2014
Comment_on_Status
not applicable

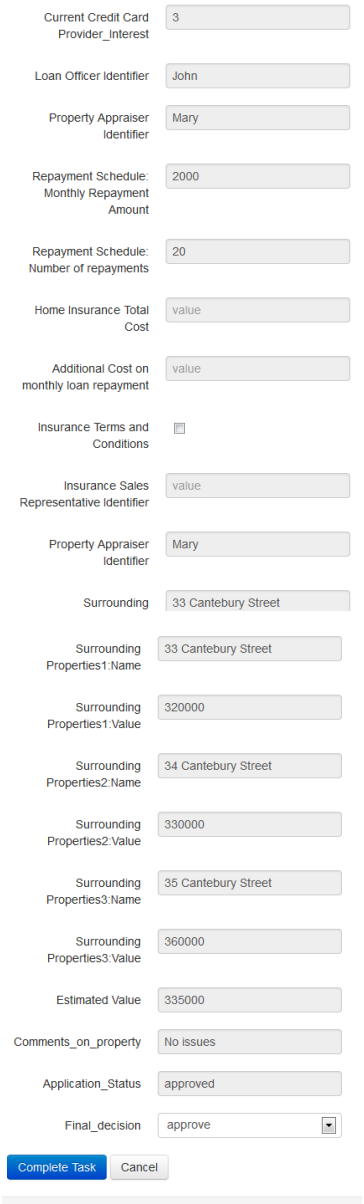

Loan Officer Identifier
John
Financial Officer Identifier
Peter
Court Judgement Information
Not Applicable
Bankruptcy Information
Not Applicable
Credit_Assessment
BBB
Risk_Weight
40
Loan Application History_Amount
20000
Loan Application History_Duration
2 years
Loan Application History_Start Date
2010-02-02T00:00:00
Loan Application History_End Date
2012-02-02T00:00:00
Loan Application History_Interest
3
Overdue Credit Accounts_Amount
2000
Overdue Credit Accounts_Amount
2000
Overdue Credit Accounts_Duration
2 months
Overdue Credit Accounts_Start Date
2010-02-02T00:00:00
Overdue Credit Accounts_End Date
2010-04-02T00:00:00
Overdue Credit Accounts_Interest
2
Current Credit Card Provider_Amount
5000
Current Credit Card Provider_Duration
3 months
Current Credit Card Provider_Start Date
2013-02-02T00:00:00
Current Credit Card Provider_End Date
2018-02-02T00:00:00
Current Credit Card Provider_Interest
3
Loan Officer Identifier
John


Provider_Interest
Loan Officer Identifier
John
Repayment Schedule: Monthly Repayment Amount
2000
Repayment Schedule: Number of repayments
20
Home Insurance Total Cost
value
Additional Cost on monthly loan repayment
value
Insurance Terms and Conditions
Insurance Sales Representative Identifier
value
Property Appraiser Identifier
Mary
Surrounding Properties1:Name
33 Cantebury Street
Surrounding Properties1:Value
320000
Surrounding Properties1:Value
320000
Surrounding Properties2:Name
34 Cantebury Street
Surrounding Properties2:Value
330000
Surrounding Properties3:Name
35 Cantebury Street
Surrounding Properties3:Value
360000
Estimated Value
335000
Comments_on_property
No issues
Application_Identifier
value
Conditions Agreed
Repayment Agreed
Application_Status
approved
Complete Task
Cancel

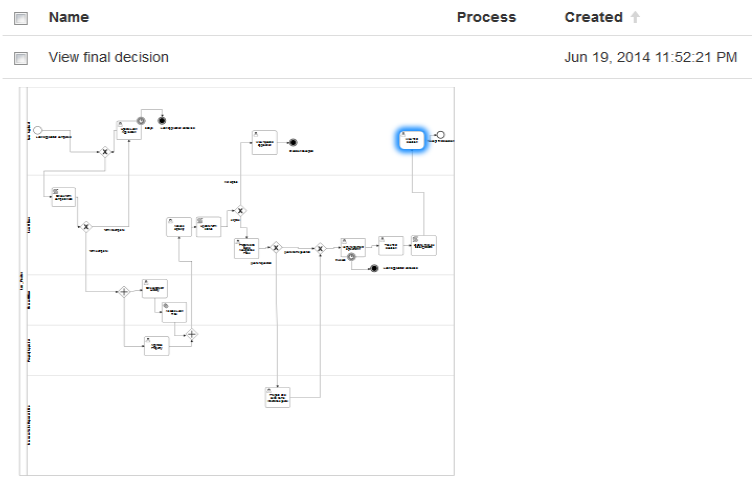
Login as peter/peter (Loan Officer) to take final decision

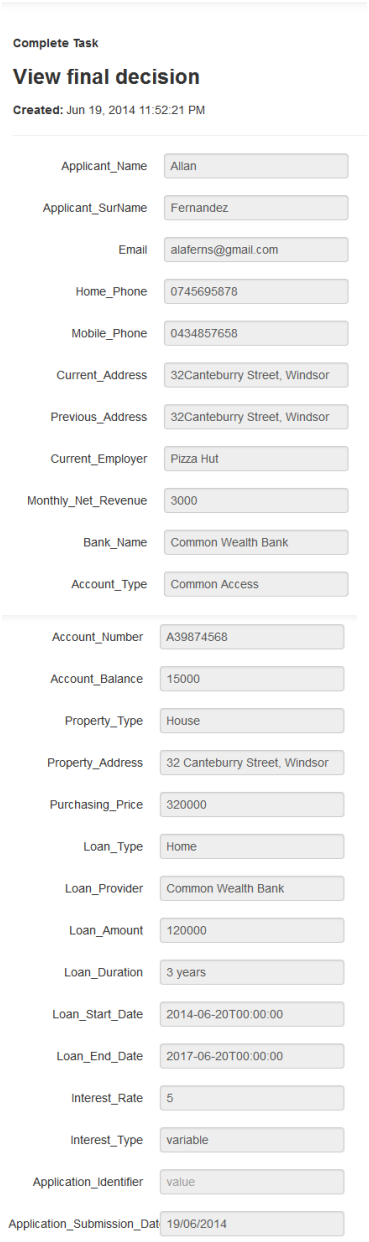


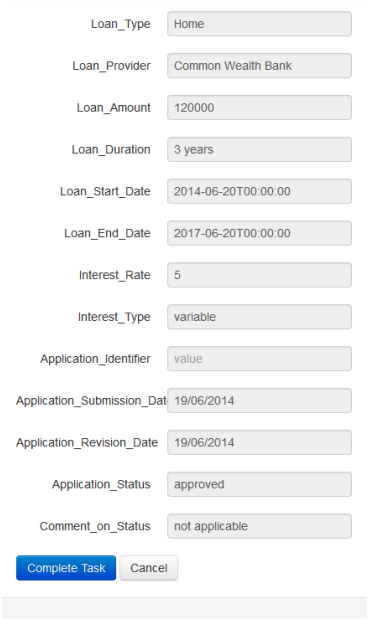

Comment_on_Status
not applicable
Loan Officer Identifier
John
Financial Officer Identifier
Peter
Court Judgement Information
Not Applicable
Bankruptcy Information
Not Applicable
Credit_Assessment
BBB
Risk_Weight
40
Loan Application History_Amount
20000
Loan Application History_Duration
2 years
Loan Application History_Start Date
2010-02-02T00:00:00
Loan Application History_End Date
2012-02-02T00:00:00
Loan Application History_Interest
3
Loan Application History_Interest
3
Overdue Credit Accounts_Amount
2000
Overdue Credit Accounts_Duration
2 months
Overdue Credit Accounts_Start Date
2010-02-02T00:00:00
Overdue Credit Accounts_End Date
2010-04-02T00:00:00
Overdue Credit Accounts_Interest
2
Current Credit Card Provider_Amount
5000
Current Credit Card Provider_Duration
3 months
Current Credit Card Provider_Start Date
2013-02-02T00:00:00
Current Credit Card Provider_End Date
2018-02-02T00:00:00
Current Credit Card Provider_Interest
3



Login as mary/mary (Loan Applicant) to view the final decision



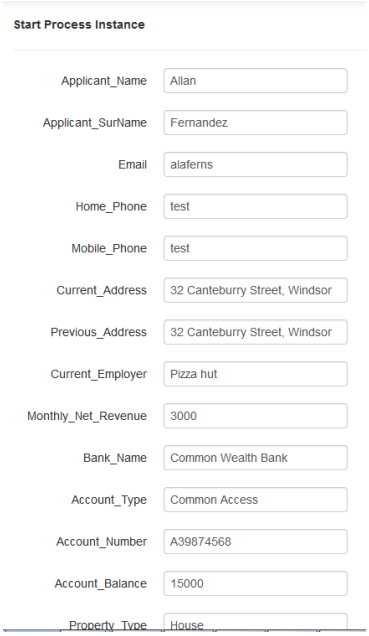

Complete Task
View final decision
Created: Jun 19, 2014 11:52:21 PM
Applicant_Name
Allan
Applicant_SurName
Fernandez
Email
alaferns@gmail.com
Home_Phone
0745695878
Mobile_Phone
0434857658
Current_Address
32Canteburry Street, Windsor
Previous_Address
32Canteburry Street, Windsor
Current_Employer
Pizza Hut
Monthly_Net_Revenue
3000
Bank_Name
Common Wealth Bank
Account_Type
Common Access
Account_Number
A39874568
Account_Balance
15000
Property_Type
House
Property_Address
32 Canteburry Street, Windsor
Purchasing_Price
320000
Loan_Type
Home
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
2014-06-20T00:00:00
Loan_End_Date
2017-06-20T00:00:00
Interest_Rate
5
Interest_Type
variable
Application_Identifier
value
Application_Submission_Dat
19/06/2014

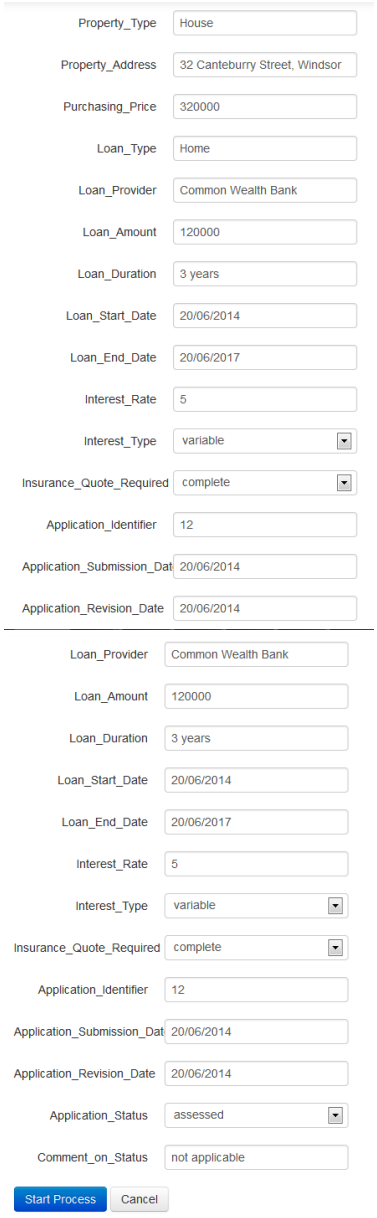


6.2 Scenario 2:

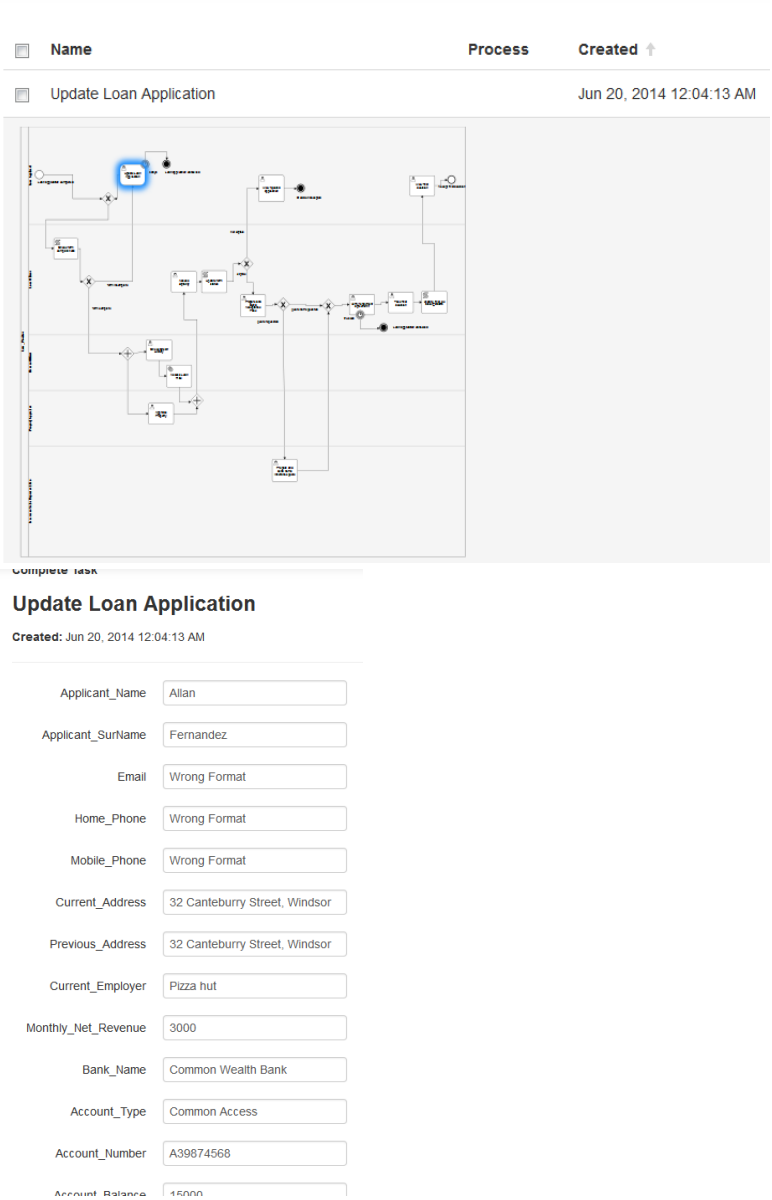
Loan application information is inaccurate and update is required

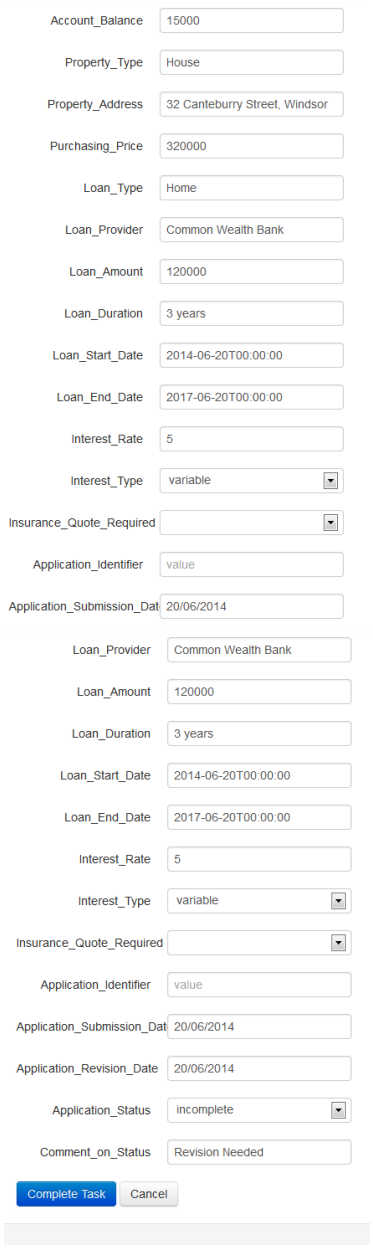
Login with the user credentials demo/demo (for loan applicant)




Property_Type
House
Property_Address
32 Canteburry Street, Windsor
Purchasing_Price
320000
Loan_Type
Home
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
20/06/2014
Loan_End_Date
20/06/2017
Interest_Rate
5
Interest_Type
variable
Insurance_Quote_Required
complete
Application_Identifier
12
Application_Submission_Dat
20/06/2014
Application_Revision_Date
20/06/2014
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
20/06/2014
Loan_End_Date
20/06/2017
Interest_Rate
5
Interest_Type
variable
Insurance_Quote_Required
complete
Application_Identifier
12
Application_Submission_Dat
20/06/2014
Application_Revision_Date
20/06/2014
Application_Status
assessed
Comment_on_Status
not applicable
Start Process
Cancel

Login as mary/mary (Loan Applicant) to update the loan application

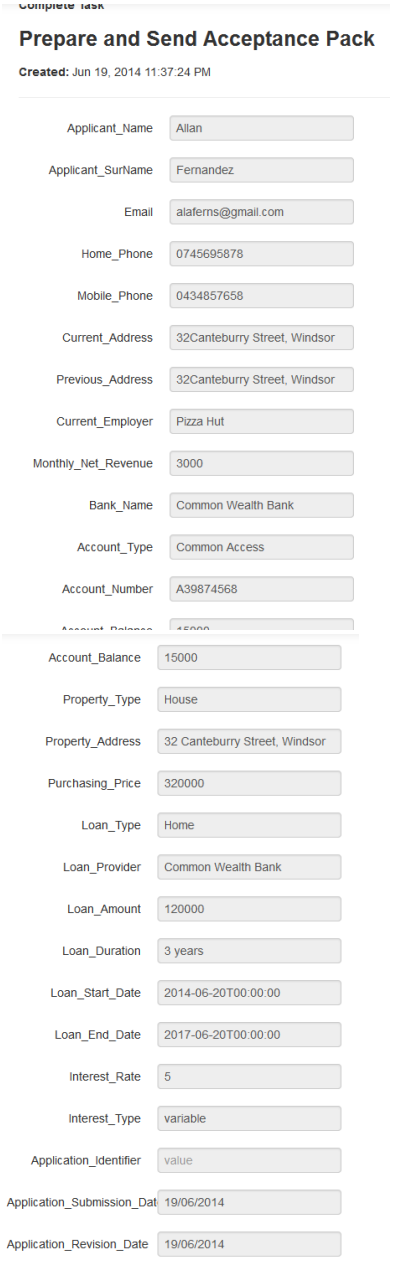


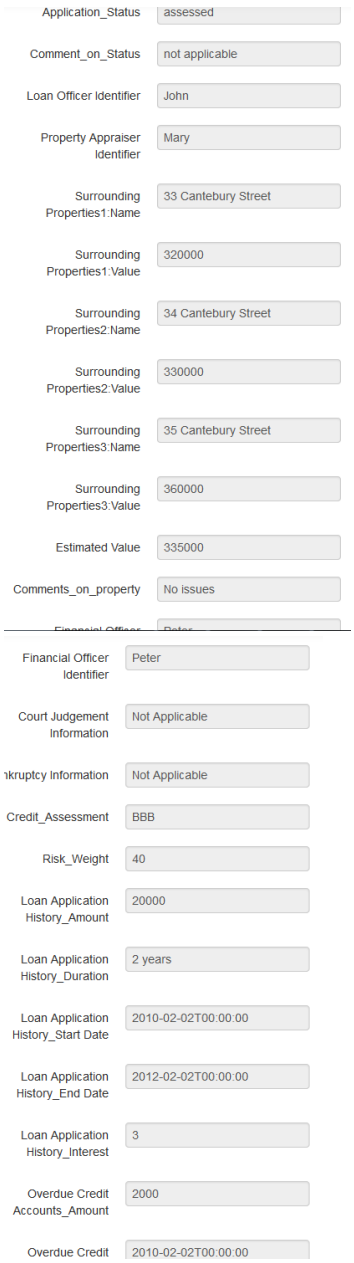

Account_Balance
15000
Property_Type
House
Property_Address
32 Canteburry Street, Windsor
Purchasing_Price
320000
Loan_Type
Home
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
2014-06-20T00:00:00
Loan_End_Date
2017-06-20T00:00:00
Interest_Rate
5
Interest_Type
variable
Insurance_Quote_Required
Application_Identifier
value
Application_Submission_Dat
20/06/2014
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
2014-06-20T00:00:00
Loan_End_Date
2017-06-20T00:00:00
Interest_Rate
5
Interest_Type
variable
Insurance_Quote_Required
Application_Identifier
value
Application_Submission_Dat
20/06/2014
Application_Revision_Date
20/06/2014
Application_Status
incomplete
Comment_on_Status
Revision Needed
Complete Task
Cancel

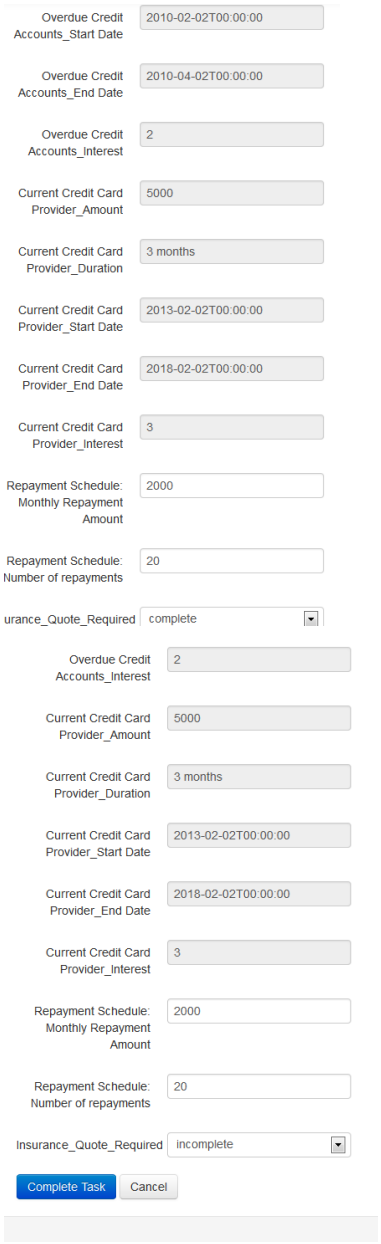
6.3 Scenario 3:

Loan application information is accurate and insurance quote is required

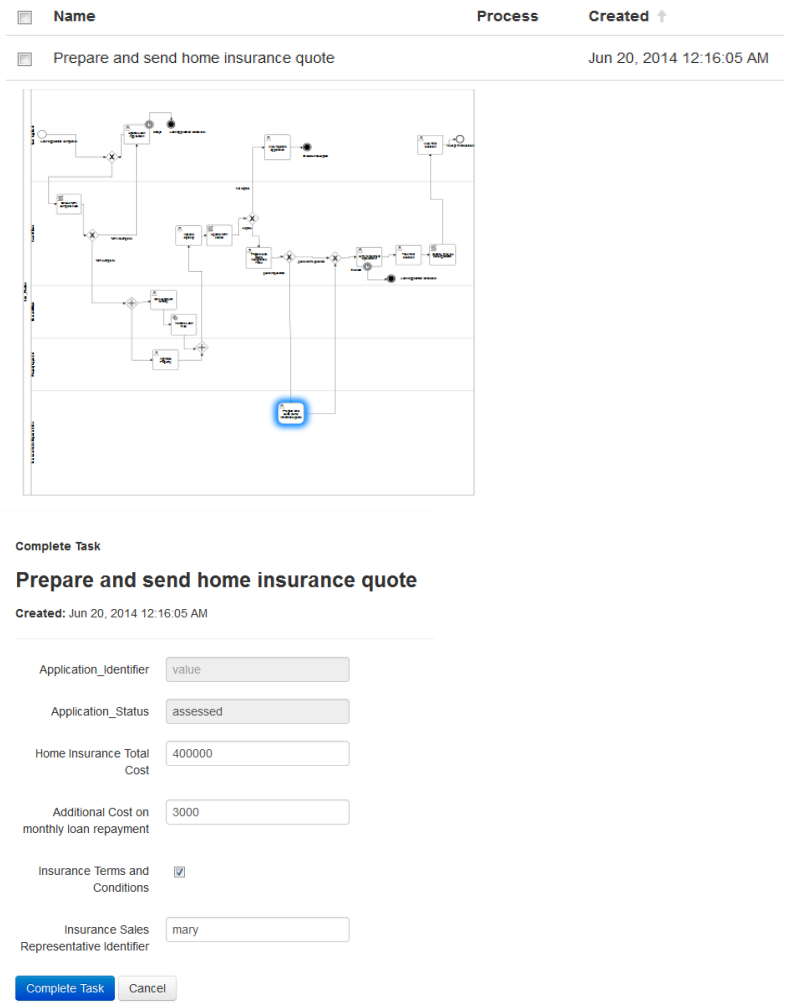
Login as john/john (Loan Officer) to prepare and send acceptance pack



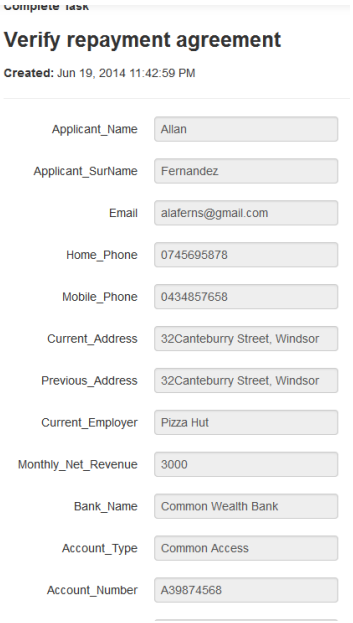

Application_Status
assessed
Comment_on_Status
not applicable
Loan Officer Identifier
John
Property Appraiser Identifier
Mary
Surrounding Properties1:Name
33 Cantebury Street
Surrounding Properties1:Value
320000
Surrounding Properties2:Name
34 Cantebury Street
Surrounding Properties2:Value
330000
Surrounding Properties3:Name
35 Cantebury Street
Surrounding Properties3:Value
360000
Estimated Value
335000
Comments_on_property
No issues
Financial Officer Identifier
Peter
Court Judgement Information
Not Applicable
ıkruptcy Information
Not Applicable
Credit_Assessment
BBB
Risk_Weight
40
Loan Application History_Amount
20000
Loan Application History_Duration
2 years
Loan Application History_Start Date
2010-02-02T00:00:00
Loan Application History_End Date
2012-02-02T00:00:00
Loan Application History_Interest
3
Overdue Credit Accounts_Amount
2000
Overdue Credit
2010-02-02T00:00:00

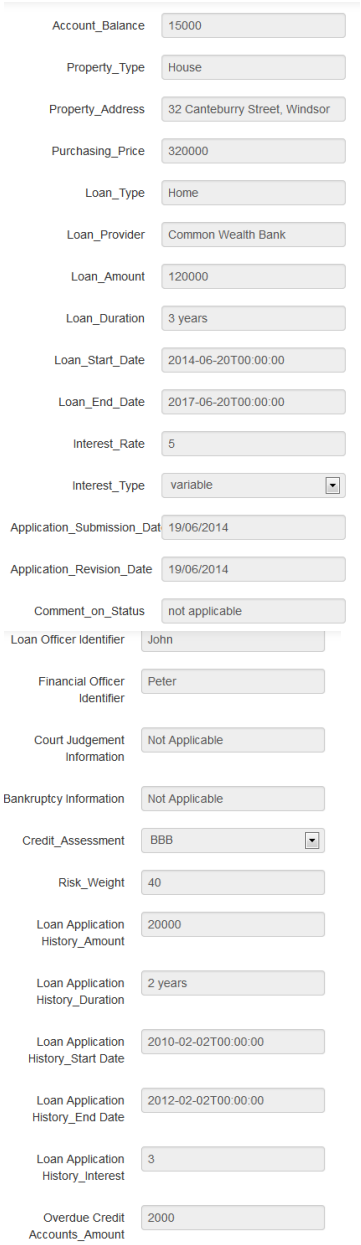

Overdue Credit Accounts_Start Date
2010-02-02T00:00:00
Overdue Credit Accounts_End Date
2010-04-02T00:00:00
Overdue Credit Accounts_Interest
2
Current Credit Card Provider_Amount
5000
Current Credit Card Provider_Duration
3 months
Current Credit Card Provider_Start Date
2013-02-02T00:00:00
Current Credit Card Provider_End Date
2018-02-02T00:00:00
Current Credit Card Provider_Interest
3
Repayment Schedule: Monthly Repayment Amount
2000
Repayment Schedule: Number of repayments
20
urance_Quote_Required
complete
Overdue Credit Accounts_Interest
2
Current Credit Card Provider_Amount
5000
Current Credit Card Provider_Duration
3 months
Current Credit Card Provider_Start Date
2013-02-02T00:00:00
Current Credit Card Provider_End Date
2018-02-02T00:00:00
Current Credit Card Provider_Interest
3
Repayment Schedule: Monthly Repayment Amount
2000
Repayment Schedule: Number of repayments
20
Insurance_Quote_Required
incomplete
Complete Task
Cancel

Login as mary/mary (Insurance Sales Representative) to prepare and send home insurance quote

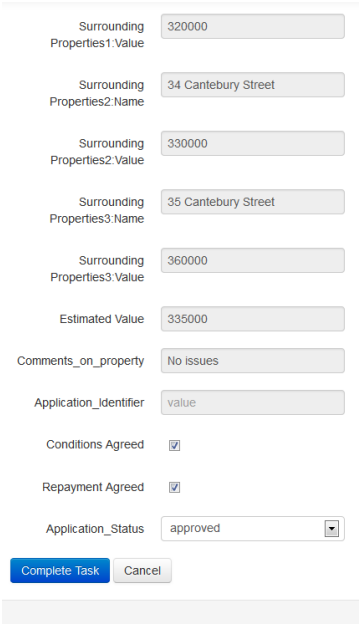


Login as john/john (Loan Officer) to verify repayment agreement




Account_Balance
15000
Property_Type
House
Property_Address
32 Canteburry Street, Windsor
Purchasing_Price
320000
Loan_Type
Home
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
2014-06-20T00:00:00
Loan_End_Date
2017-06-20T00:00:00
Interest_Rate
5
Interest_Type
variable
Application_Submission_Dat
19/06/2014
Application_Revision_Date
19/06/2014
Comment_on_Status
not applicable
Loan Officer Identifier
John
Financial Officer Identifier
Peter
Court Judgement Information
Not Applicable
Bankruptcy Information
Not Applicable
Credit_Assessment
BBB
Risk_Weight
40
Loan Application History_Amount
20000
Loan Application History_Duration
2 years
Loan Application History_Start Date
2010-02-02T00:00:00
Loan Application History_End Date
2012-02-02T00:00:00
Loan Application History_Interest
3
Overdue Credit Accounts_Amount
2000

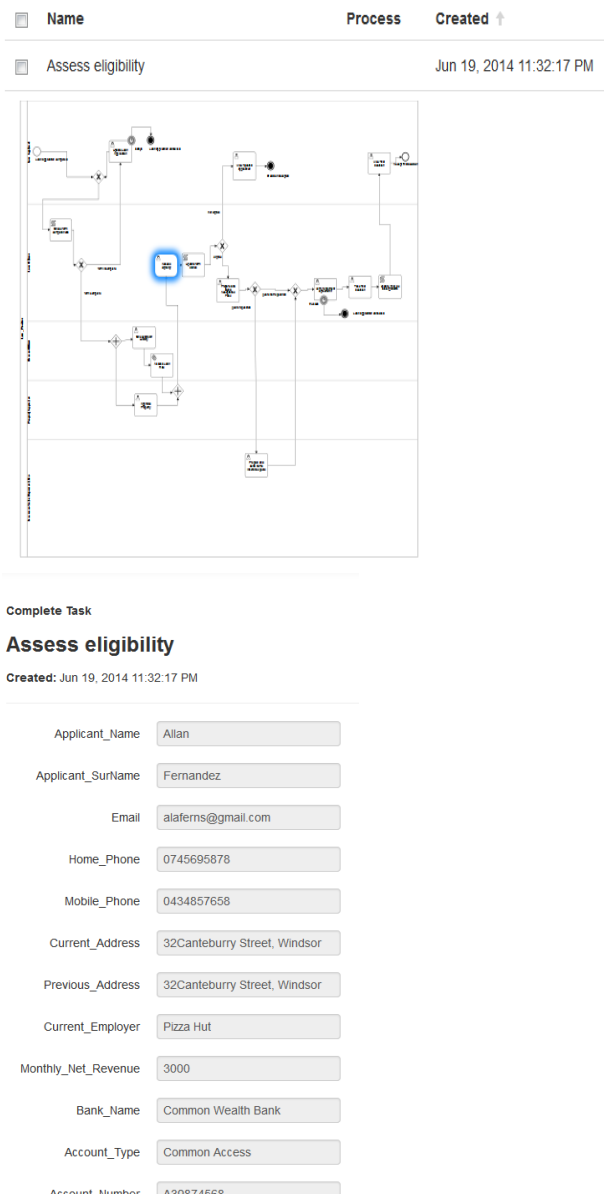
Overdue Credit Accounts_Amount	2000
Overdue Credit Accounts_Duration	2 months
Overdue Credit Accounts_Start Date	2010-02-02T00:00:00
Overdue Credit Accounts_End Date	2010-04-02T00:00:00
Overdue Credit Accounts_Interest	2
Current Credit Card Provider_Amount	5000
Current Credit Card Provider_Duration	3 months
Current Credit Card Provider_Start Date	2013-02-02T00:00:00
Current Credit Card Provider_End Date	2018-02-02T00:00:00
Current Credit Card Provider_Interest	3
Loan Officer Identifier	John
Loan Officer Identifier	John
Repayment Schedule: Monthly Repayment Amount	2000
Repayment Schedule: Number of repayments	20
Home Insurance Total Cost	400000
Additional Cost on monthly loan repayment	3000
Insurance Terms and Conditions	☑
Insurance Sales Representative Identifier	mary
Property Appraiser Identifier	Mary
Surrounding Properties1:Name	33 Cantebury Street
Surrounding Properties1:Value	320000
Surrounding Properties2:Name	34 Cantebury Street

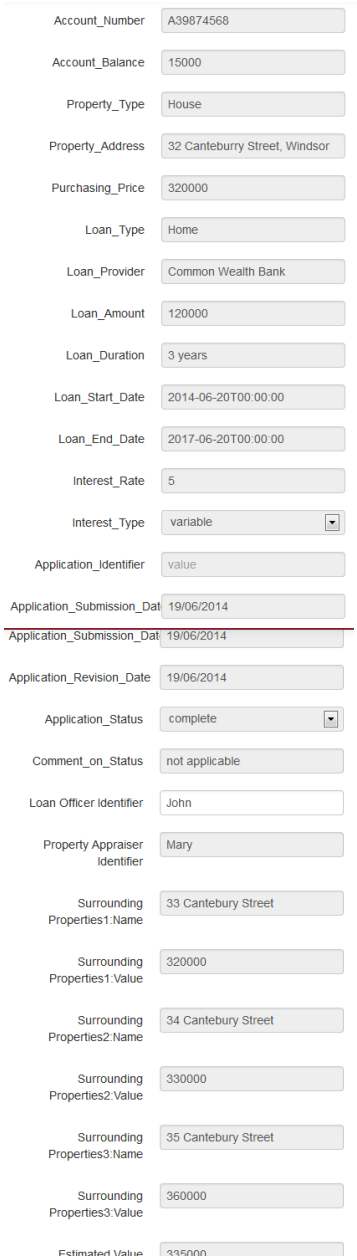

Surrounding Properties1:Value
320000
Surrounding Properties2:Name
34 Cantebury Street
Surrounding Properties2:Value
330000
Surrounding Properties3:Name
35 Cantebury Street
Surrounding Properties3:Value
360000
Estimated Value
335000
Comments_on_property
No issues
Application_Identifier
value
Conditions Agreed
Repayment Agreed
Application_Status
approved
Complete Task
Cancel

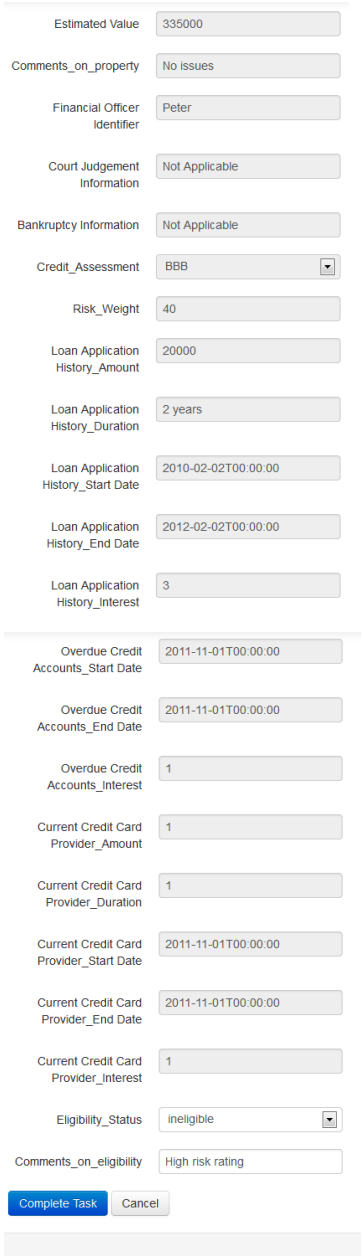
6.4 Scenario 4:

Loan application is rejected during eligibility check

Login as john/john (Loan Officer) to check eligibility

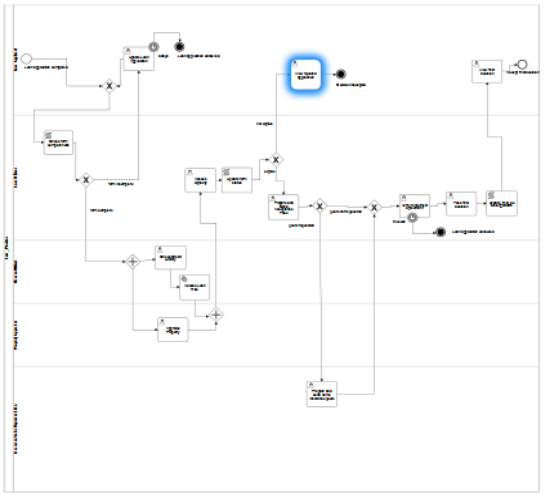



Account_Number
A39874568
Account_Balance
15000
Property_Type
House
Property_Address
32 Canteburry Street, Windsor
Purchasing_Price
320000
Loan_Type
Home
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
2014-06-20T00:00:00
Loan_End_Date
2017-06-20T00:00:00
Interest_Rate
5
Interest_Type
variable
Application_Identifier
value
Application_Submission_Dat
19/06/2014
Application_Submission_Dat
19/06/2014
Application_Revision_Date
19/06/2014
Application_Status
complete
Comment_on_Status
not applicable
Loan Officer Identifier
John
Property Appraiser Identifier
Mary
Surrounding Properties1:Name
33 Cantebury Street
Surrounding Properties1:Value
320000
Surrounding Properties2:Name
34 Cantebury Street
Surrounding Properties2:Value
330000
Surrounding Properties3:Name
35 Cantebury Street
Surrounding Properties3:Value
360000
Estimated Value
335000


Estimated Value
335000
Comments_on_property
No issues
Financial Officer Identifier
Peter
Court Judgement Information
Not Applicable
Bankruptcy Information
Not Applicable
Credit_Assessment
BBB
Risk_Weight
40
Loan Application History_Amount
20000
Loan Application History_Duration
2 years
Loan Application History_Start Date
2010-02-02T00:00:00
Loan Application History_End Date
2012-02-02T00:00:00
Loan Application History_Interest
3
Overdue Credit Accounts_Start Date
2011-11-01T00:00:00
Overdue Credit Accounts_End Date
2011-11-01T00:00:00
Overdue Credit Accounts_Interest
1
Current Credit Card Provider_Amount
1
Current Credit Card Provider_Duration
1
Current Credit Card Provider_Start Date
2011-11-01T00:00:00
Current Credit Card Provider_End Date
2011-11-01T00:00:00
Current Credit Card Provider_Interest
1
Eligibility_Status
ineligible
Comments_on_eligibility
High risk rating
Complete Task
Cancel

Login as mary/mary (Loan Applicant) to view rejected loan application

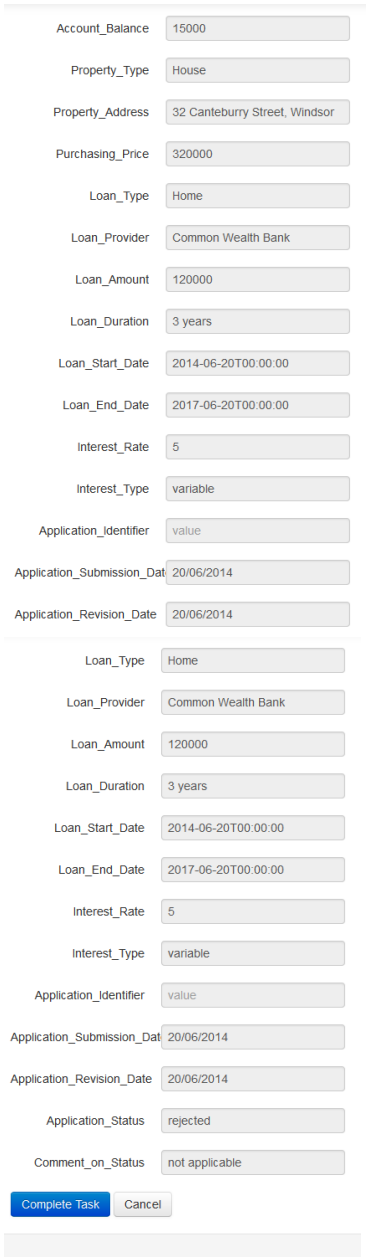
Name	Process	Created
View rejected application		Jun 20, 2014 12:35:11 AM



View rejected application

Created: Jun 20, 2014 12:35:11 AM

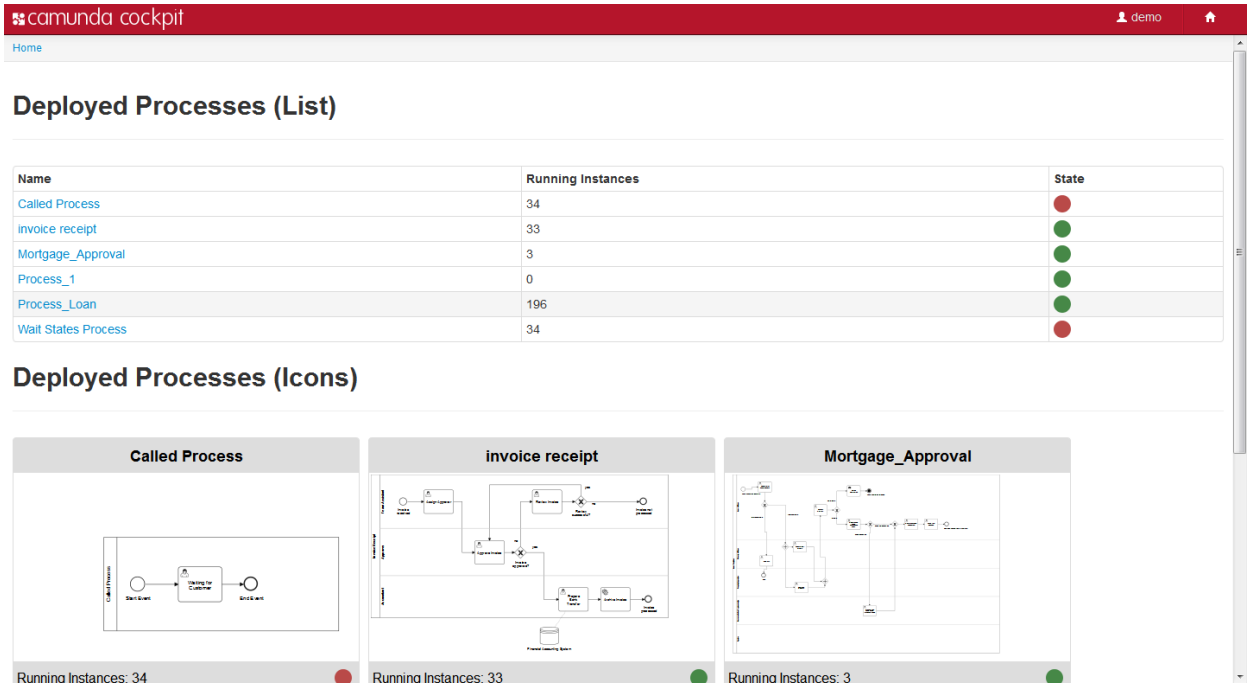
Applicant_Name	Allan
Applicant_SurName	Fernandez
Email	alaferns@gmail.com
Home_Phone	0745695878
Mobile_Phone	0434857658
Current_Address	32 Canteburry Street, Windsor
Previous_Address	32 Canteburry Street, Windsor
Current_Employer	Pizza Hut
Monthly_Net_Revenue	3000
Bank_Name	Common Wealth Bank
Account_Type	Common Access
Account_Number	A39874568


Account_Balance
15000
Property_Type
House
Property_Address
32 Canteburry Street, Windsor
Purchasing_Price
320000
Loan_Type
Home
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
2014-06-20T00:00:00
Loan_End_Date
2017-06-20T00:00:00
Interest_Rate
5
Interest_Type
variable
Application_Identifier
value
Application_Submission_Dat
20/06/2014
Application_Revision_Date
20/06/2014
Loan_Type
Home
Loan_Provider
Common Wealth Bank
Loan_Amount
120000
Loan_Duration
3 years
Loan_Start_Date
2014-06-20T00:00:00
Loan_End_Date
2017-06-20T00:00:00
Interest_Rate
5
Interest_Type
variable
Application_Identifier
value
Application_Submission_Dat
20/06/2014
Application_Revision_Date
20/06/2014
Application_Status
rejected
Comment_on_Status
not applicable
Complete Task
Cancel

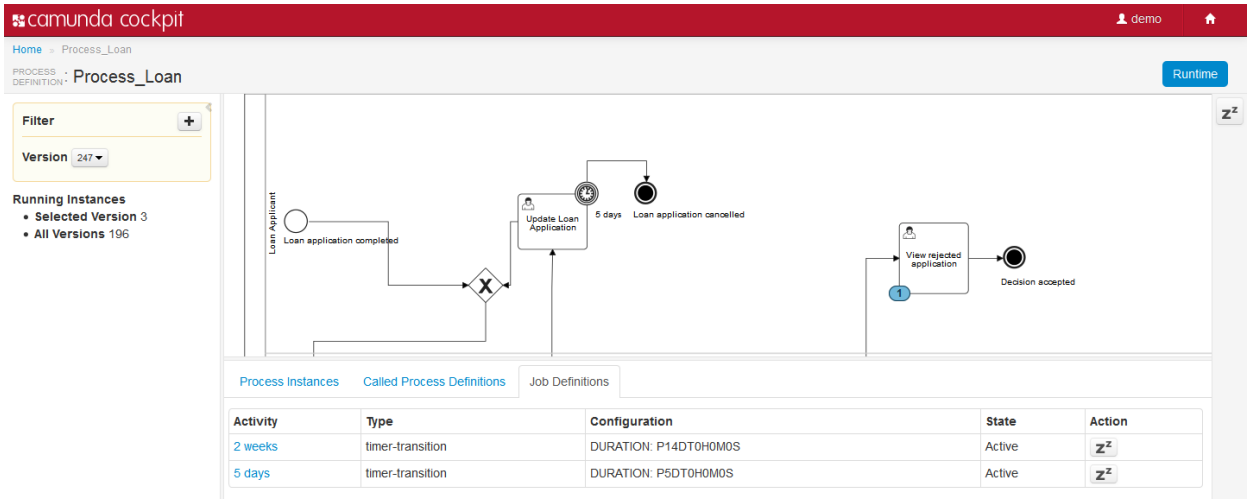
6.5 Scenario 5:

Loan application is cancelled after the 5 day wait period for update loan application

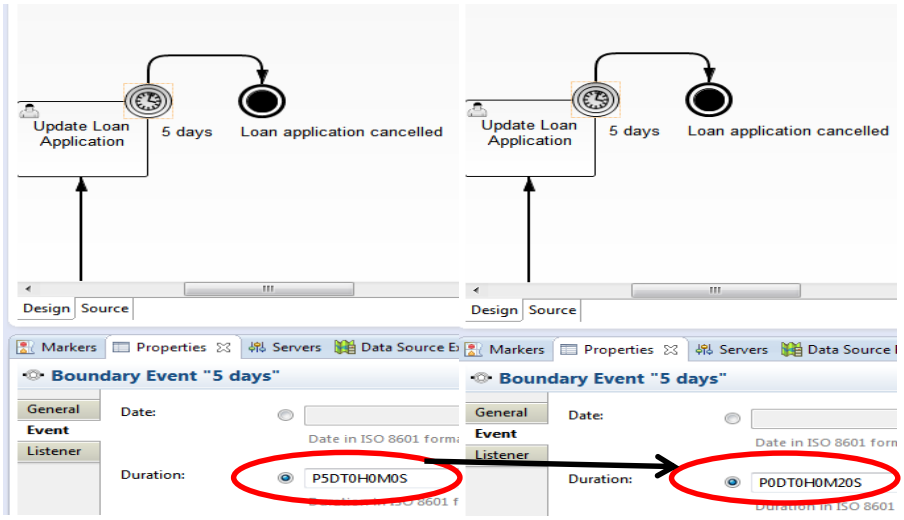
Login as demo/demo into Camunda Cockpit and choose instance Process_Loan

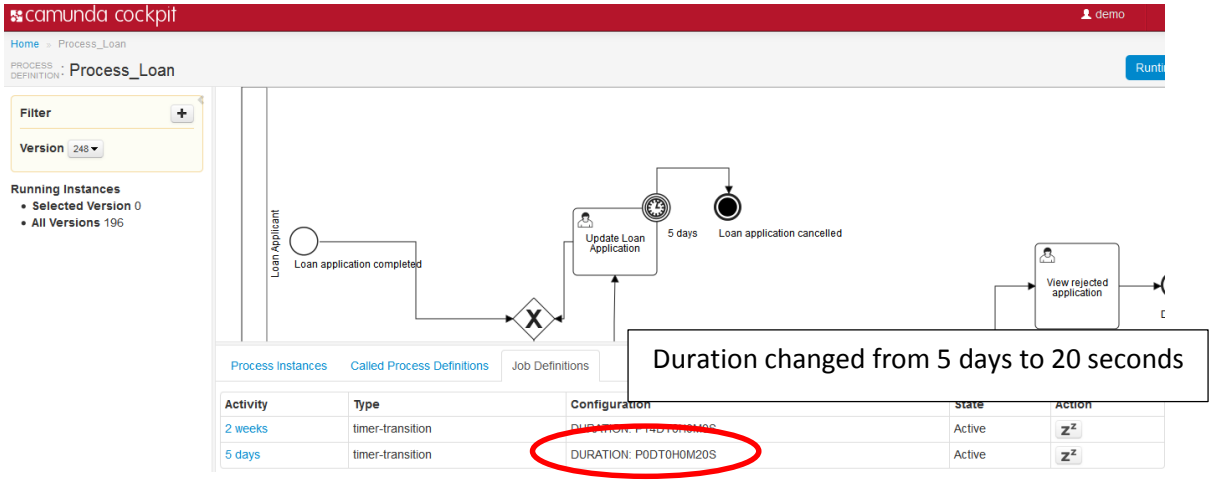


Click on Job Definitions to check the timer-transition events



Change the timer-transition time from 5 days to 20 seconds

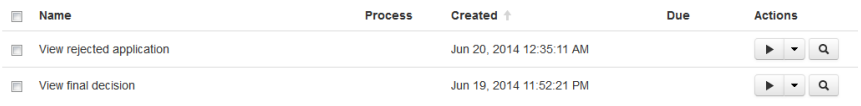




Update loan application is in 20 second wait state

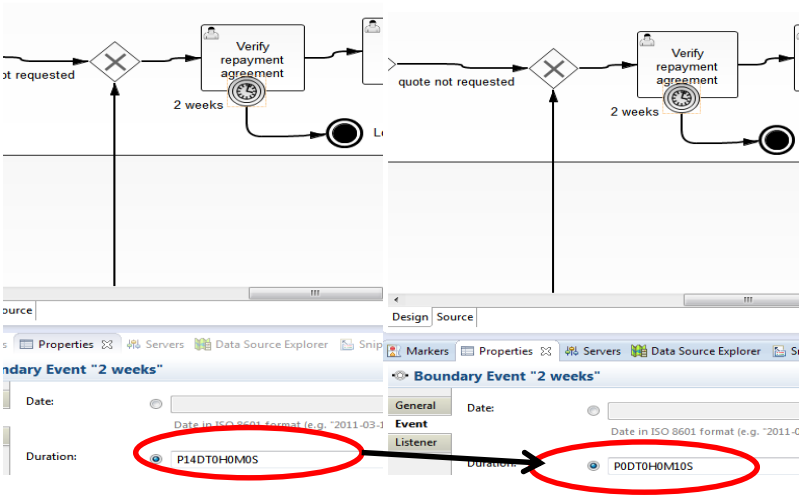


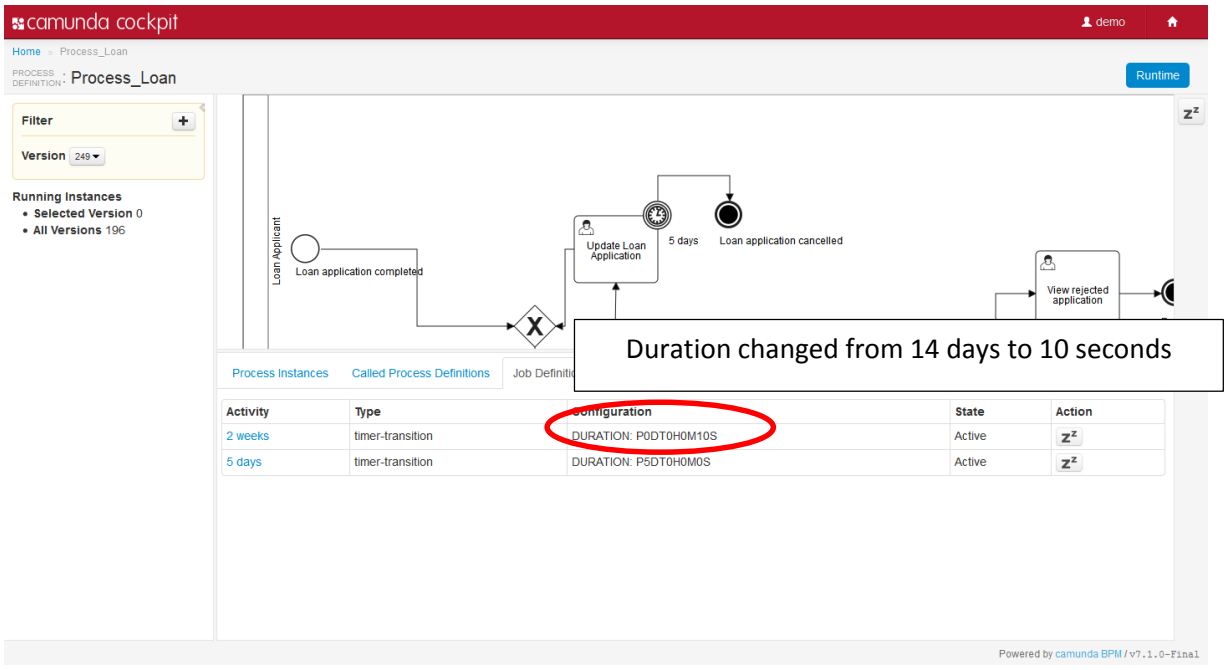
After 20 second wait state



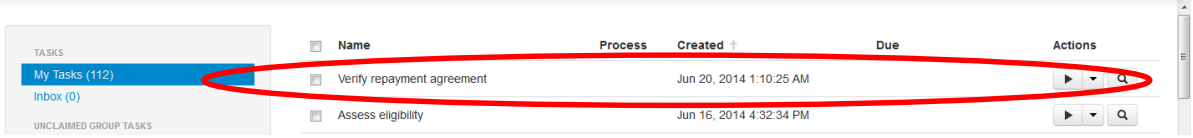
6.6 Scenario 6:

Loan application is cancelled after the 14 day wait period for verify repayment agreement

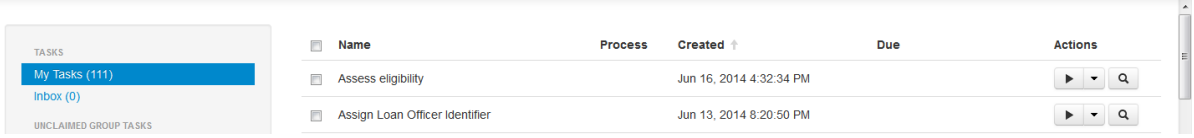




Verify repayment agreement user task is in 10 second wait state

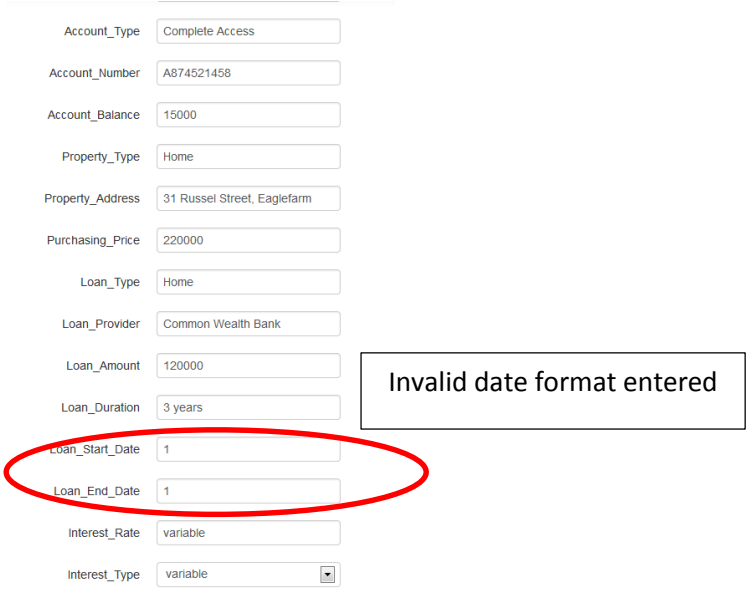


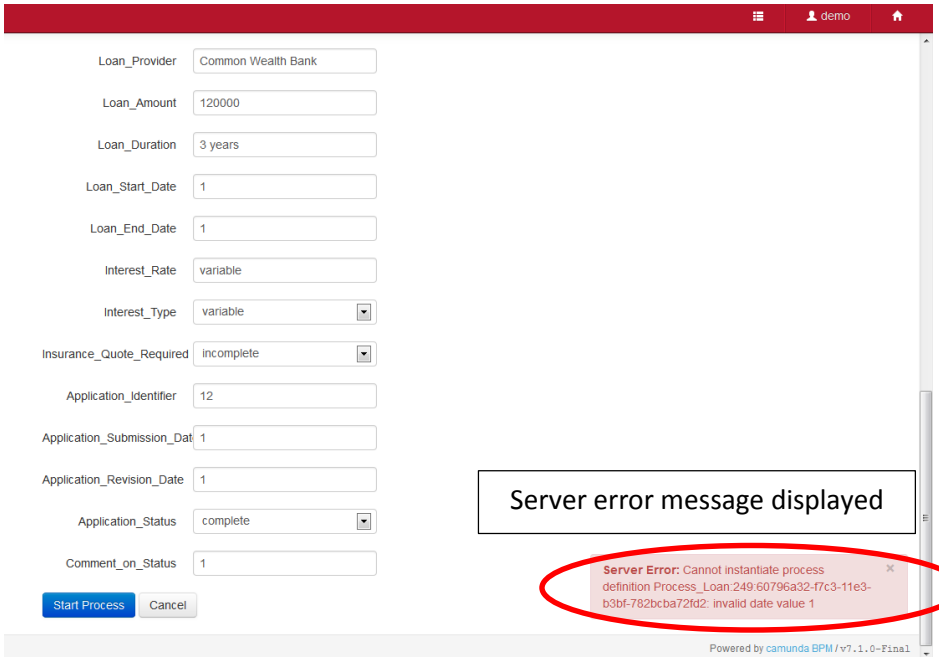
After 10 second wait state



6.7 Scenario 7:

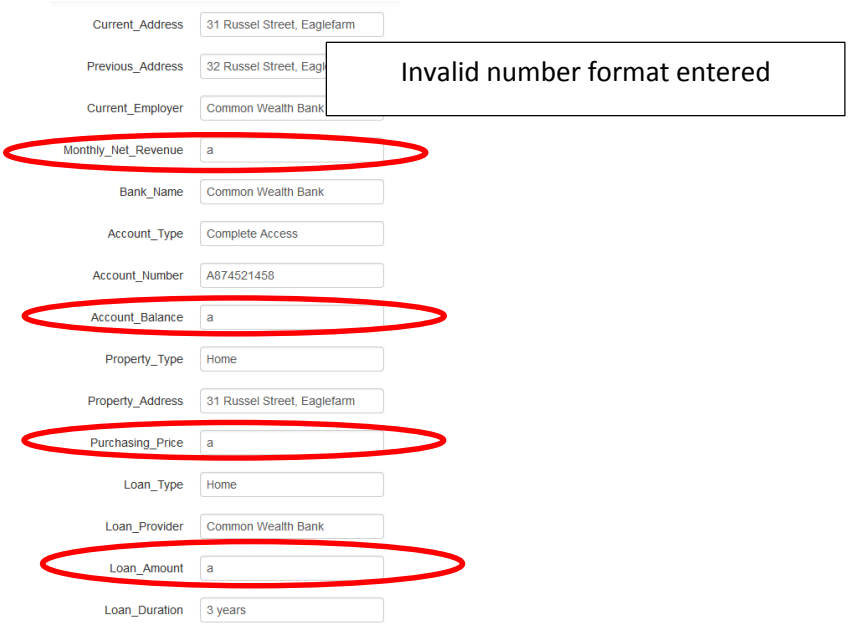
Date fields format check in Loan Application form

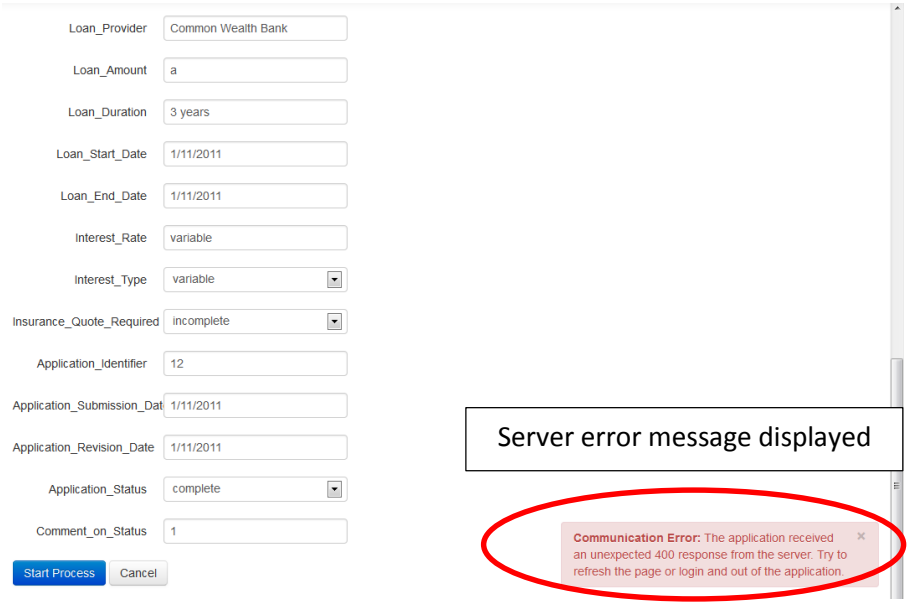




6.8 Scenario 8:

Long fields format check in Loan Application form





7.0 Further Readings

Programming resources:

Groovy

http://www.javabeat.net/introduction-to-groovy-scripting-language/

Groovy Beginner Tutorial

http://groovy.codehaus.org/Beginners+Tutorial

Introduction to Java

http://www.vogella.com/tutorials/JavaIntroduction/article.html

Java Tutorial

http://www.tutorialspoint.com/java/

Online Java Simulator

http://www.learnjavaonline.org/en/Variables_and_Types

Technical resources:

Camunda User Guide

http://docs.camunda.org/latest/guides/user-guide/

Camunda BPMN 2.0 Implementation Reference

http://docs.camunda.org/latest/api-references/bpmn20/

Camunda Rest API

http://docs.camunda.org/latest/api-references/rest/

Camunda Java doc

http://docs.camunda.org/latest/api-references/javadoc/

Camunda Deployment Descriptors

http://docs.camunda.org/latest/api-references/deployment-descriptors/

Camunda examples on GitHub

https://github.com/camunda/camunda-bpm-examples

Camunda Tutorial

http://docs.camunda.org/latest/real-life/how-to/

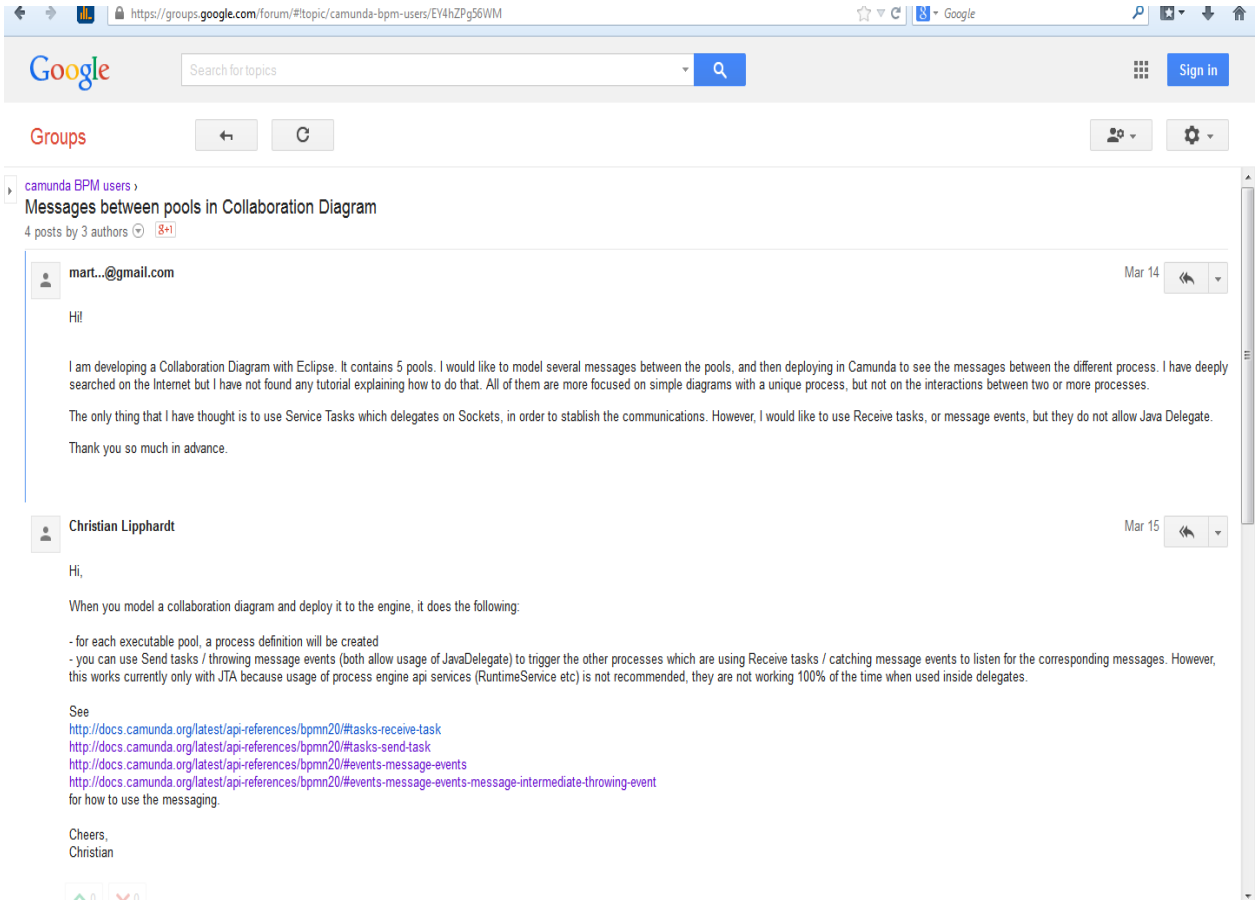
Appendix A: Constraints observed in Camunda BPM Platform

Constraint1:

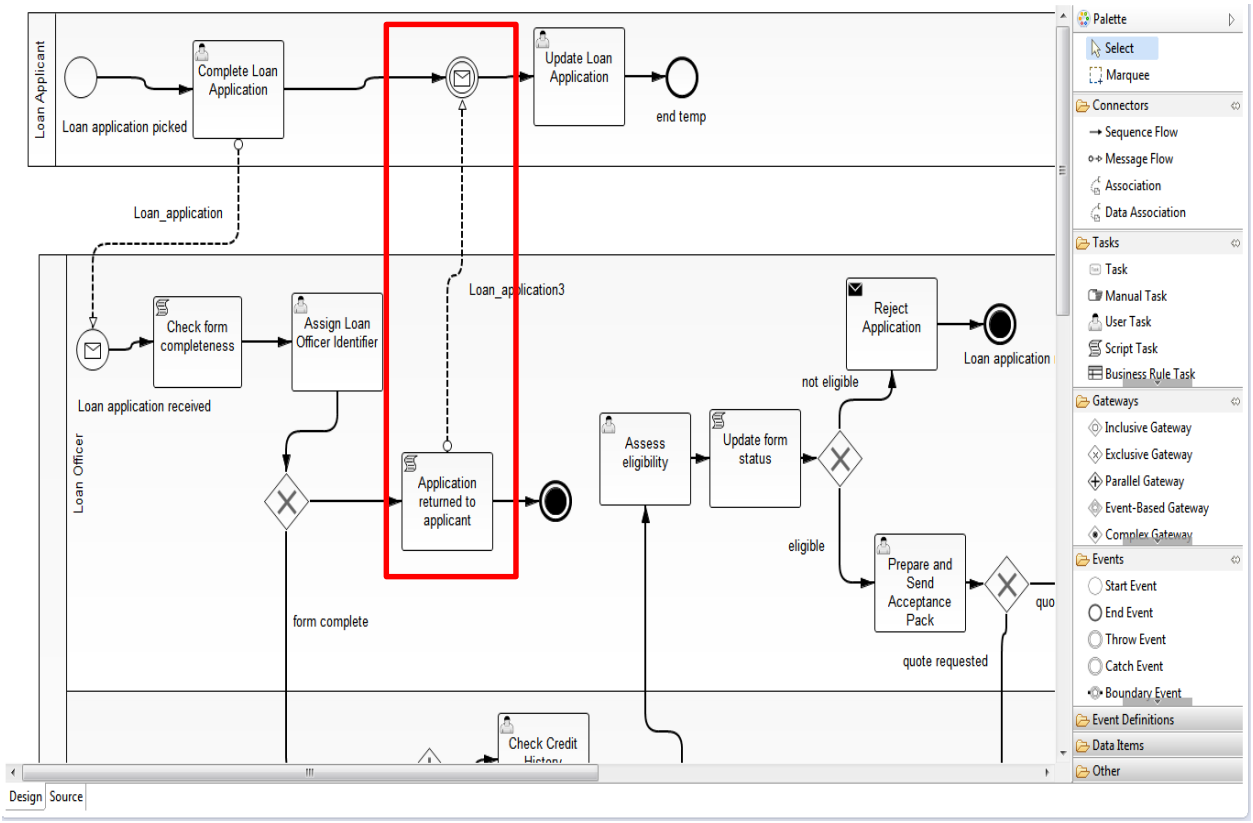
Problem of outgoing message transfer between two different pools

The interaction between the 'Loan Provider' and 'Loan Applicant' pools for task "Application returned to applicant" and catching message event prior to 'Update Loan Application' task is not possible, even though the documented methods on Camunda user guide of assigning events and task notations have been followed.

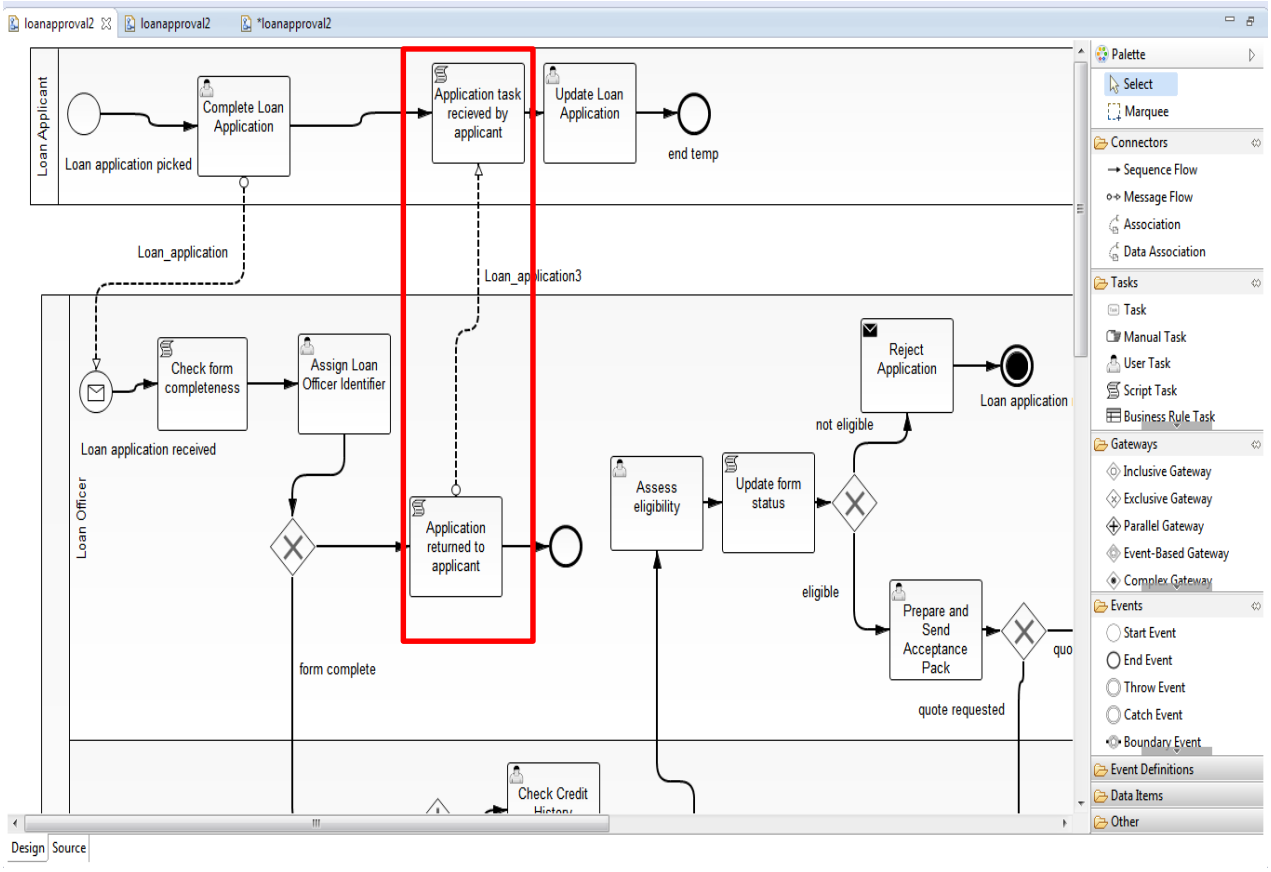
Documentation Link: Camunda User BPM Forum Google Group (A forum for interaction between Camunda users and developers)
https://groups.google.com/forum/#!topic/camunda-bpm-users/EY4hZPg56WM



Attempted Method 1:



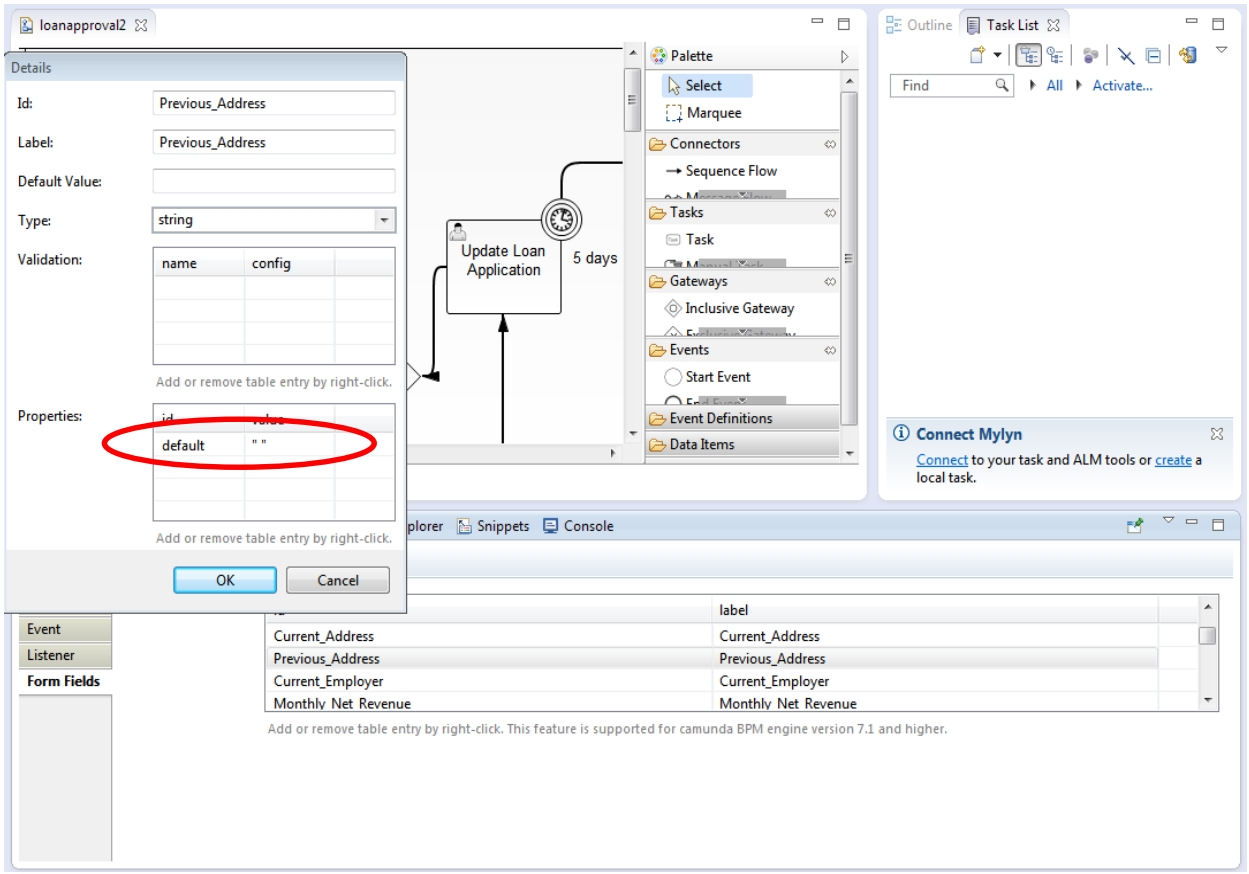
Attempted Method 2:

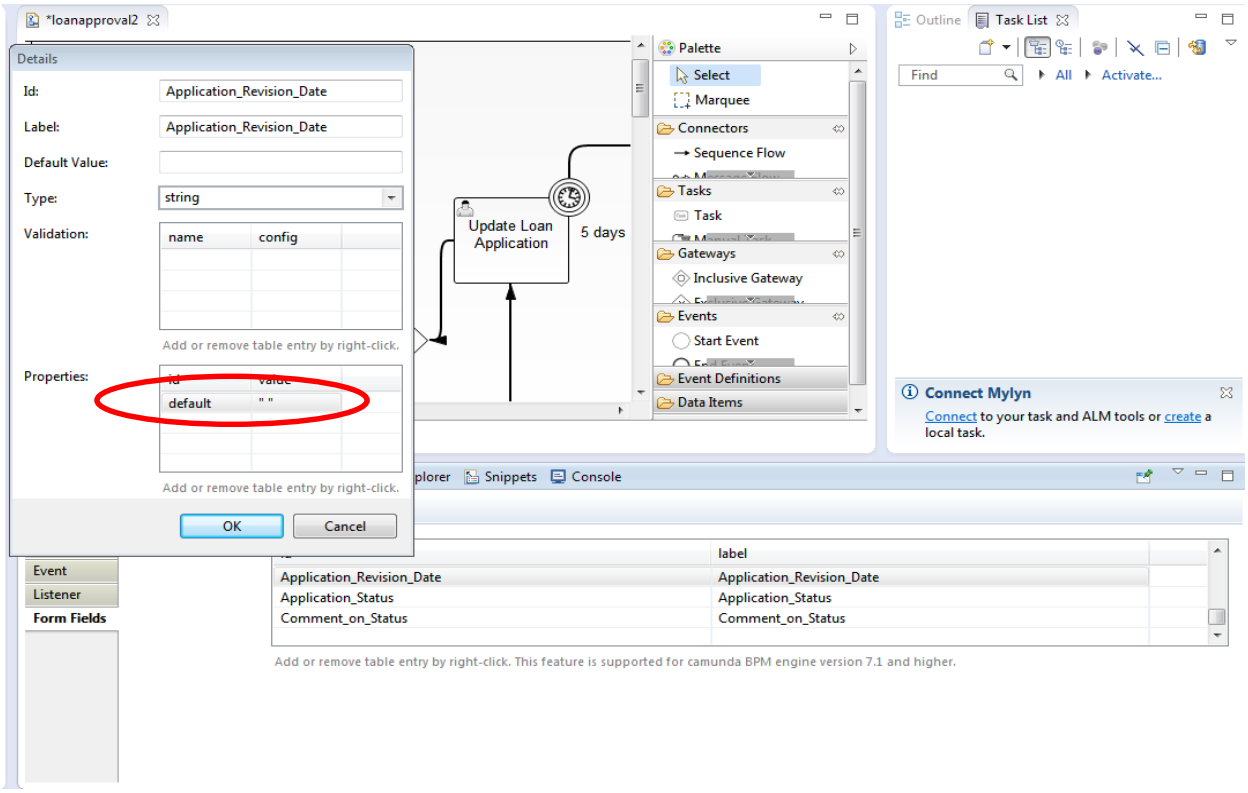


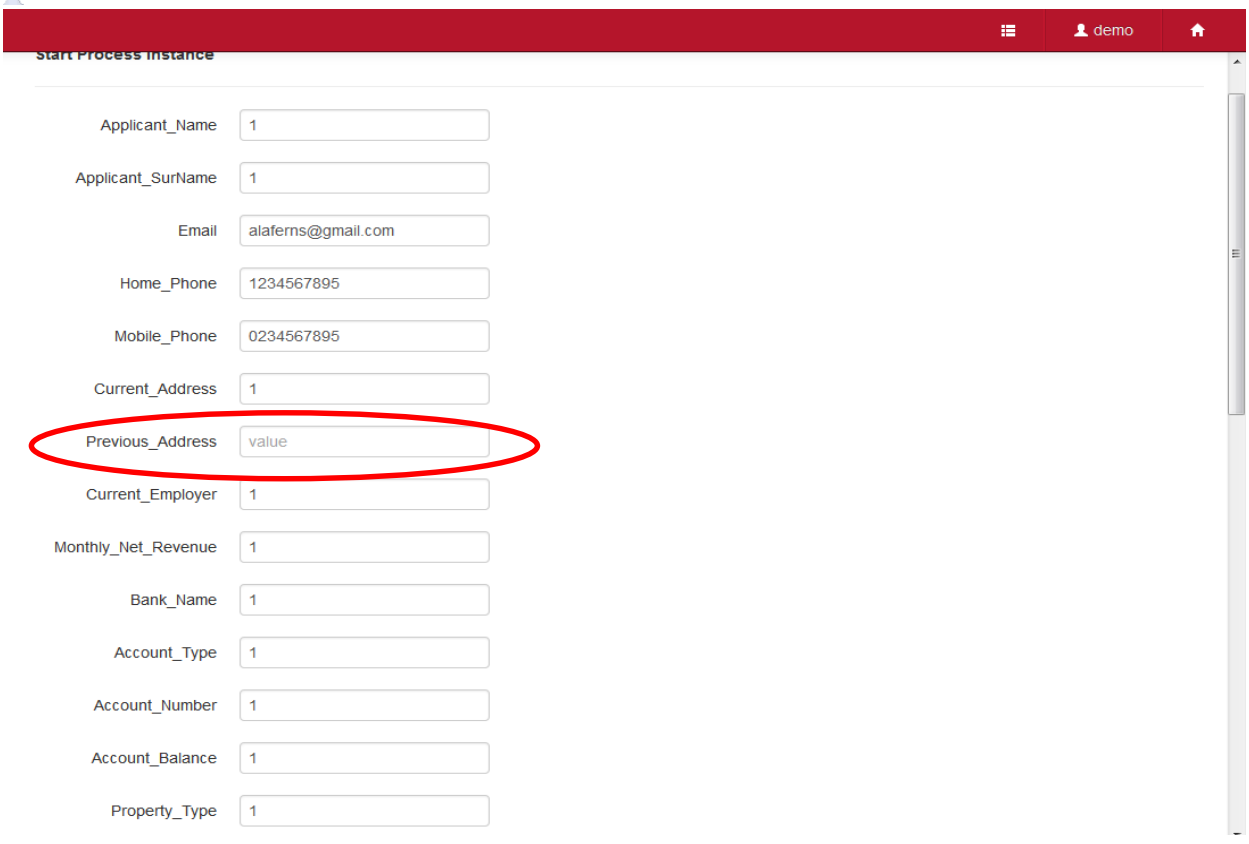
Constraint 2:

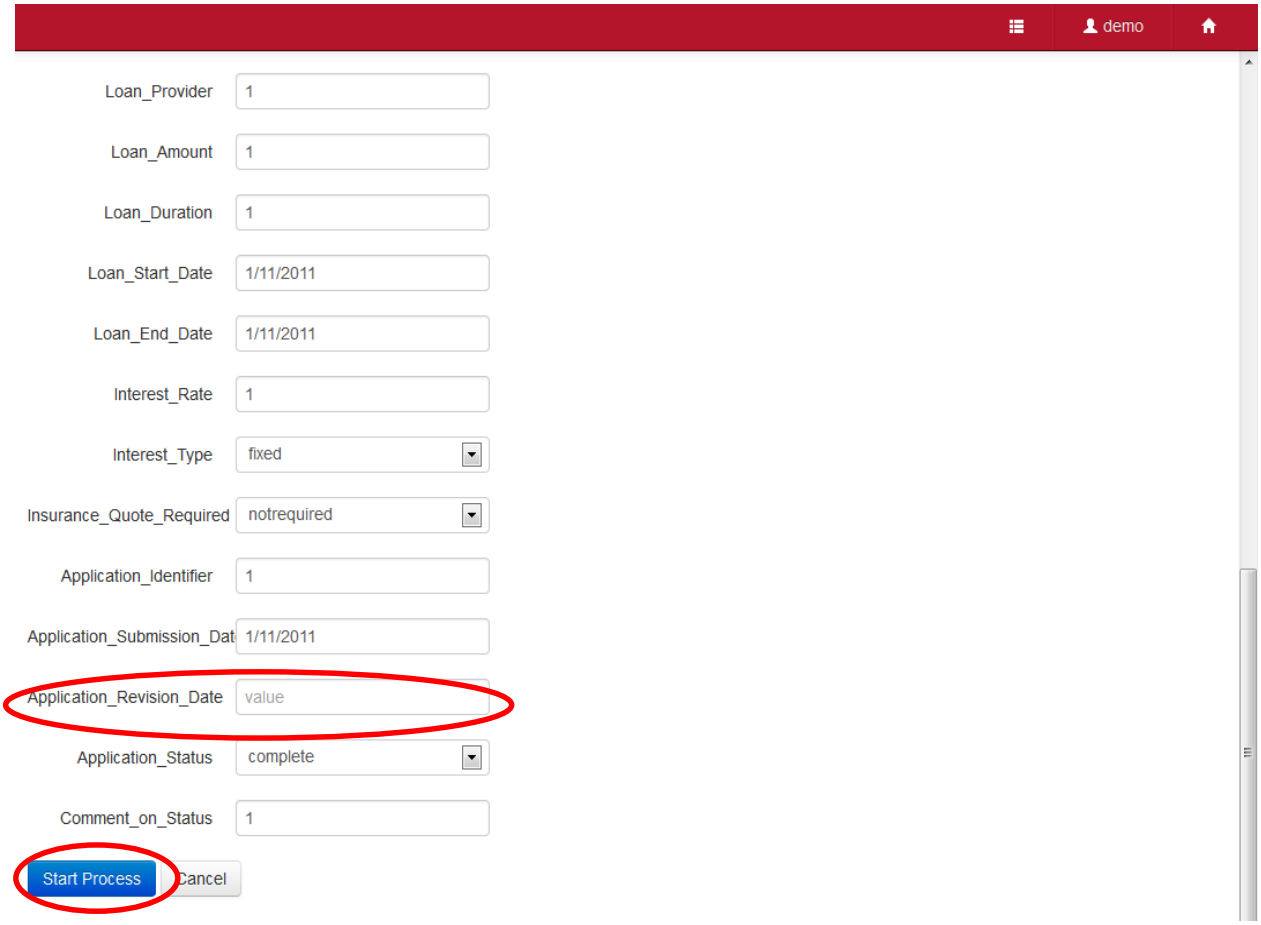
Non-mandatory fields in Camunda

Even if a field is set as ‘non-mandatory’ by entering the condition default “ ” in the properties applet of form field. Camunda Process Engine does not consider it. It works only if one field is marked as default “ “, but if two fields are marked with the same statement, it fails to execute.



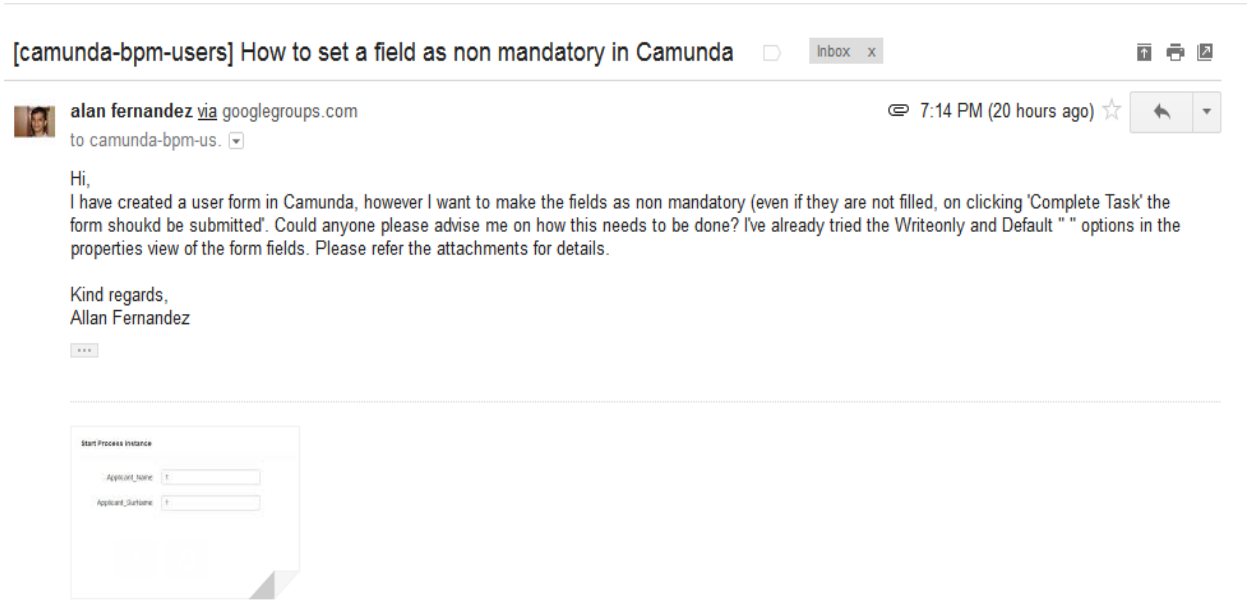

*loanapproval2
Details
Id:
Application_Revision_Date
Label:
Application_Revision_Date
Default Value:
Type:
string
Validation:
name
config
Add or remove table entry by right-click.
Properties:
default
" "
Add or remove table entry by right-click.
OK
Cancel
Update Loan Application
5 days
Palette
Select
Marquee
Connectors
Sequence Flow
Tasks
Task
Gateways
Inclusive Gateway
Events
Start Event
Event Definitions
Data Items
Outline
Task List
Find
All
Activate...
Connect Mylyn
Connect to your task and ALM tools or create a local task.
Snippets
Console
Event
Listener
Form Fields
label
Application_Revision_Date
Application_Revision_Date
Application_Status
Application_Status
Comment_on_Status
Comment_on_Status
Add or remove table entry by right-click. This feature is supported for camunda BPM engine version 7.1 and higher.


demo
Start Process Instance
Applicant_Name
1
Applicant_SurName
1
Email
alaferns@gmail.com
Home_Phone
1234567895
Mobile_Phone
0234567895
Current_Address
1
Previous_Address
value
Current_Employer
1
Monthly_Net_Revenue
1
Bank_Name
1
Account_Type
1
Account_Number
1
Account_Balance
1
Property_Type
1

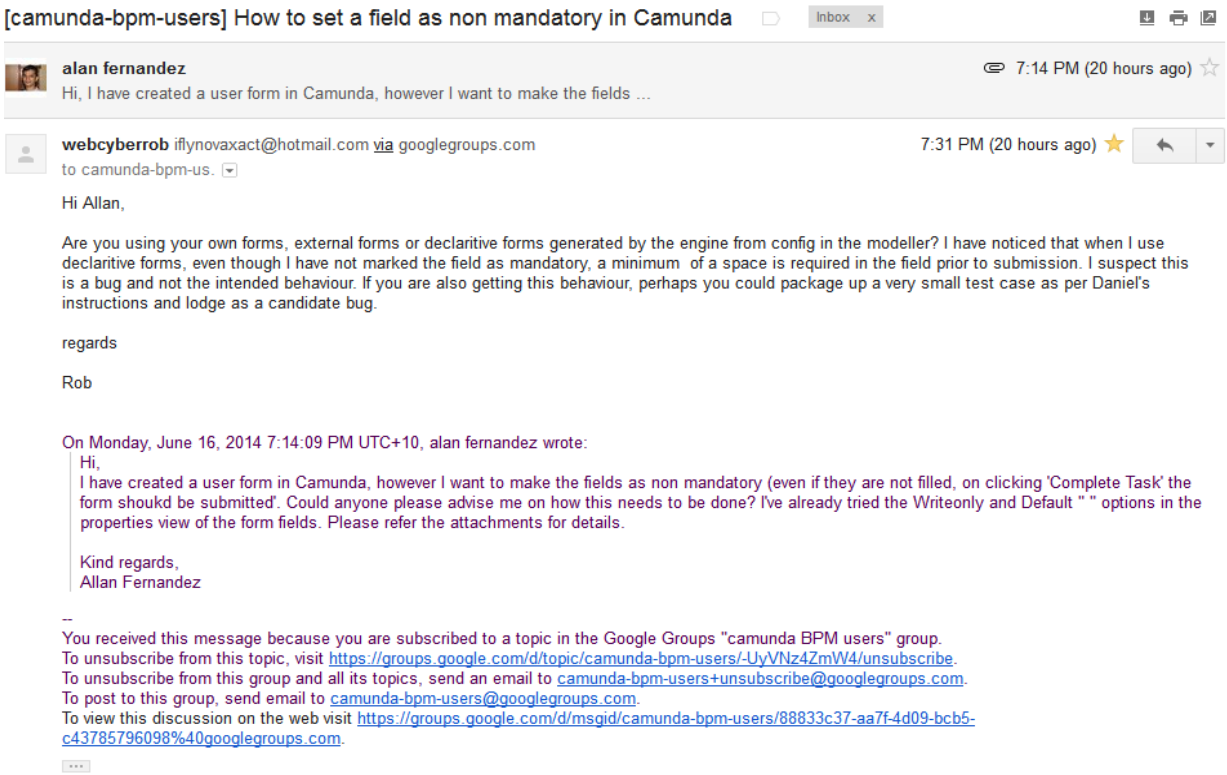


Response from Camunda User BPM Forum Google Group (A forum for interaction between Camunda users and developers):

Question posted:



Response:



[camunda-bpm-users] How to set a field as non mandatory in Camunda

Inbox x

alan fernandez 7:14 PM (20 hours ago)

Hi, I have created a user form in Camunda, however I want to make the fields ...

webcyberrob iflynovaxact@hotmail.com via googlegroups.com 7:31 PM (20 hours ago)

to camunda-bpm-us.

Hi Allan,

Are you using your own forms, external forms or declaritive forms generated by the engine from config in the modeller? I have noticed that when I use declaritive forms, even though I have not marked the field as mandatory, a minimum of a space is required in the field prior to submission. I suspect this is a bug and not the intended behaviour. If you are also getting this behaviour, perhaps you could package up a very small test case as per Daniel's instructions and lodge as a candidate bug.

regards

Rob

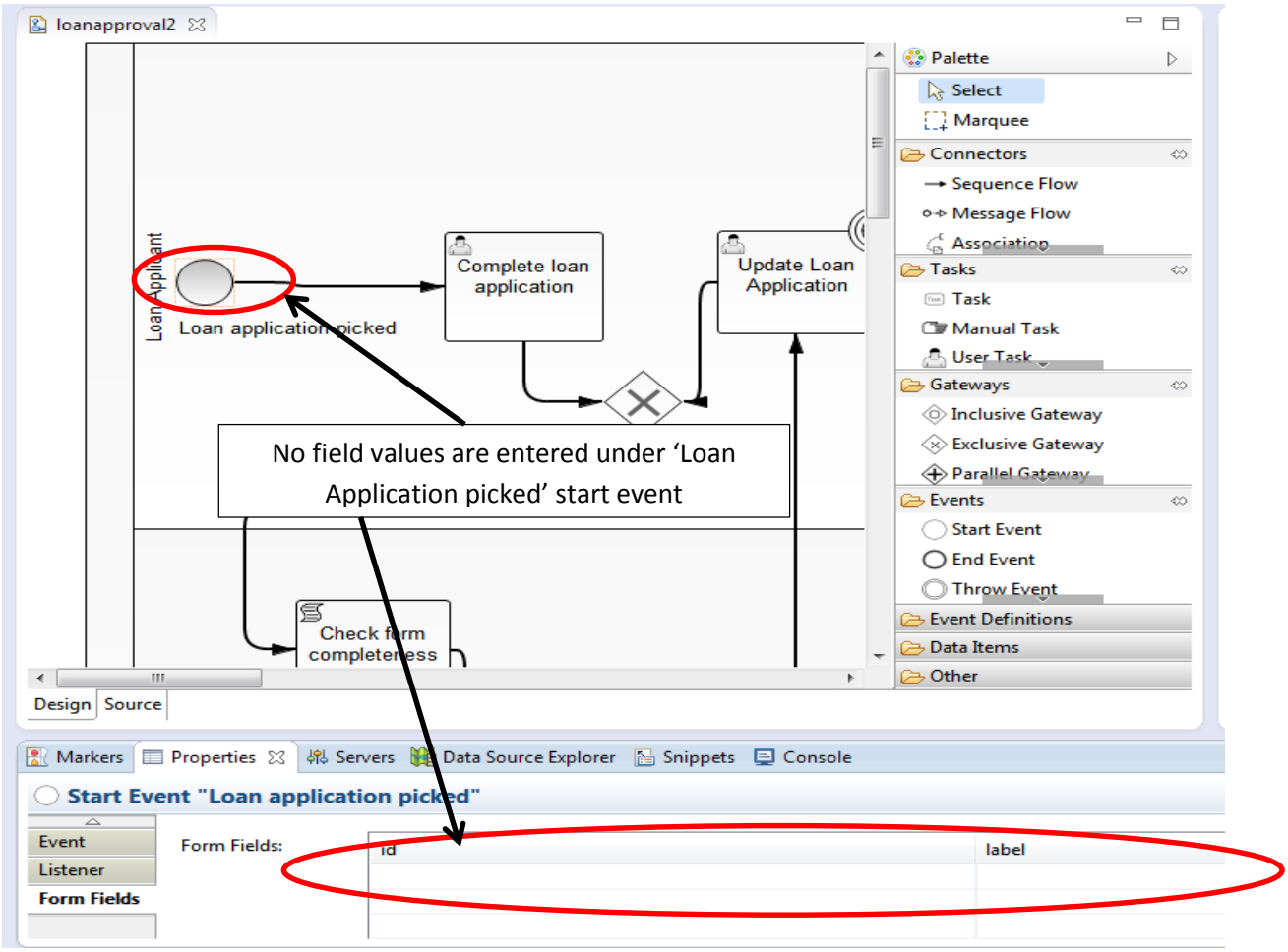
On Monday, June 16, 2014 7:14:09 PM UTC+10, alan fernandez wrote:

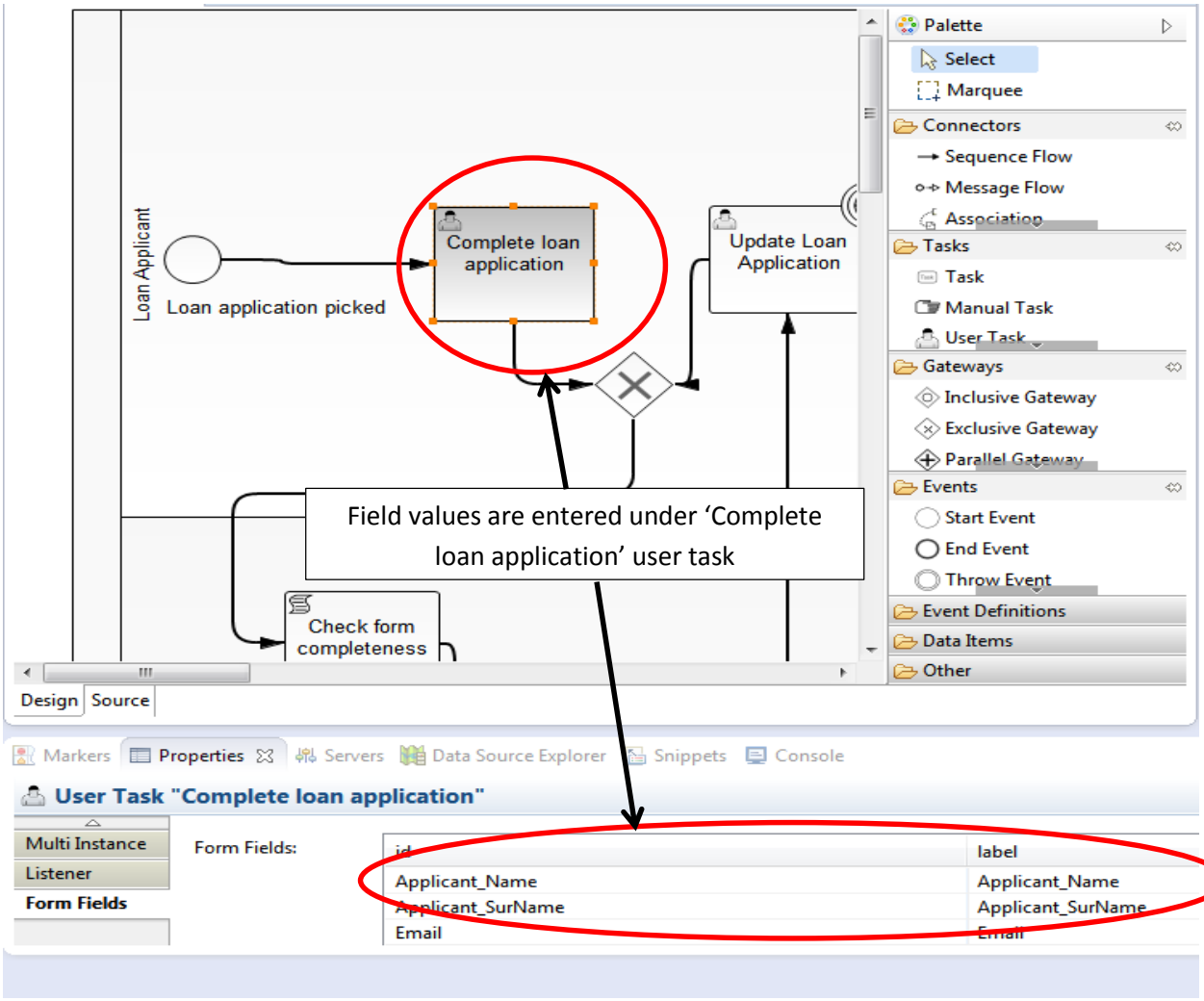
> Hi,
> I have created a user form in Camunda, however I want to make the fields as non mandatory (even if they are not filled, on clicking 'Complete Task' the form shoukd be submitted'. Could anyone please advise me on how this needs to be done? I've already tried the Writeonly and Default " " options in the properties view of the form fields. Please refer the attachments for details.
>
> Kind regards,
> Allan Fernandez

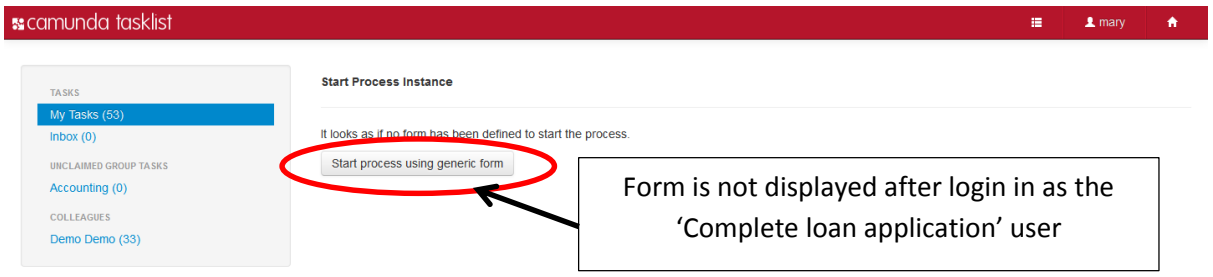
--
You received this message because you are subscribed to a topic in the Google Groups "camunda BPM users" group.
To unsubscribe from this topic, visit https://groups.google.com/d/topic/camunda-bpm-users/-UyVNz4ZmW4/unsubscribe.
To unsubscribe from this group and all its topics, send an email to camunda-bpm-users+unsubscribe@googlegroups.com.
To post to this group, send email to camunda-bpm-users@googlegroups.com.
To view this discussion on the web visit https://groups.google.com/d/msgid/camunda-bpm-users/88833c37-aa7f-4d09-bcb5-c43785796098%40googlegroups.com.

Constraint 3:

If the loan application form is not included in the Start event and instead added under user task: 'Complete Loan Application', the field names are not displayed while running the .war file in Camunda Process Engine

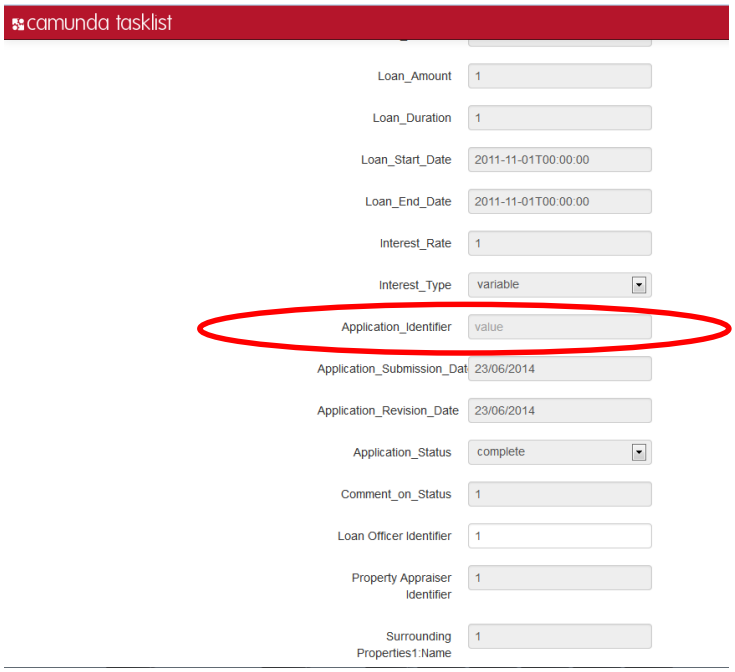


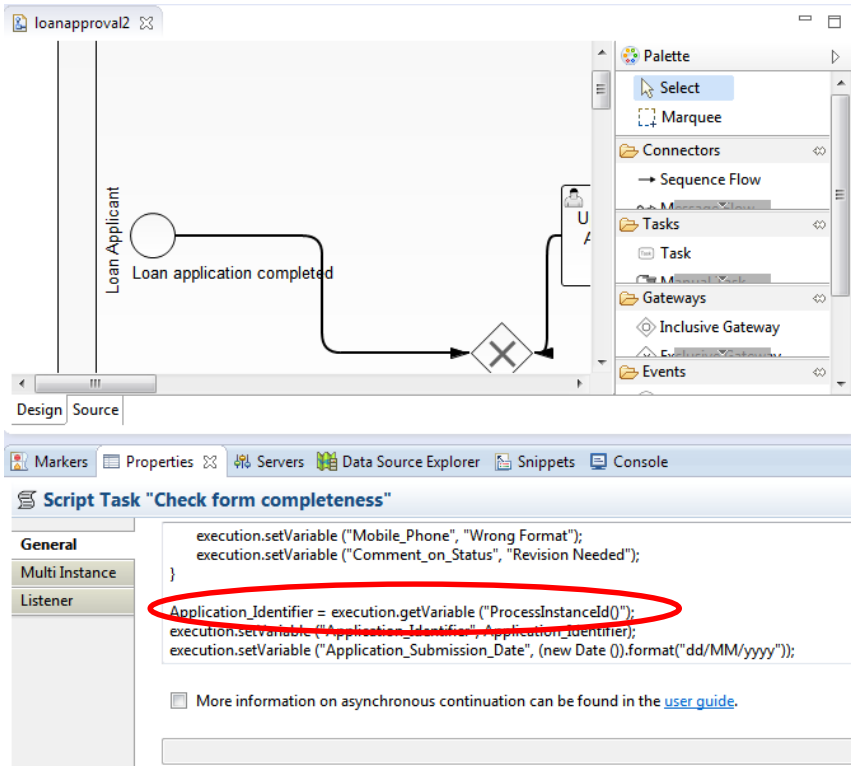

Loan Applicant
Loan application picked
Complete loan application
Update Loan Application
Check form completeness
Palette
Select
Marquee
Connectors
Sequence Flow
Message Flow
Association
Tasks
Task
Manual Task
User Task
Gateways
Inclusive Gateway
Exclusive Gateway
Parallel Gateway
Events
Start Event
End Event
Throw Event
Event Definitions
Data Items
Other
Design
Source
Field values are entered under 'Complete loan application' user task
Markers
Properties
Servers
Data Source Explorer
Snippets
Console
User Task "Complete loan application"
Multi Instance
Listener
Form Fields
Form Fields:
id
label
Applicant_Name
Applicant_Name
Applicant_SurName
Applicant_SurName
Email
Email


camunda tasklist
mary
TASKS
My Tasks (53)
Inbox (0)
UNCLAIMED GROUP TASKS
Accounting (0)
COLLEAGUES
Demo Demo (33)
Start Process Instance
It looks as if no form has been defined to start the process.
Start process using generic form
Form is not displayed after login in as the 'Complete loan application' user

Constraint 4:

Unable to automatically assign unique ids for Application Identifiers through Camunda Process Id field. In order to assign Application Identifiers, the Camunda developers were contacted through Camunda user BPM forum google group. Their suggestions were implemented, however the issue is unresolved.





Response from Camunda User BPM Forum Google Group (A forum for interaction between Camunda users and developers):

Question posted:

[camunda-bpm-users] How to assign Process ID and user name to script task fields in camunda

Inbox x

alan fernandez via googlegroups.com Jun 16 (8 days ago)
to camunda-bpm-us.

Hi guys,

I am attempting to write a script task where I need to process ID and User name (Eg:John) to be automatically assigned to field names (Eg: John must be assigned to field name:Loan_Identifier,
Process ID must be assigned to field name: Application_Identifier). Could anyone please advise me on what queries I must input to execute this?
I am working on Eclipse Kepler using Camunda Modeler plugin.

Kind regards,
Allan Fernandez

alan fernandez via googlegroups.com Jun 16 (8 days ago)
to camunda-bpm-us.

I am running a groovy svript for the above tasks

On Monday, June 16, 2014 6:29:27 PM UTC+10, alan fernandez wrote:
> Hi guys,
>
> I am attempting to write a script task where I need to process ID and User name (Eg:John) to be automatically assigned to field names (Eg: John must be assigned to field name:Loan_Identifier,
> Process ID must be assigned to field name: Application_Identifier). Could anyone please advise me on what queries I must input to execute this?
> I am working on Eclipse Kepler using Camunda Modeler plugin.
>
> Kind regards,
> Allan Fernandez

--

Response:

webcyberrob iflynovaxact@hotmail.com via googlegroups.com Jun 16 (8 days ago)
to camunda-bpm-us.

Hi Alan,

In your scrip task, you can use a 'special' variable called execution which gives you access to the engine API. Hence you could call execution.getVariable("name") to get the value of a process variable or execution.setVariable(name, value) to set a process variable. From the execution object you should be able to lookup the process instance id etc.

Im a bit puzzled what you meant by fields as I would normally associate fields with a form associated with a user task. hence do you mean you want to set process variables in a script task prior to displaying in a user task?

regards

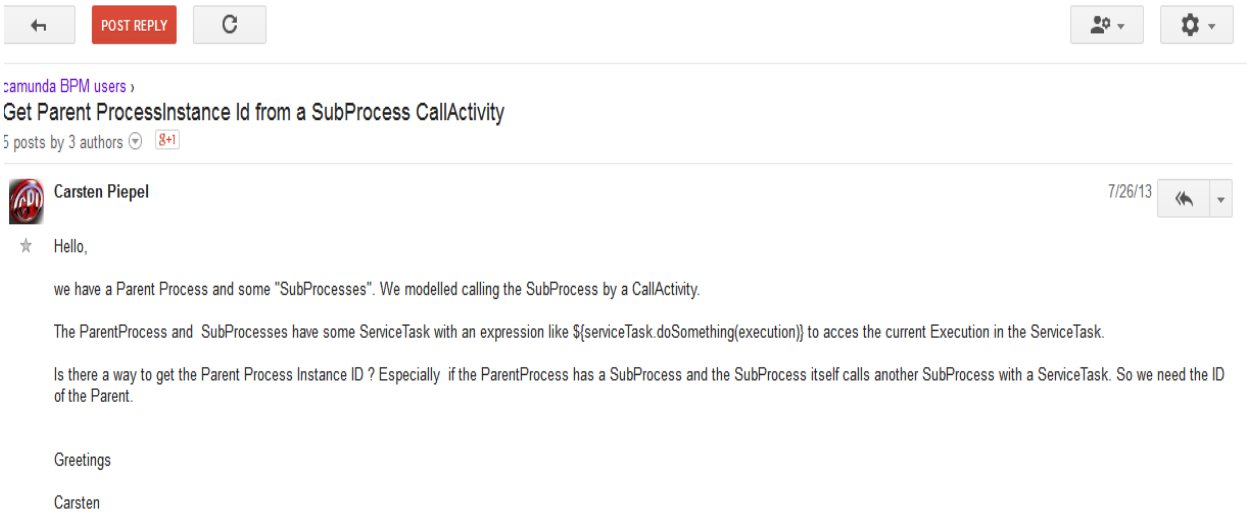
Rob

Another similar issue raised in the forum

Documentation Link: Camunda User BPM Forum Google Group (A forum for interaction between Camunda users and developers)

https://groups.google.com/forum/#!topic/camunda-bpm-users/W0JgTJ3SWHY

Question posted:



POST REPLY

camunda BPM users ›

Get Parent ProcessInstance Id from a SubProcess CallActivity

5 posts by 3 authors

Carsten Piepel 7/26/13

Hello,

we have a Parent Process and some "SubProcesses". We modelled calling the SubProcess by a CallActivity.

The ParentProcess and SubProcesses have some ServiceTask with an expression like ${serviceTask.doSomething(execution)} to acces the current Execution in the ServiceTask.

Is there a way to get the Parent Process Instance ID ? Especially if the ParentProcess has a SubProcess and the SubProcess itself calls another SubProcess with a ServiceTask. So we need the ID of the Parent.

Greetings

Carsten

Response:

Daniel Meyer

Hi Carsten,

I just checked, in org.camunda.bpm.engine.delegate.DelegateExecution there seems to be a method missing "getSuperProcessInstanceId()"

Would that satisfy your requirements?

I created a Jira: https://app.camunda.com/jira/browse/CAM-1042

Until we provide such a method, you would have to cast Execution to internal implementation class ExecutionEntity and get the Id from there:

```
ExecutionEntity executionEntity = (ExecutionEntity) delegateExecution;
ExecutionEntity processInstance = executionEntity.getProcessInstance();

ExecutionEntity superExecution = processInstance.getSuperExecution();
if(superExecution != null) {
  String superProcessInstanceId = superExecution.getProcessInstanceId();
}
```

(untested, I hope it works J)

Modified the code as per the above sample, same issue exists.

Constraint 5:

Unable to send email by selecting the variable entered from “Email” field in the loan application form. Contacted Camunda developers through Camunda user BPM forum google group. Attempted the suggestions posted, but issue exists. The same code executes perfectly in eclipse.

Question posted:

alan fernandez via googlegroups.com 2:17 PM (12 hours ago)
to camunda-bpm-us.

Hi,

I am currently attempting to send an email to a user whenever a user task is assigned. I used the codes from the following blog: http://blog.camunda.org/2013/10/how-to-send-email-when-usertask-is.html and modified it as per the requirements. However, the following error message is being displayed whenever I try to executed it:

I am working on Eclipse Kepler Maven Project with Camunda BPM Modeller installed.

Error Message:
Server Error: Cannot submit task form 32fbe808-fa89-11e3-ba8f-782bcba72fd2: Exception while invoking TaskListener: couldn't instantiate class org.camunda.bpm.getstarted.TaskAssignmentListener

Email Task Code:

```
package org.camunda.bpm.getstarted.TaskAssignmentListener;


   import java.util.logging.Level;
   import java.util.logging.Logger;

   import org.apache.commons.mail.Email;
   import org.apache.commons.mail.SimpleEmail;
   import org.camunda.bpm.engine.IdentityService;
   import org.camunda.bpm.engine.delegate.DelegateTask;
   import org.camunda.bpm.engine.delegate.TaskListener;
   import org.camunda.bpm.engine.identity.User;
   import org.camunda.bpm.engine.impl.context.Context;

   public class TaskAssignmentListener implements TaskListener {

     // TODO: Set Mail Server Properties
     private static final String HOST = "smtp.gmail.com";
     private static final String USER = "camunda01@gmail.com";
     private static final String PWD = "Camunda1234";


     private final static Logger LOGGER = Logger.getLogger(TaskAssignmentListener.class.getName());
```

Response:

to camunda-bpm-us:

Hi, an alternative is the javamail-runtime.
pom.xml:

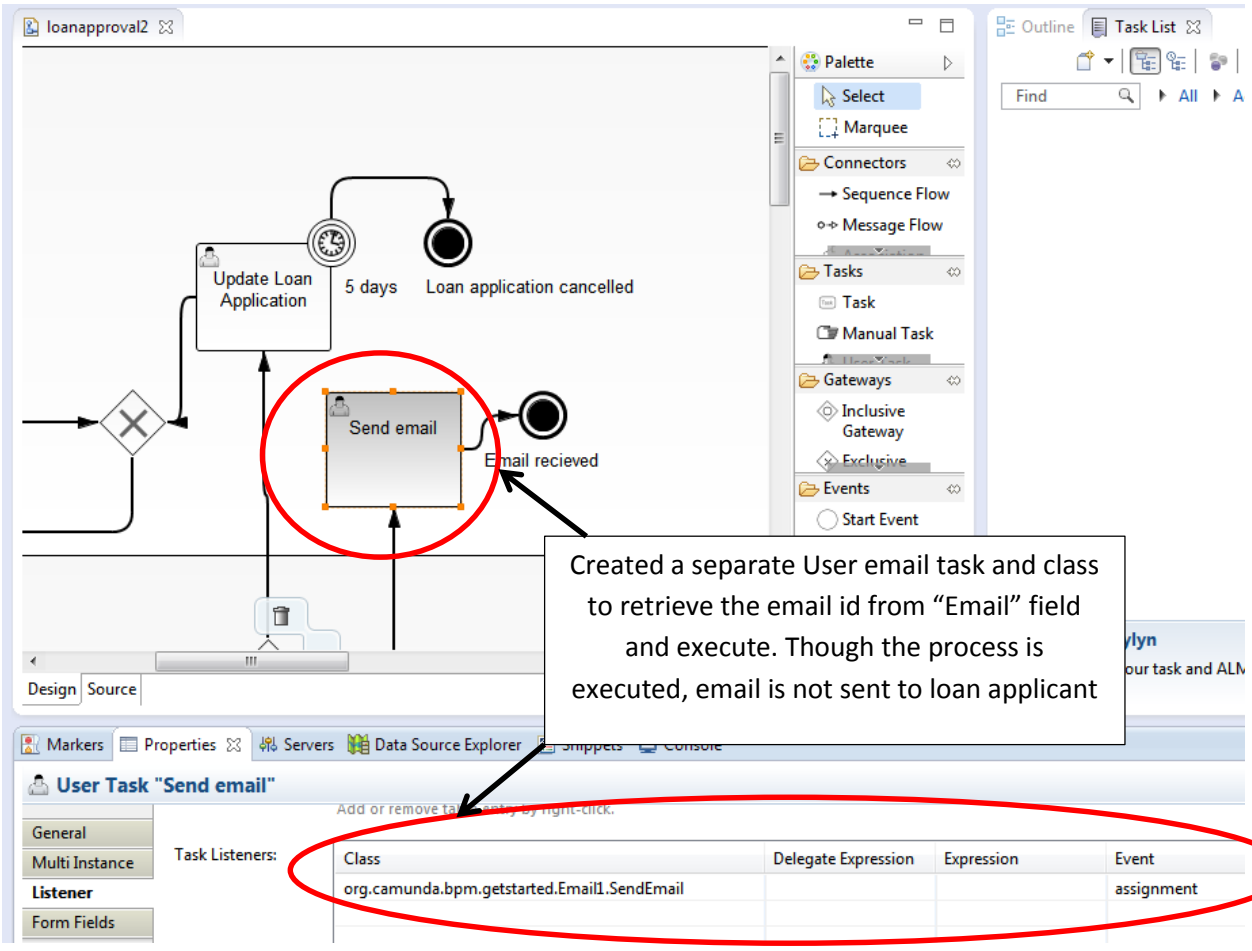
```
<dependency>
    <groupId>org.glassfish.main.resources</groupId>
    <artifactId>javamail-runtime</artifactId>
    <version>4.0-b33</version>
    <classifier>sources</classifier>
</dependency>
```

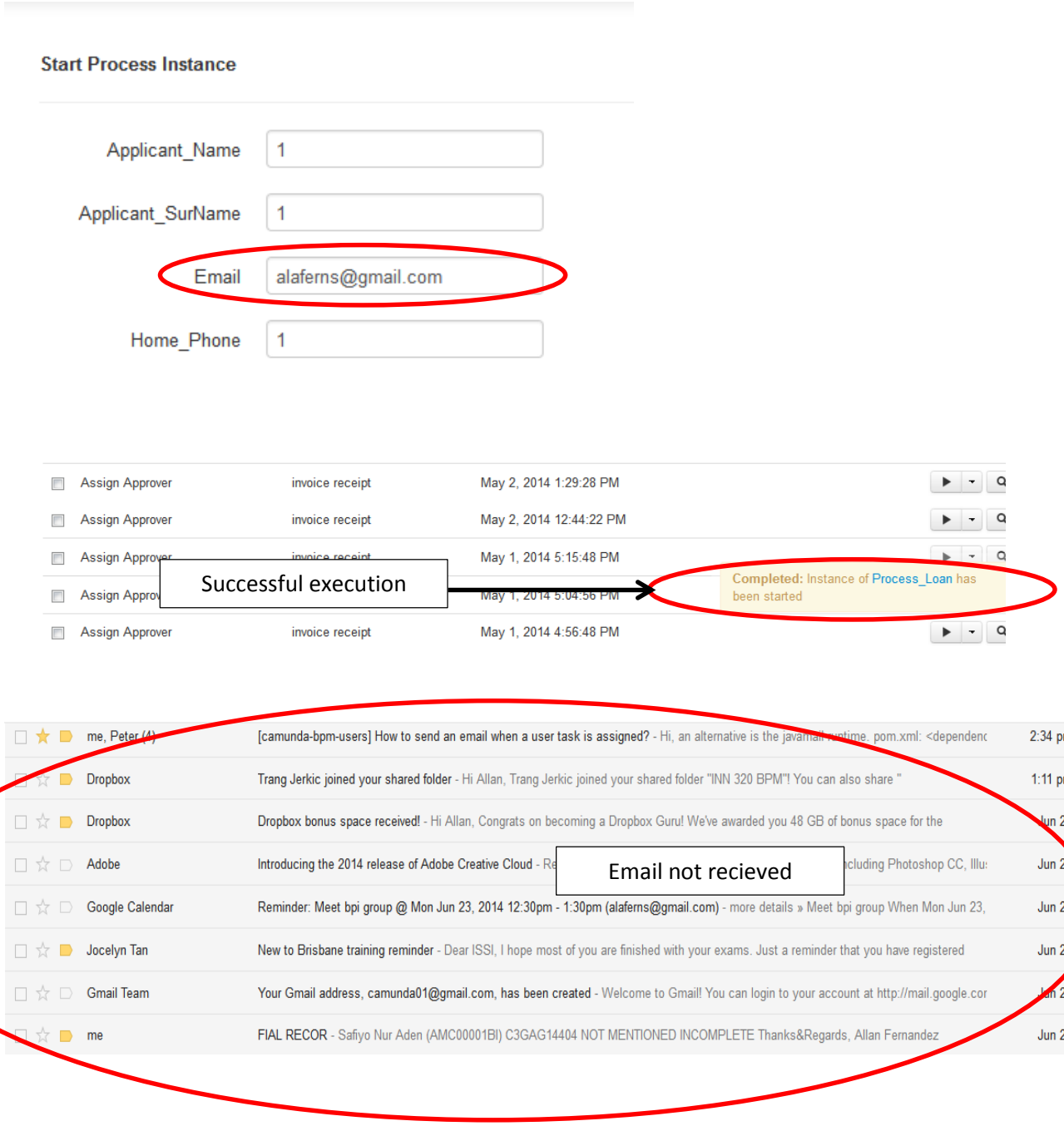
Your email code (there are quite good examples in the web):

```
...
import javax.mail.*;
import javax.mail.internet.*;
...
```

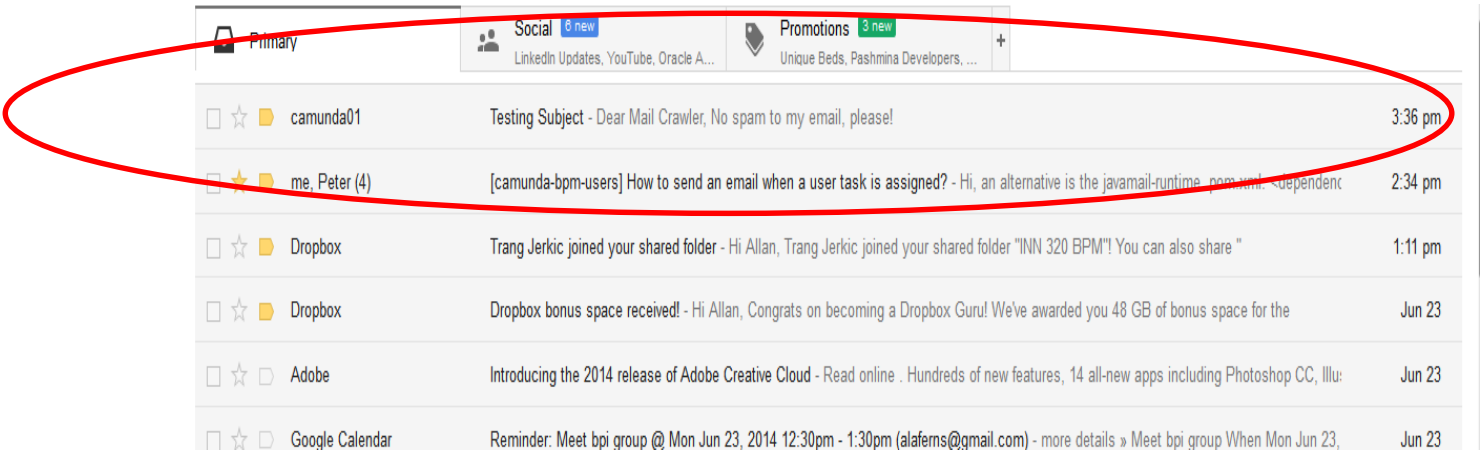
Regards, Peter

Attempted Method 1:





However, when the same Java code is executed in Eclipse, the email is being delivered to the loan applicant's inbox



Java Code:

```
package org.camunda.bpm.getstarted.Email1;

import java.util.*;

import javax.mail.*;
import javax.mail.internet.*;

import org.camunda.bpm.engine.delegate.DelegateTask;
import org.camunda.bpm.engine.delegate.TaskListener;


public class SendEmail implements TaskListener {

        static String host1 = "smtp.gmail.com";
        static String user = "camunda01@gmail.com";
        static String pass = "Camunda1234";

        public void notify(DelegateTask delegateTask) {

          String email= getVariable ("Email");
    // Recipient's email ID needs to be mentioned.
    String to = email;

    // Sender's email ID needs to be mentioned
    String from = "Camunda01@gmail.com";


    // Get system properties
    Properties properties = System.getProperties();


        // Setup mail server

    properties.put("mail.smtp.socketFactory.port", "465");
          properties.put("mail.smtp.socketFactory.class", "javax.net.ssl.SSLSocketFactory");
    properties.put("mail.smtp.starttls.enable", "true");
    properties.put("mail.smtp.host", host1);
    properties.put("mail.smtp.user", user);
    properties.put("mail.smtp.password", pass);
    properties.put("mail.smtp.port", "465");
    properties.put("mail.smtp.auth", "true");
```

```
    Session session = Session.getDefaultInstance(properties,
                        new javax.mail.Authenticator() {

                                protected PasswordAuthentication getPasswordAuthentication() {
                                        return new PasswordAuthentication(user,pass);
                                }
                        });

                try {

                        Message message = new MimeMessage(session);
                        message.setFrom(new InternetAddress(from));
                        message.setRecipients(Message.RecipientType.TO,
                                        InternetAddress.parse(to));
                        message.setSubject("Testing Subject");
                        message.setText("Dear Mail Crawler," +
                                        "\n\n No spam to my email, please!");

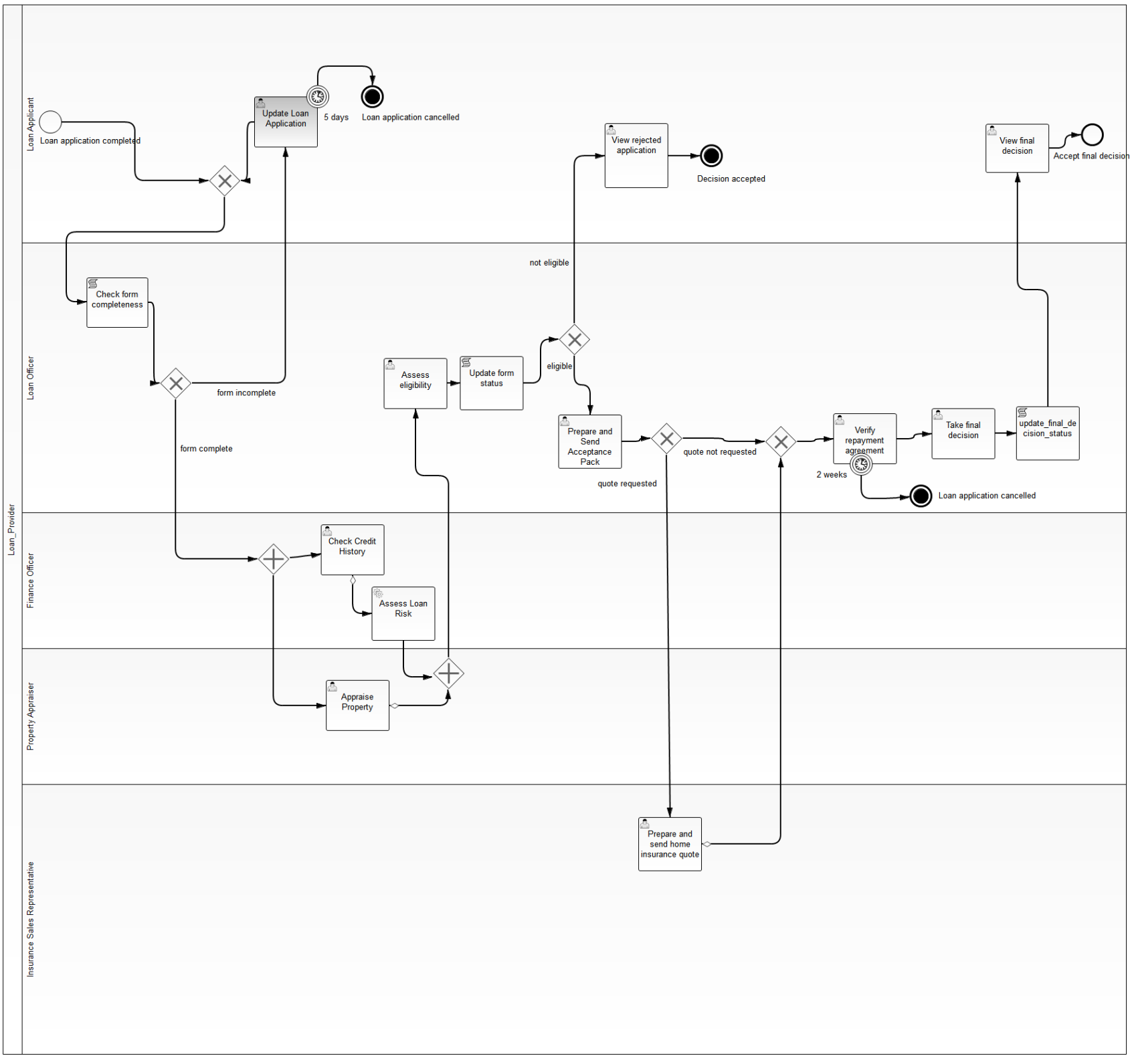
                        Transport.send(message);

                        System.out.println("Done");

                } catch (MessagingException e) {
                        throw new RuntimeException(e);
                }
  }

private static String getVariable(String string) {
        // TODO Auto-generated method stub
        return null;
}
}
```

Appendix B: Loan Assessment Process without Email Task



Appendix C: Loan Assessment Process with Email Task (Execution failure in Camunda BPM Platform)

